PIC – Workflow Flow Audit: full flow or broken?

Date: 7 November 2026, updated 8 November 2026 after the fixes. Code baseline of the audit: `main` at `f354bbe` (migration head 0022). The fixes are migrations 0023 to 0025 on top of it. Written for: the product owner, as the answer to "is every workflow implemented as a full flow, or does it break?", with the evidence.

0. Status after the fixes (8 November 2026)

All 16 defects in section 3 are fixed, and so are the gaps in section 4, except four items that need inputs from the client (listed at the end of this section). Every step the audit graded BROKEN, PARTIAL or NOT BUILT was driven again through the API on fresh scratch databases and now passes.

How the fixes were checked

- **Regression tests.** Each fix has one or more new tests:
  - `test_flow_fixes_plan.py` (18 tests, M01 to M04);
  - `test_flow_fixes_order.py` (21 tests, M05 to M08);
  - `test_flow_fixes_pay.py` (28 tests, M09 to M12 and Platform).
- **Re-audit.** The audit harnesses were run again on fresh tenants, with the same steps as on 7 November plus new steps for the new commands: `FA9` for plan to order, `FX2` for sourcing to PO, `flowb3` for receive to pay. Results are compared step by step with the original run.
- **Full test suite.** The whole backend suite was run on the combined code with the CI settings (English collation database, CI keys, coverage gate). Result: 1,167 passed, 1 skipped, 0 failed (1 h 15 min), including the 68 new flow-fix tests (plan 19, order 21, pay 28). The coverage gate passes: total 89.6% (minimum 85%); api 85.9%, services 89.8%, platform 86.1%, events 76.1%, models 100%, seed 95.5%.
- **Frontend.** The type check and the CI-scope lint pass (type check: 0 errors; lint: 0 errors, 4 older warnings not touched by these fixes).

Defects, before and after

#	Defect (section 3)	Status	What changed	Evidence
1	SOW text over 200 characters gives HTTP 500	**Fixed**	The line description is limited and validated (422 with a clear message instead of 500). The SOW attributes are stored on the requisition line and carried to the PO and sourcing lines.	Re-audit F2 A12: 3,782 characters, 422. Attributes stored.
2	Custom roles refused payment-block release and savings validation	**Fixed**	The checks use the scopes `payment:run` and `savings:validate`, not the role name. Four other role-name checks were converted the same way: retention release, consignment settlement, reconciliation and the early-payment buyer side. Negotiation is open to `savings:manage` holders for their categories, bound by SOD-07.	F7 8a: BROKEN, now PASS (custom role PAY_DESK, 200). F9 4c is covered by a regression test.
3	Second bank change, and a bank change on a BLOCKED supplier, accepted	**Fixed**	One shared guard is used for onboarding, the manager route and the new portal route. It answers 409 BANK_CHANGE_PENDING, SUPPLIER_BLOCKED or SUPPLIER_INACTIVE.	F8: both steps BROKEN, now PASS.
4	MRP plans converted demand again; a PIC run deletes ERP planned orders	**Fixed**	Open stock requisitions and draft or in-approval PO lines count as supply. Requisitions raised to source a year's volume are left out, because the call-offs against the resulting agreement are planned one by one. The purposes left out are set in rule MRP-SUPPLY (default ANNUAL_SOURCING). Unfirmed planned orders imported from the ERP are kept.	F1 net-change: BROKEN, now PASS. The ERP-imported order is kept (PARTIAL, now PASS).
5	Return deliveries never close	**Fixed**	A return delivery has its own state machine. A credit note moves it to CREDITED, the replacement receipt to REPLACED, and a reversal of that receipt back to REPLACEMENT_DUE. A manual close needs a reason (CLOSED). New list, detail and close routes, and a Returns view on the receiving screen.	F6 5e (RD-4411 REPLACED) and F7 2e (RD-4410 CREDITED): PARTIAL, now PASS.
6	Rejected service entry sheet cannot be resubmitted	**Fixed**	New routes for submit/resubmit, reject, revoke, read and create as draft. Every state of the sheet can be reached, with actions on the receiving screen.	F6 9f: NOT BUILT, now PASS.
7	Copilot misses invoice numbers without an INV, HTS or KD prefix	**Fixed**	Words in the question are looked up against the tenant's own document numbers instead of fixed prefixes.	M12: `SSPL-A-0001`, `MTW-0001`, `KAC/26-27/0311`, RFP-3101 and QC-6103 found.
8	New stock material gets MRP type ND; MRP-type change ungoverned	**Fixed**	A new stock material gets PD by default (a tenant rule can change it). Changing the MRP type is now a material change request: 202, approved by the plant steward, the version goes up and the ERP message is sent. Type VM is refused, because neither the Technical Document nor the Blueprint defines it.	M03 and F1 VB: PARTIAL, now PASS.
9	Source determination takes an unordered first row	**Fixed**	Contracts are ranked by lowest valid net price, then earliest validity end, then number. The determination records the candidates and the basis.	F5 B4: three POs in a row went to the cheaper contract.
10	No PO from a contract or planned order, no manual PO, no PO currency	**Fixed**	New PO sources: CONTRACT (release order), PLANNED_ORDER (through the MRP requisition, as in BP P02) and MANUAL (needs a reason and purpose code, approved through the bands; the creator cannot approve). A foreign-currency PO fixes a reference FX rate, or answers 409 FX_RATE_MISSING. The purchasing organisation is resolved from the organisation masters. New screen `/pic/purchase-orders/new`.	F5 CONTRACT and MANUAL: NOT BUILT, now PASS. Currency covered by a regression test.
11	No ABAC on MRP, PO create/release or supplier changes	**Fixed**	New ABAC actions MRP_RUN, MRP_PLAN, PO_CREATE, PO_RELEASE, SUPPLIER_BANK_CHANGE and SUPPLIER_MASTER_CHANGE. A supplier's scope comes from its live categories and its company codes. The task inbox no longer lists tasks the approver is not allowed to act on.	F1 A5 (403), F5 B1 (403), F8 out-of-scope change (403). The inbox is covered by a regression test.
12	No scheduled ERP sync, KPI snapshot, drift or retrain jobs	**Fixed**	New jobs: six ERP sync jobs (off until the tenant switches them on with rule ERP-SYNC), `kpi-snapshot-monthly`, `ml-drift-check` and `ml-retrain`. A retrained model waits for an administrator unless rule AI-RETRAIN allows it to activate itself. Also added: `supplier-risk-scan`, `sanctions-rescreen`, `rebate-accrual` and `bid-offer-expiry`. The nightly `erp-reconciliation` now covers POs.	Platform P12c and M12 5a: PARTIAL, now PASS. All new jobs pass a dry run.
13	No invoice approval chain or delegation	**Fixed**	Rule INVOICE-APPROVAL defines value bands per company code. Steps go to the task inbox with delegation, SLA and value limits, under SOD-08 and SOD-09. A rejection comes back to AP as an exception. The shipped default has no bands, so the single AP approval is unchanged until a tenant adds bands.	Re-audit fix phase 16 of 16: the chain opens, a second approve gets 409, posting waits, finance approves, and the payment run applies SOD-09.
14	Invited suppliers not notified at publish	**Fixed**	Suppliers are notified at publish, when a new round opens, and when the deadline is extended (new command `extend`).	F3: PARTIAL, now PASS.

#	Defect (section 3)	Status	What changed	Evidence
15	No volume-shortfall leakage	**Fixed**	Contracted and posted invoiced quantities are compared per contract line and month. The shortfall appears as leakage of kind VOLUME_SHORTFALL, with a volume compliance table on the savings screen.	Re-audit: about 331 units expected, 300 invoiced, leakage ₹1,575.35.
16	Cosmetic items	**Fixed**	Separate number ranges per event and contract type (RFI-, RFP-, AUC-, QC-, VC-, RC-, SOW-, MA-, NDA-; RFQ- and SA- unchanged, so RFQ-2231 and SA-5520 are intact). SOD-11 pair at user creation answers 409. Replays set the `Idempotent-Replayed: true` header. A duplicate hold emits `invoice.exception.raised`. The inventory screens are labelled STK-01 to STK-05.	Inventory 5j now 409. Platform P9a has the header. F7 duplicate hold emits the event. RFP-3101, QC-6103.

The Workflow Guide's gap list (section 4), after the fixes

Item	Status
A1 to A14	Fixed. A4 (PERIOD lot size) and A6 (Kanban flags and goods-receipt refill) are now built and tested; the goods-receipt refill is covered by a regression test, not by the harness.
B1 to B9	Fixed. For B5, the contract is posted to the ERP as an OUTLINE_AGREEMENT message, mapped for the ERP simulator only (see below).
C1, C4 to C8	Fixed.
C2	Partly fixed. The 102 reversal now carries its lines, the SAP goods-receipt mapping takes the movement type from the document, and a SAP supplier debit memo mapping was added. PAYMENT_REVERSAL and SCF_FINANCING stay mapped for the simulator only (see below).
C3	Approval chain and delegation fixed. Form 16A PDF, partial payments and s.195 TDS are not built (see below).

Found and fixed along the way

- Creating a delegation between users with long e-mail addresses failed with HTTP 500, because the stored object reference overflowed its 60-character column. It is now cut to fit.
- Lists of award decisions and scenarios on a sourcing event had no defined order. Two decisions made at the same business time could come back in either order. They are now ordered by time, then creation id.
- `ml-retrain` no longer registers an empty forecast candidate when there is no demand history.
- One test depended on the order the test files ran in (`test_new_tables_are_tenant_isolated`). It now creates its own second tenant.

Still open: needs the client or a decision

Item	Why it is open	What is needed
Live SAP mapping for PAYMENT_REVERSAL, SCF_FINANCING and OUTLINE_AGREEMENT (contracts)	Mapped and tested against the ERP simulator only	A SAP sandbox and the client's document types and design
Form 16A PDF	The statutory certificate is issued from TRACES; PIC exports the data	Decide whether a PIC-generated PDF is wanted
Partial payments, s.195 TDS	Not asked for by the Technical Document or the Blueprint	A product decision
Default invoice approval bands	Shipped without bands, so today's single AP approval is unchanged	The client's value bands (for example ₹10 lakh and above to the Finance Controller)
Award with an expired bid offer	The bid is flagged and the event owner told; the award is not blocked	Decide whether to block it (recommended: block, with a reasoned override)

The steps that still show as not PASS in the raw re-audit logs are of two kinds. Some are results the original audit already graded correct on review, such as the rule-correct duplicate hold on SSPL-Q-0001, and the reconciliation with no differences because the simulator posts the reversal itself. The others are mistakes in the audit harness itself: a wrong inspection-lot order, a wrong ledger reference, a query on a column named `utr` instead of `bank_utr`, and phases that repeat steps already done. None of them is a product defect.

Sections 1 to 5 and the appendix below are the original audit of 7 November, kept unchanged as the baseline.

1. How this was checked

Every workflow was driven end to end through the real HTTP API, step by step, as the role that does it in practice (requester, buyer, approver, supplier through the portal, receiver, AP clerk, finance controller, administrator). Nothing was taken from the code's description or from earlier tests: each step was run and its result checked, including the status of the document, whether the next step became available, the events, the audit entry, the ERP integration message, the task in the inbox and the effect on stock or GR/IR.

- Two scratch databases on the local test Postgres, never production. Fresh tenants were built the way a new client would be (`FA7` with 32 users for plan-to-order, `FLB` with 21 users and 6 supplier portal users for receive-to-pay, `PLB` provisioned with the tenant tool); the AXN demo seed was used only where a worklist needs history (maverick and steward queues).
- The business clock was pinned and moved forward where a flow needs time to pass (approvals SLAs, due dates, renewals), and the background worker was run so events reached their consumers.
- Each step was graded: **PASS** (completes with the right result), **PARTIAL** (completes, but something is missing: an event, audit entry, ERP message, task or stock effect), **BROKEN** (cannot be completed or gives a wrong result) or **NOT BUILT** (no route or command exists). The 31 gaps the Workflow Guide recorded from reading the code (its Appendix A) were each confirmed or refuted at runtime.
- No code was changed during the audit. The harness scripts and raw results stay in the session scratchpad.

2. Verdict per flow

Flow	Module	Steps	Pass	Partial	Broken	Not built	Verdict
F1 MRP to requisition	M02	53	44	5	1	3	Works with gaps. Wrong result: demand already converted to a requisition is planned again
Material master	M03	46	40	4	0	2	Works with gaps. A new stock material gets MRP type ND, so MRP ignores it; no withdraw
F2 Requisition	M04	58	54	2	1	1	Works with gaps. Broken: an SOW-form text over 200 characters gives HTTP 500
F3 Strategic sourcing	M05	52	50	2	0	0	**Full flow.** Invited suppliers get no notification at publish
F4 Contract	M07	61	57	4	0	0	**Full flow.** `RENEWED` unused, no ERP posting of contracts
F5 Purchase order	M08	63	59	2	0	2	**Full flow from a requisition.** No PO from a contract and no manual PO; no plant check on create
F8 Supplier lifecycle	M06	92	86	2	2	2	Works with gaps. Broken: two bank-change controls missing
Spend governance	M01	35	35	0	0	0	**Full flow** (for what was driven)
F6 Receiving and quality	M09	49	47	1	0	1	Works with gaps. Return deliveries never close; a rejected service entry cannot be resubmitted
Inventory	M09	32	32	0	0	0	**Full flow**
F7 Invoice to pay	M10	57	54	1	1	1	Works with gaps. Broken: a custom finance role is refused the payment-block release; no API to set a PO currency
Spend feed from invoices	M01 / M10	5	5	0	0	0	**Full flow** (invoices, credit notes negative, debit notes positive, USD in group currency)
F9 Savings	M11	13	12	0	1	0	Works with gaps. Broken: a custom role with `savings:validate` is refused
AI, Copilot and KPIs	M12	22	19	1	2	0	Works with gaps. Broken: Copilot misses invoice numbers without an INV/HTS/KD prefix
Platform administration	Platform	27	26	1	0	0	Works with gaps. Inbound ERP sync is not scheduled
Total		**665**	**620**	**25**	**8**	**12**	

Answer. No workflow is broken end to end: every one of the fifteen flows completes its main path through the API, and 620 of the 665 steps passed as they should. Six flows are full flows with no broken step (sourcing, contract, purchase order from a requisition, spend governance, inventory, and the spend feed). The other ten work but have one or more defects: 8 steps are broken (5 distinct defects), 25 steps complete with something missing, and 12 steps have no route or command at all.

3. What breaks, in priority order

#	Severity	Flow	Defect	Where in the code
1	Blocks the step	F2	A category-SOW description longer than 200 characters fails with HTTP 500 (database column overflow); the API sets no length limit, and the structured SOW attributes are dropped	`api/plan_request.py` (`PrLineIn.description`), `models/plan_request.py:129`, `category-sow-forms.tsx`
2	Blocks the step for custom roles	F7, F9	Payment-block release and savings validation check for the built-in role name FINANCE_CONTROLLER instead of the scope, so a custom role holding `payment:run` or `savings:validate` gets 403	`services/payables.py:865`, `services/savings.py:175`
3	Missing control	F8	A second bank change is accepted while one is pending, and a bank change is accepted on a BLOCKED supplier; both guards exist only in the onboarding path	`services/supplier.py` (`request_bank_change` vs `submit_profile`)
4	Wrong result	F1	After a planned order is converted to a requisition, the next MRP run plans the same demand again (open requisition quantity is not counted as supply); a PIC run also deletes planned orders imported from the ERP	`services/planning.py` (`_supply`, `_assemble_supply`, `_replace_unfirmed` lines 340-349)
5	Wrong result	F6, F7	Return deliveries never leave `CREATED` / `REPLACEMENT_DUE`, even after the credit note or the replacement receipt; there is no route to list returns	`services/receiving.py`, credit-note resolution in `services/payables.py`
6	Blocks a branch	F6	A rejected service entry sheet cannot be resubmitted: `submit_ses` exists but has no route	`api/order_pay.py:287-309`, `services/receiving.py:1004`
7	Wrong result	M12	Copilot recognises only invoice numbers starting INV, HTS or KD (real invoices such as `SSPL-A-0001` are not found)	`services/copilot.py:27`
8	Wrong result	M03 to F1	A new stock material created through MDM gets MRP type ND, so MRP ignores it until someone changes it; changing the MRP type bypasses steward approval, versioning and the ERP message	`services/mdm.py`, `services/ind_manufacturing.set_mrp_type`
9	Wrong result	F5	With two live contracts and no quota, all POs went to the dearer contract (₹2,100 against ₹1,990): source determination takes the first row of an unordered query	`services/source.py`
10	Not built	F5, F7	No PO from a contract or a planned order, no manual PO, and no API to set a PO currency (every new PO is INR with purchasing org 1000), so a foreign-currency flow cannot be started through the API	`services/order.py:216`, `api/order_pay.py`
11	Missing control	F1, F5, F8	Plant and company scoping (ABAC) is not applied to MRP run/firm/convert, PO create/release, or supplier master changes: a P200 user ran MRP and created a PO for P100; a supplier manager scoped to MRO approved a change on a FORGINGS supplier	`api/plan_request.py`, `services/order.py`, `platform/abac.py` (`_document`)
12	Missing automation	Platform, M12	No scheduled jobs for inbound ERP sync, month-end KPI snapshots, model drift checks or retraining; they run only when started by hand	`services/erp_sync.py`, `services/kpi.py`, job registry
13	Missing control	F7	Invoice approval is one command with a per-user value limit: no approval chain by value band and no delegation	`services/payables.py`
14	Missing notification	F3	Invited suppliers get no notification when an RFQ is published	`services/sourcing.py` (`publish`)
15	Missing reporting	F9	No volume-shortfall leakage (only 100 of 1,000 contracted units invoiced shows no leakage)	`services/savings.py`
16	Cosmetic	various	Every sourcing event is numbered `RFQ-` and every contract `SA-`; user creation with the SOD-11 role pair answers 422 instead of 409; idempotent replays set `"replayed": true` in the body but not the documented response header; a duplicate hold emits no `invoice.exception.raised` event; the inventory page is labelled "INV-01", the Blueprint ID of the Invoice Inbox	various

Numbers 1 to 3 are the ones that stop real users or weaken a financial control and should be fixed first. Numbers 4, 5, 8 and 9 produce wrong data that users will notice in a pilot.

4. The Workflow Guide's gap list, checked at runtime

The Workflow Guide (Appendix A) recorded 31 gaps by reading the code. At runtime:

Verdict	Items
Confirmed (the gap is real)	A1, A2, A3, A5, A7, A8, A9 (the missing demo case), A10, A11, A12, A13, A14, B1, B2, B4, B5, B6, B7, B8, B9, C1, C3, C4, C5, C6, C7, C8 (source only)
Not testable in this audit	A4 (open supply covered the test demand), A6 (the goods-receipt refill half needs a receipt on a Kanban material), B3 (contract quantity at release), C2 (the simulator posts all four message types; only a live SAP system can show the gap)
Refuted	none

The audit also found three defects that were not on that list: the duplicate MRP planning after conversion (#4), the two missing bank-change guards (#3) and the rejected service entry that cannot be resubmitted (#6).

5. Not covered

These parts were not driven and remain unproven: reverse auctions with extra rounds, punch-out, CAPA and the scheduled risk scan, the savings initiative created automatically from a sourcing award, lot recall, supply-chain finance, MT940 import, TDS challans in this run (they are covered by the finance tests), rebate settlement with money (it needs posted invoices on the same tenant), and the goods-receipt refill of a Kanban card.

Appendix. Step-by-step evidence

Every step that was run, with the call made, what was expected, what happened and the verdict.

A. Plan to order (M01 to M08)

Each row is one call or one check, in the order run. "Evidence" is the response snippet and/or the DB rows.

F1 MRP (M02)

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
1	material master planning fields	mrp.controller	`GET /v1/materials/M-400001`	ACTIVE, mrp_type PD, safety 200 (after MCR)	200: ok ACTIVE; ok mrp_type PD	**PASS**	{"number": "M-400001", "description": "Forged steel weld neck flange DN100 PN16 ASTM A105", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuation_class": "3000", "hsn_code": ...
2	F1-01 demand: reservation	receiver	`POST /v1/reservations`	201 reservation OPEN -> RESERVATION demand element	201	**PASS**	{"id": "01a1b6b9-32ba-7068-9601-ac727ab47a7c", "number": "RSV-1001", "status": "OPEN", "companyCode": "1000", "material": "M-400001", "description": "Forged steel weld neck flange DN100 PN16 ASTM A105", "uom": "EA", "pla...
3	F1-01 demand element written	system	`demand_element`	RESERVATION demand row	[('RESERVATION', 'RSV-1001', Decimal('800.000'))]	**PASS**	
4	A5 MRP run outside the controller's plants	mrp.other (plants P200)	`POST /v1/mrp/runs {plant P100}`	403/404 if ABAC applied	201 {"number": "MR-1043", "mode": "SIMULATION", "plant": "P100", "scope": {}, "run_date": "2026-11-09", ...	**PARTIAL**	missing control: no ABAC plant check on MRP run (A5)
5	F1-02 simulation run	mrp.controller	`POST /v1/mrp/runs`	201 SIMULATED, no planned orders written	201: ok SIMULATED	**PASS**	{"number": "MR-1044", "mode": "SIMULATION", "plant": "P100", "scope": {}, "run_date": "2026-11-09", "status": "SIMULATED", "materials_planned": 1, "planned_orders_created": 0, "exceptions_raised": 0, "results": [{"materi...
6	F1-02 simulation writes nothing	system	`planned_order count`	unchanged	0 -> 0	**PASS**	
7	F1-02..06 net-change run (first run: all)	mrp.controller	`POST /v1/mrp/runs`	201 COMPLETED, planned orders, exceptions, mrp.run.completed	201: ok COMPLETED	**PASS**	{"number": "MR-1045", "mode": "NET_CHANGE", "plant": "P100", "scope": {}, "run_date": "2026-11-09", "status": "COMPLETED", "materials_planned": 1, "planned_orders_created": 1, "exceptions_raised": 1, "results": [{"materi...
8	F1-05 planned orders of the run	mrp.controller	`GET /v1/mrp/runs/MR-1045/planned-orders`	planned order(s) for the material	200: ok has planned orders	**PASS**	{"items": [{"number": "PLO-55821", "mrp_run_id": "01a1b6b9-36be-76f8-a415-f2f1bce8049e", "material_id": "01a1b6b9-0359-73e2-94d9-9d26b1d5cbf1", "plant": "P100", "storage_location": "0001", "qty": "1000", "need_date": "20...
9	F1-04 netting trace	mrp.controller	`GET /v1/mrp/runs/MR-1045/results`	netting rows	200	**PASS**	{"run": "MR-1045", "total": 1, "items": [{"net": "1000.000", "gross": "800.000", "plant": "P100", "on_hand": "0", "open_po": "0", "material": "M-400001", "proposals": [{"net": "1000.000", "qty": "1000.000", "need_date": ...
10	F1-06 exceptions	mrp.controller	`GET /v1/mrp/runs/MR-1045/exceptions`	list (by value at risk)	200	**PASS**	[{"mrp_run_id": "01a1b6b9-36be-76f8-a415-f2f1bce8049e", "material_id": "01a1b6b9-0359-73e2-94d9-9d26b1d5cbf1", "planned_order_id": null, "code": "BELOW_SAFETY", "message": "On-hand 0 is below safety stock 200.000", "valu...
11	F1-06 run completed event + notification	system	`outbox/notification`	mrp.run.completed -> MRP controller notified	outbox=[('mrp.run.completed', 'MR-1045', True)] notif=[('MRP_RUN', 'MR-1045', 'mrp.controller@fa7.example')]	**PASS**	
12	F1-05 planned order detail	mrp.controller	`GET /v1/mrp/runs/MR-1045/planned-orders`	qty = demand + safety, source determined	PLO-55821 qty=1000 status=CREATED source={"method": "NONE", "excluded": [], "allocations": []}	**PASS**	
13	F1-07 firm planned order	mrp.controller	`POST /v1/planned-orders/PLO-55821/firm`	200 FIRMED, planned_order.firmed	200: ok FIRMED; ok event	**PASS**	{"number": "PLO-55821", "mrp_run_id": "01a1b6b9-36be-76f8-a415-f2f1bce8049e", "material_id": "01a1b6b9-0359-73e2-94d9-9d26b1d5cbf1", "plant": "P100", "storage_location": "0001", "qty": "1000", "need_date": "2026-12-09", ...
14	F1-07 requester cannot firm (scope)	requester	`POST /v1/planned-orders/PLO-55821/firm`	403	403 FORBIDDEN	**PASS**	{"type": "https://errors.alphahelion.io/pic/FORBIDDEN", "title": "FORBIDDEN", "status": 403, "detail": "Missing scope(s): mrp:plan", "code": "FORBIDDEN", "instance": "/v1/planned-orders/PLO-55821/firm", "traceId": "d5b28...

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
15	F1-02 net-change with no change	mrp.controller	`POST /v1/mrp/runs`	201 COMPLETED, no material in scope, firmed order untouched	201	**PASS**	{"number": "MR-1046", "mode": "NET_CHANGE", "plant": "P100", "scope": {}, "run_date": "2026-11-09", "status": "COMPLETED", "materials_planned": 0, "planned_orders_created": 0, "exceptions_raised": 0, "results": [], "star...
16	F1 firmed order kept	system	`planned_order`	FIRMED stays	[('FIRMED',)]	**PASS**	
17	A2 planned order cancel	mrp.controller	`POST /v1/planned-orders/{ref}/cancel`	command exists	404	**NOT BUILT**	no command (Appendix A2 CONFIRMED)
18	A2 planned order unfirm	mrp.controller	`POST /v1/planned-orders/{ref}/unfirm`	command exists	404	**NOT BUILT**	no command (Appendix A2 CONFIRMED)
19	A1 close an MRP exception	mrp.controller	`POST /v1/mrp/exceptions/{id}/close | PATCH`	no command (A1)	404/404; statuses in db=[('OPEN', 1)]	**NOT BUILT**	no command (Appendix A1 CONFIRMED)
20	F1-08 convert planned order to requisition	mrp.controller	`POST /v1/planned-orders/PLO-55821/convert`	200 REQ origin MRP, purpose MRP_REPLENISHMENT, submitted (IN_APPROVAL/APPROVED)	200: ok origin MRP; ok submitted; ok purpose MRP_REPLENISHMENT	**PASS**	{"number": "REQ-2311", "origin": "MRP", "origin_ref": "PLO-55821", "purpose_code": "MRP_REPLENISHMENT", "purpose_text": "MRP replenishment from PLO-55821", "requester_id": "01a1b6b8-8b2b-76fc-be89-82bd334a31dc", "company...
21	F1-08 convert replay (Idempotency-Key)	mrp.controller	`POST /v1/planned-orders/PLO-55821/convert`	same REQ	200 REQ-2311	**PASS**	
22	F1-08 PR price and chain	system	`requisition view`	price from source else standard price; DMN-01 chain	total=125000000 chain= ['COST_OBJECT_OWNER', 'DEPT_HEAD', 'PROCUREMENT_HEAD', 'FINANCE_CONTROLLER'] lines=[[null, "1000", 125000]]	**PASS**	
23	F1-08 MRP PR approval: REQ_APPROVAL task (COST_OBJECT_OWNER) APPROVE	cc.owner	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-3d59-7561-8e2e-0a2dd70d60cb", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2311 \u00b7 Cost Object Own...
24	F1-08 MRP PR approval: REQ_APPROVAL task (DEPT_HEAD) APPROVE	dept.head	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-3d59-7561-8e2e-0a2dd70d60cb", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2311 \u00b7 Dept Head", "ob...
25	F1-08 MRP PR approval: REQ_APPROVAL task (PROCUREMENT_HEAD) APPROVE	proc.head	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-3d59-7561-8e2e-0a2dd70d60cb", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2311 \u00b7 Procurement Hea...
26	F1-08 MRP PR approval: REQ_APPROVAL task (FINANCE_CONTROLLER) APPROVE	fin1	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-3d59-7561-8e2e-0a2dd70d60cb", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2311 \u00b7 Finance Control...
27	F1-09 MRP PR approved	buyer	`GET /v1/requisitions/REQ-2311`	APPROVED	200: ok APPROVED	**PASS**	{"number": "REQ-2311", "origin": "MRP", "origin_ref": "PLO-55821", "purpose_code": "MRP_REPLENISHMENT", "purpose_text": "MRP replenishment from PLO-55821", "requester_id": "01a1b6b8-8b2b-76fc-be89-82bd334a31dc", "company...
28	F1-09/12 buyer workbench origin=MRP	buyer	`GET /v1/buyer-workbench`	the MRP line is listed	200: ok listed	**PASS**	[{"requisition": "REQ-2311", "line": 10, "origin": "MRP", "description": "Forged steel weld neck flange DN100 PN16 ASTM A105", "itemType": "MATERIAL", "purchaseType": "STOCK", "material": "M-400001", "category": "FORGING...
29	F1-10 process instance	system	`process_instance pic.mrp-to-pr`	instance of the run ENDED once no CREATED/FIRMED left	[('pic.mrp-to-pr', 'COMPLETED', 'F1-10')]	**INFO**	
30	VB set MRP type	mrp.controller	`PUT /v1/materials/M-400002/mrp-type`	200 VB	200	**PASS**	{"material": "M-400002", "mrpType": "VB"}
31	VB mrp-type change governance	system	`material / integration_message`	master field change versioned and sent to ERP like an MCR (MDM governance)	version 1->1 mrp_type=VB ERP messages 1->1	**PARTIAL**	
32	VB reorder point parameters	mrp.controller	`PUT /v1/planning/parameters`	200	200	**PASS**	{"material_id": "01a1b6b9-0c91-700c-9ac7-6d7efbf04b67", "plant": "P100", "reorder_point": "100", "max_stock": "500", "note": "go-live", "id": "01a1b6b9-5a13-764b-b0f4-1aae6960423f", "created_at": "2026-11-09T09:00:00+05:...

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
33	VB regenerative run (reorder point)	mrp.controller	`POST /v1/mrp/runs`	201 COMPLETED, order up to max (500)	201	**PASS**	{"number": "MR-1047", "mode": "REGENERATIVE", "plant": "P100", "scope": {"materials": ["M-400002"]}, "run_date": "2026-11-09", "status": "COMPLETED", "materials_planned": 1, "planned_orders_created": 1, "exceptions_raise...
34	VB planned order qty	system	`planned orders`	one order, qty 500 (max stock, stock 0)	[('PLO-55822', '500')]	**PASS**	
35	KB set MRP type	mrp.controller	`PUT /v1/materials/M-400003/mrp-type`	200 KB	200	**PASS**	{"material": "M-400003", "mrpType": "KB"}
36	KB create card	mrp.controller	`POST /v1/kanban/cards`	201 card FULL	201: ok FULL	**PASS**	{"id": "01a1b6b9-5d34-7236-9908-def4a87e8cf8", "number": "KB-1001", "material": "M-400003", "description": "Hex bolt M12x60 grade 8.8 zinc plated", "plant": "P100", "storageLocation": "0001", "binQty": "500", "uom": "EA"...
37	KB signal empty -> requisition	mrp.controller	`POST /v1/kanban/cards/KB-1001/signal`	200 REPLENISHING + REQ purpose KANBAN_REPLENISHMENT submitted	200: ok REPLENISHING	**PASS**	{"id": "01a1b6b9-5d34-7236-9908-def4a87e8cf8", "number": "KB-1001", "material": "M-400003", "description": "Hex bolt M12x60 grade 8.8 zinc plated", "plant": "P100", "storageLocation": "0001", "binQty": "500", "uom": "EA"...
38	KB requisition raised	system	`requisition`	REQ KANBAN_REPLENISHMENT, submitted	[('REQ-2312', 'IN_APPROVAL', 'KANBAN_REPLENISHMENT', 'MRP')]	**PASS**	
39	KB signal again refused	mrp.controller	`POST /v1/kanban/cards/KB-1001/signal`	409	409 ILLEGAL_TRANSITION	**PASS**	{"type": "https://errors.alphahelion.io/pic/ILLEGAL_TRANSITION", "title": "ILLEGAL_TRANSITION", "status": 409, "detail": "Kanban card KB-1001 is REPLENISHING; 'signal' is not allowed from this state", "code": "ILLEGAL_TR...
40	KB refill (manual)	mrp.controller	`POST /v1/kanban/cards/KB-1001/refill`	200 FULL, cycles 1	200: ok FULL	**PASS**	{"id": "01a1b6b9-5d34-7236-9908-def4a87e8cf8", "number": "KB-1001", "material": "M-400003", "description": "Hex bolt M12x60 grade 8.8 zinc plated", "plant": "P100", "storageLocation": "0001", "binQty": "500", "uom": "EA"...
41	KB suspend	mrp.controller	`POST /v1/kanban/cards/KB-1001/suspend`	200	200	**PASS**	{"id": "01a1b6b9-5d34-7236-9908-def4a87e8cf8", "number": "KB-1001", "material": "M-400003", "description": "Hex bolt M12x60 grade 8.8 zinc plated", "plant": "P100", "storageLocation": "0001", "binQty": "500", "uom": "EA"...
42	KB resume	mrp.controller	`POST /v1/kanban/cards/KB-1001/resume`	200	200	**PASS**	{"id": "01a1b6b9-5d34-7236-9908-def4a87e8cf8", "number": "KB-1001", "material": "M-400003", "description": "Hex bolt M12x60 grade 8.8 zinc plated", "plant": "P100", "storageLocation": "0001", "binQty": "500", "uom": "EA"...
43	A6 Kanban flags purposeCode/autoRequisition	system	`rule KANBAN`	flags read by the code	refill is manual only (no GR hook observed); flags verified by code read only	**NOT TESTABLE**	
44	MRP-04 forecast generate (new tenant, no history)	mrp.controller	`POST /v1/forecasts/generate`	200 (0 created: no consumption history)	200	**PASS**	{"created": 0, "recommendations": 0}
45	F1 alt: ERP planned-order import	mrp.controller	`POST /v1/integration/planned-orders/import`	201 1 accepted, 1 rejected; ERP_IMPORT run	201	**PASS**	{"id": "01a1b6b9-6201-75f2-a571-709c13cbce6e", "number": "IB-101", "sourceSystem": "SAP_S4", "kind": "PLANNED_ORDERS", "batchRef": "MRP-ERP-FA7-1", "rows": 2, "accepted": 1, "rejected": 1, "mrpRuns": ["MR-1048"], "result...
46	F1 alt: import replay	mrp.controller	`POST /v1/integration/planned-orders/import`	200 replay	200	**PASS**	
47	F1 demand 2: reservation	receiver	`POST /v1/reservations`	201	201	**PASS**	{"id": "01a1b6b9-63ff-771c-bef0-ede1ce07f802", "number": "RSV-1002", "status": "OPEN", "companyCode": "1000", "material": "M-400001", "description": "Forged steel weld neck flange DN100 PN16 ASTM A105", "uom": "EA", "pla...
48	F1 net-change picks the new demand	mrp.controller	`POST /v1/mrp/runs`	201, material in scope (demand changed)	201	**PASS**	{"number": "MR-1049", "mode": "NET_CHANGE", "plant": "P100", "scope": {}, "run_date": "2026-11-09", "status": "COMPLETED", "materials_planned": 1, "planned_orders_created": 2, "exceptions_raised": 1, "results": [{"materi...
49	F1 net-change result	system	`planned orders`	one new order for the 300 demand only (800 already converted to an approved MRP requisition)	[('PLO-55825', '300', 'CREATED'), ('PLO-55824', '1000', 'CREATED')]	**BROKEN**	
50	F1 alt: ERP-imported planned order after a PIC run	system	`planned_order`	kept (ERP owns it) or replaced explicitly	[('PLO-55823', 'DELETED', Decimal('250.000'))]	**PARTIAL**	
51	F1 demand withdrawn (reservation cancel)	receiver	`POST /v1/reservations/{n}/cancel`	200	200	**PASS**	{"id": "01a1b6b9-63ff-771c-bef0-ede1ce07f802", "number": "RSV-1002", "status": "CANCELLED", "companyCode": "1000", "mate...

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
52	F1 regenerative run after demand withdrawn	mrp.controller	`POST /v1/mrp/runs`	201; CANCEL/RESCHEDULE message on firmed order	201	**PASS**	{"number": "MR-1050", "mode": "REGENERATIVE", "plant": "P100", "scope": {}, "run_date": "2026-11-09", "status": "COMPLETED", "materials_planned": 2, "planned_orders_created": 2, "exceptions_raised": 2, "results": [{"mate...
53	F1 CANCEL message on firmed order	system	`mrp_exception`	CANCEL message	['BELOW_SAFETY', 'RESCHEDULE_IN']	**PARTIAL**	No CANCEL message because the converted 800 EA demand is planned again (see F1 net-change result): the firmed order is told to RESCHEDULE_IN instead.
54	A2 firmed order after CANCEL message	mrp.controller	`-`	controller can cancel/unfirm it	order PLO-55825 status [('FIRMED',)]; no cancel/unfirm route (see A2 probes)	**CONFIRMED**	
55	A1 exception backlog over runs	system	`mrp_exception`	older exceptions closed	[('OPEN', 4)]	**CONFIRMED**	
56	A3 net-change after a supply-only change (new PO)	mrp.controller	`POST /v1/mrp/runs`	material with new open PO re-planned	201	**PASS**	{"number": "MR-1051", "mode": "NET_CHANGE", "plant": "P100", "scope": {}, "run_date": "2026-11-15", "status": "COMPLETED", "materials_planned": 0, "planned_orders_created": 0, "exceptions_raised": 0, "results": [], "star...
57	A3 net-change scope after PO only	system	`mrp run scope`	MAT-A in scope (supply changed)	scope={} results=0	**PARTIAL**	wrong scope: supply-only change not re-planned (A3)
58	A4 PERIOD lot size	mrp.controller	`MCR lot_size_policy=PERIOD + 2 October demands + regenerative run`	one order per period (100+150 in one bucket)	policy=PERIOD run=201 october orders=[]	**NOT TESTABLE**	

M03 MDM

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
1	[MAT-A] request	requester	`POST /v1/materials/requests`	201 MRQ, duplicate check, status IN_STEWARDSHIP (or DUPLICATE_BLOCKED)	201: ok status IN_STEWARDSHIP	**PASS**	{"number": "MRQ-7731", "status": "IN_STEWARDSHIP", "description": "Forged steel weld neck flange DN100 PN16 ASTM A105", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuation_...
2	[MAT-A] F9-05 four steward tasks	system	`task`	4 MDM_STEWARD tasks	4 tasks	**PASS**	
3	[MAT-A] requester cannot decide	requester	`POST /v1/materials/requests/MRQ-7731/decision`	403 NOT_YOUR_TASK	403 NOT_YOUR_TASK	**PASS**	{"type": "https://errors.alphahelion.io/pic/NOT_YOUR_TASK", "title": "NOT_YOUR_TASK", "status": 403, "detail": "You have no open steward task on this request", "code": "NOT_YOUR_TASK", "instance": "/v1/materials/requests...
4	[MAT-A] steward approve (mdm.purchasing)	mdm.purchasing	`POST /v1/materials/requests/MRQ-7731/decision`	200	200	**PASS**	{"number": "MRQ-7731", "status": "IN_STEWARDSHIP", "description": "Forged steel weld neck flange DN100 PN16 ASTM A105", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuation_...
5	[MAT-A] steward approve (mdm.quality)	mdm.quality	`POST /v1/materials/requests/MRQ-7731/decision`	200	200	**PASS**	{"number": "MRQ-7731", "status": "IN_STEWARDSHIP", "description": "Forged steel weld neck flange DN100 PN16 ASTM A105", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuation_...
6	[MAT-A] steward approve (mdm.finance)	mdm.finance	`POST /v1/materials/requests/MRQ-7731/decision`	200	200	**PASS**	{"number": "MRQ-7731", "status": "IN_STEWARDSHIP", "description": "Forged steel weld neck flange DN100 PN16 ASTM A105", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuation_...
7	[MAT-A] steward approve (mdm.plant)	mdm.plant	`POST /v1/materials/requests/MRQ-7731/decision`	200	200	**PASS**	{"number": "MRQ-7731", "status": "SYNCED_TO_ERP", "description": "Forged steel weld neck flange DN100 PN16 ASTM A105", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuation_c...
8	[MAT-A] APPROVED + material created	requester	`GET /v1/materials/requests/MRQ-7731`	APPROVED, material M-..., MATERIAL_MASTER queued	200: ok APPROVED/SYNCED; ok material set	**PASS**	{"number": "MRQ-7731", "status": "SYNCED_TO_ERP", "description": "Forged steel weld neck flange DN100 PN16 ASTM A105", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuation_c...
9	[MAT-A] F9-06 ERP sync -> SYNCED_TO_ERP	requester	`GET /v1/materials/requests/MRQ-7731`	SYNCED_TO_ERP after worker tick	200: ok SYNCED_TO_ERP	**PASS**	{"number": "MRQ-7731", "status": "SYNCED_TO_ERP", "description": "Forged steel weld neck flange DN100 PN16 ASTM A105", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuation_c...
10	[MAT-A] new STOCK material planning type	system	`GET /v1/materials/M-400001`	a plannable MRP type (PD) for a STOCK material	mrp_type=ND safety_stock=0 standard_price_minor=None	**PARTIAL**	
11	[MAT-A] MATERIAL_MASTER message	system	`integration_message`	MAT| POSTED	[('MATERIAL_MASTER', 'MRQ-7731', 'POSTED', 'MAT|M-400001', '100000331')]	**PASS**	outbox=['material_request.approved', 'task.assigned'] audit=['APPROVAL_APPROVE', 'APPROVED', 'CREATED', 'ERP_POSTED', 'SYNCED_TO_ERP', 'TASK_CREATED'] erp=['MATERIAL_MASTER:POSTED'] tasks=['MDM_STEWARD/MDM_STEWARD/COMPLETED']

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
12	[SoD-06] request	steward.req	`POST /v1/materials/requests`	201 MRQ, duplicate check, status IN_STEWARDSHIP (or DUPLICATE_BLOCKED)	201	**PASS**	{"number": "MRQ-7732", "status": "IN_STEWARDSHIP", "description": "Hot forged alloy steel crankshaft blank 4-cylinder", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuation_...
13	[SoD-06] requester-steward approves own request	steward.req	`POST /v1/materials/requests/MRQ-7732/decision`	refused (403 excluded / 409 SOD_CONFLICT)	403 NOT_YOUR_TASK	**PASS**	{"type": "https://errors.alphahelion.io/pic/NOT_YOUR_TASK", "title": "NOT_YOUR_TASK", "status": 403, "detail": "You have no open steward task on this request", "code": "NOT_YOUR_TASK", "instance": "/v1/materials/requests...
14	[SoD-06] violation recorded	system	`sod_violation`	SOD-06 row	[]	**PASS**	Re-graded: the requester-steward is excluded from the task (403 NOT_YOUR_TASK), so no violation row is needed; the SOD-06 row is written on the override path (next rows).
15	[MAT-B] steward approve (mdm.purchasing)	mdm.purchasing	`POST /v1/materials/requests/MRQ-7732/decision`	200	200	**PASS**	{"number": "MRQ-7732", "status": "IN_STEWARDSHIP", "description": "Hot forged alloy steel crankshaft blank 4-cylinder", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuation_...
16	[MAT-B] steward approve (mdm.quality)	mdm.quality	`POST /v1/materials/requests/MRQ-7732/decision`	200	200	**PASS**	{"number": "MRQ-7732", "status": "IN_STEWARDSHIP", "description": "Hot forged alloy steel crankshaft blank 4-cylinder", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuation_...
17	[MAT-B] steward approve (mdm.finance)	mdm.finance	`POST /v1/materials/requests/MRQ-7732/decision`	200	200	**PASS**	{"number": "MRQ-7732", "status": "IN_STEWARDSHIP", "description": "Hot forged alloy steel crankshaft blank 4-cylinder", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuation_...
18	[MAT-B] steward approve (mdm.plant)	mdm.plant	`POST /v1/materials/requests/MRQ-7732/decision`	200	200	**PASS**	{"number": "MRQ-7732", "status": "SYNCED_TO_ERP", "description": "Hot forged alloy steel crankshaft blank 4-cylinder", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuation_c...
19	[MAT-B] APPROVED + material created	requester	`GET /v1/materials/requests/MRQ-7732`	APPROVED, material M-..., MATERIAL_MASTER queued	200: ok APPROVED/SYNCED; ok material set	**PASS**	{"number": "MRQ-7732", "status": "SYNCED_TO_ERP", "description": "Hot forged alloy steel crankshaft blank 4-cylinder", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuation_c...
20	[MAT-C] request	requester2	`POST /v1/materials/requests`	201 MRQ, duplicate check, status IN_STEWARDSHIP (or DUPLICATE_BLOCKED)	201	**PASS**	{"number": "MRQ-7733", "status": "IN_STEWARDSHIP", "description": "Hex bolt M12x60 grade 8.8 zinc plated", "type": "STOCK", "material_group": "FASTENERS", "category": "FASTENERS", "base_uom": "EA", "valuation_class": "30...
21	[MAT-C] steward approve (mdm.purchasing)	mdm.purchasing	`POST /v1/materials/requests/MRQ-7733/decision`	200	200	**PASS**	{"number": "MRQ-7733", "status": "IN_STEWARDSHIP", "description": "Hex bolt M12x60 grade 8.8 zinc plated", "type": "STOCK", "material_group": "FASTENERS", "category": "FASTENERS", "base_uom": "EA", "valuation_class": "30...
22	[MAT-C] steward approve (mdm.quality)	mdm.quality	`POST /v1/materials/requests/MRQ-7733/decision`	200	200	**PASS**	{"number": "MRQ-7733", "status": "IN_STEWARDSHIP", "description": "Hex bolt M12x60 grade 8.8 zinc plated", "type": "STOCK", "material_group": "FASTENERS", "category": "FASTENERS", "base_uom": "EA", "valuation_class": "30...
23	[MAT-C] steward approve (mdm.finance)	mdm.finance	`POST /v1/materials/requests/MRQ-7733/decision`	200	200	**PASS**	{"number": "MRQ-7733", "status": "IN_STEWARDSHIP", "description": "Hex bolt M12x60 grade 8.8 zinc plated", "type": "STOCK", "material_group": "FASTENERS", "category": "FASTENERS", "base_uom": "EA", "valuation_class": "30...
24	[MAT-C] steward approve (mdm.plant)	mdm.plant	`POST /v1/materials/requests/MRQ-7733/decision`	200	200	**PASS**	{"number": "MRQ-7733", "status": "SYNCED_TO_ERP", "description": "Hex bolt M12x60 grade 8.8 zinc plated", "type": "STOCK", "material_group": "FASTENERS", "category": "FASTENERS", "base_uom": "EA", "valuation_class": "300...
25	[MAT-C] APPROVED + material created	requester	`GET /v1/materials/requests/MRQ-7733`	APPROVED, material M-..., MATERIAL_MASTER queued	200: ok APPROVED/SYNCED; ok material set	**PASS**	{"number": "MRQ-7733", "status": "SYNCED_TO_ERP", "description": "Hex bolt M12x60 grade 8.8 zinc plated", "type": "STOCK", "material_group": "FASTENERS", "category": "FASTENERS", "base_uom": "EA", "valuation_class": "300...
26	[DUP] request	requester2	`POST /v1/materials/requests`	201 MRQ, duplicate check, status IN_STEWARDSHIP (or DUPLICATE_BLOCKED)	201: ok status DUPLICATE_BLOCKED	**PASS**	{"number": "MRQ-7734", "status": "DUPLICATE_BLOCKED", "description": "Forged steel weld neck flange DN100 PN16 ASTM A105", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuati...
27	A8 withdraw a blocked request	requester2	`POST /v1/materials/requests/MRQ-7734/withdraw`	no such command (A8)	404	**NOT BUILT**	no command (Appendix A8 CONFIRMED)

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
28	A8 withdraw a blocked request	requester2	`POST /v1/materials/requests/MRQ-7734/cancel`	no such command (A8)	404	**NOT BUILT**	no command (Appendix A8 CONFIRMED)
29	[DUP] merge into existing	requester2	`POST /v1/materials/requests/MRQ-7734/merge`	200 MERGED	200: ok MERGED	**PASS**	{"number": "MRQ-7734", "status": "MERGED", "description": "Forged steel weld neck flange DN100 PN16 ASTM A105", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuation_class": ...
30	[DUP2] request	steward.req	`POST /v1/materials/requests`	201 MRQ, duplicate check, status IN_STEWARDSHIP (or DUPLICATE_BLOCKED)	201: ok status DUPLICATE_BLOCKED	**PASS**	{"number": "MRQ-7735", "status": "DUPLICATE_BLOCKED", "description": "Forged steel weld neck flange DN100 PN16 ASTM A105", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuati...
31	[DUP2] override by the requester-steward (SoD-06)	steward.req	`POST /v1/materials/requests/MRQ-7735/override`	409 SOD_CONFLICT	409 SOD_CONFLICT	**PASS**	{"type": "https://errors.alphahelion.io/pic/SOD_CONFLICT", "title": "SOD_CONFLICT", "status": 409, "detail": "SOD-06: Requester may not approve the same material as steward", "code": "SOD_CONFLICT", "instance": "/v1/mate...
32	[DUP2] override by another steward	mdm.purchasing	`POST /v1/materials/requests/MRQ-7735/override`	200 IN_STEWARDSHIP	200: ok IN_STEWARDSHIP	**PASS**	{"number": "MRQ-7735", "status": "IN_STEWARDSHIP", "description": "Forged steel weld neck flange DN100 PN16 ASTM A105", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuation_...
33	[DUP2] reject by a steward	mdm.quality	`POST /v1/materials/requests/MRQ-7735/decision`	200 REJECTED	200: ok REJECTED	**PASS**	{"number": "MRQ-7735", "status": "REJECTED", "description": "Forged steel weld neck flange DN100 PN16 ASTM A105", "type": "STOCK", "material_group": "FORGINGS", "category": "FORGINGS", "base_uom": "EA", "valuation_class"...
34	[SoD-06] violation recorded after override attempt	system	`sod_violation`	SOD-06 row	[('SOD-06', 'MATERIAL_STEWARD_APPROVE', 'BLOCKED')]	**PASS**	
35	[MCR] preview	requester	`POST /v1/materials/M-400001/change-requests/preview`	200 classification + impact (guide: body {changes})	422 VALIDATION_FAILED: status 422 not in (200,)	**PARTIAL**	Preview refuses a body without `reason` (422); the guide documents `{changes}`; the UI sends a placeholder reason. Cosmetic contract mismatch. {"type": "https://errors.alphahelion.io/pic/VALIDATION_FAILED", "title": "VALIDATION_FAILED", "status": 422, "detail": "reason: Field required", "code": "VAL
36	[MCR] preview (with reason)	requester	`POST /v1/materials/M-400001/change-requests/preview`	200 classification + impact	200	**PASS**	{"material": "M-400001", "version": 1, "changes": [{"field": "safety_stock", "label": "Safety stock", "from": "0", "to": "200", "domain": "PLANT", "alsoApprove": [], "routed": true}, {"field": "standard_price_minor", "la...
37	A7 MCR mrp_type PW	requester	`POST /v1/materials/M-400001/change-requests/preview`	PW/PM/KB accepted, VM refused (planner set)	422 {"type": "https://errors.alphahelion.io/pic/INVALID_FIELD", "title": "INVALID_FIELD", "status": 422, "detail": "MRP type: must be one of PD,...	**PARTIAL**	MCR cannot set the weekly/monthly/Kanban MRP types the engine plans (A7)
38	A7 MCR mrp_type PM	requester	`POST /v1/materials/M-400001/change-requests/preview`	PW/PM/KB accepted, VM refused (planner set)	422 {"type": "https://errors.alphahelion.io/pic/INVALID_FIELD", "title": "INVALID_FIELD", "status": 422, "detail": "MRP type: must be one of PD,...	**INFO**	
39	A7 MCR mrp_type KB	requester	`POST /v1/materials/M-400001/change-requests/preview`	PW/PM/KB accepted, VM refused (planner set)	422 {"type": "https://errors.alphahelion.io/pic/INVALID_FIELD", "title": "INVALID_FIELD", "status": 422, "detail": "MRP type: must be one of PD,...	**INFO**	
40	A7 MCR mrp_type VM	requester	`POST /v1/materials/M-400001/change-requests/preview`	PW/PM/KB accepted, VM refused (planner set)	200 {"material": "M-400001", "version": 1, "changes": [{"field": "mrp_type", "label": "MRP type", "from": "ND", "to": "VM", "domain": "PLANT", "...	**PARTIAL**	MCR accepts VM, which the engine never plans (A7)
41	[MCR] submit change (safety stock, price, lead time)	requester	`POST /v1/materials/M-400001/change-requests`	201 MCR IN_APPROVAL, tasks per owning domain	201	**PASS**	{"number": "MCR-101", "material_id": "01a1b6b9-0359-73e2-94d9-9d26b1d5cbf1", "status": "IN_APPROVAL", "material_type": "STOCK", "changes": [{"field": "safety_stock", "label": "Safety stock", "from": "0", "to": "200", "do...
42	[MCR] tasks	system	`task`	MATERIAL_CHANGE tasks per domain (PLANT, FINANCE)	[('MATERIAL_CHANGE', 'MDM_STEWARD', 'OPEN'), ('MATERIAL_CHANGE', 'MDM_STEWARD', 'OPEN')]	**PASS**	
43	[MCR] second change while pending refused	requester2	`POST /v1/materials/M-400001/change-requests`	409 CHANGE_PENDING	409 CHANGE_PENDING	**PASS**	{"type": "https://errors.alphahelion.io/pic/CHANGE_PENDING", "title": "CHANGE_PENDING", "status": 409, "detail": "MCR-101 for M-400001 is still in approval", "code": "CHANGE_PENDING", "instance": "/v1/materials/M-400001/...
44	[MCR] decision (mdm.plant)	mdm.plant	`POST /v1/material-change-requests/MCR-101/decision`	200	200	**PASS**	{"number": "MCR-101", "material_id": "01a1b6b9-0359-73e2-94d9-9d26b1d5cbf1", "status": "IN_APPROVAL", "material_type": "STOCK", "changes": [{"to": "200", "from"...

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
45	[MCR] decision (mdm.finance)	mdm.finance	`POST /v1/material-change-requests/MCR-101/decision`	200	200	**PASS**	{"number": "MCR-101", "material_id": "01a1b6b9-0359-73e2-94d9-9d26b1d5cbf1", "status": "APPLIED", "material_type": "STOCK", "changes": [{"to": "200", "from": "0...
46	[MCR] APPLIED + version 2	requester	`GET /v1/material-change-requests/MCR-101`	APPLIED, material.changed, MAT|n|2 to ERP	200: ok APPLIED; ok ERP versioned message	**PASS**	{"number": "MCR-101", "material_id": "01a1b6b9-0359-73e2-94d9-9d26b1d5cbf1", "status": "APPLIED", "material_type": "STOCK", "changes": [{"to": "200", "from": "0", "field": "safety_stock", "label": "Safety stock", "domain...
47	[MCR2] submit hsn change	requester	`POST /v1/materials/M-400001/change-requests`	201	201	**PASS**	{"number": "MCR-102", "material_id": "01a1b6b9-0359-73e2-94d9-9d26b1d5cbf1", "status": "IN_APPROVAL", "material_type": "STOCK", "changes": [{"field": "hsn_code", "label": "HSN / SAC code", "from": "73079190", "to": "7307...
48	[MCR2] requester withdraws	requester	`POST /v1/material-change-requests/MCR-102/cancel`	200 CANCELLED	200: ok CANCELLED	**PASS**	{"number": "MCR-102", "material_id": "01a1b6b9-0359-73e2-94d9-9d26b1d5cbf1", "status": "CANCELLED", "material_type": "STOCK", "changes": [{"to": "73079990", "from": "73079190", "field": "hsn_code", "label": "HSN / SAC co...
49	A9 demo MCR in AXN seed	system	`db count`	no seeded material change request	AXN MCR rows=0	**CONFIRMED**	

F2 requisition (M04)

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
1	F2-02 guided-buying search	requester	`GET /v1/guided-buying/search`	material found, recommended channel MATERIAL, freeTextAllowed false	200: ok material in results; ok freeTextAllowed false	**PASS**	{"query": "weld neck flange", "searchOrder": ["CONTRACT", "CATALOGUE", "MATERIAL", "FREE_TEXT"], "recommendedChannel": "MATERIAL", "results": {"contract": [], "catalogue": [], "material": [{"channel": "MATERIAL", "materi...
2	F2-02 search with no hit	requester	`GET /v1/guided-buying/search`	freeTextAllowed true	200: ok freeTextAllowed	**PASS**	{"query": "zebra crossing paint", "searchOrder": ["CONTRACT", "CATALOGUE", "MATERIAL", "FREE_TEXT"], "recommendedChannel": "FREE_TEXT", "results": {"contract": [], "catalogue": [], "material": []}, "freeTextAllowed": tru...
3	REQ template catalogue	requester	`GET /v1/request-templates`	templates incl. SVC-PROFESSIONAL	200	**PASS**	[{"code": "FAC-MAINTENANCE", "name": "Facilities, maintenance and housekeeping", "group": "FACILITIES", "guidance": "State the site, the service (AMC, repair, housekeeping) and the period; recurring services use a rate c...
4	REQ template codes	requester	`GET /v1/request-templates`	SOW form codes exist	['FAC-MAINTENANCE', 'IT-HARDWARE-SOFTWARE', 'MKT-CAMPAIGN-EVENT', 'SVC-PROFESSIONAL']	**PASS**	
5	F2-02 create material requisition (DRAFT)	requester	`POST /v1/requisitions`	201 DRAFT REQ-..., process pic.requisition	201: ok DRAFT	**PASS**	{"number": "REQ-2313", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": "Line 2 tooling stock", "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant"...
6	F2-02 create replay	requester	`POST /v1/requisitions`	same REQ	201 REQ-2313	**PASS**	
7	F2-07 submit by someone else refused	requester2	`POST /v1/requisitions/REQ-2313/submit`	403	403 FORBIDDEN	**PASS**	{"type": "https://errors.alphahelion.io/pic/FORBIDDEN", "title": "FORBIDDEN", "status": 403, "detail": "Only the requester can submit this requisition", "code": "FORBIDDEN", "instance": "/v1/requisitions/REQ-2313/submit"...
8	F2-07 submit (budget n/a for stock, DMN-01)	requester	`POST /v1/requisitions/REQ-2313/submit`	IN_APPROVAL, chain [COST_OBJECT_OWNER] (₹62,500)	200: ok IN_APPROVAL; ok chain COO; ok event submitted	**PASS**	{"number": "REQ-2313", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": "Line 2 tooling stock", "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant"...
9	F2-08 approver inbox	cc.owner	`GET /v1/tasks`	REQ_APPROVAL task	[('REQ_APPROVAL', 'COST_OBJECT_OWNER')]	**PASS**	
10	F2-08 reject without comment refused	cc.owner	`POST /v1/tasks/01a1b6b9-6c0a-7033-afc3-7f87dd406885/complete`	422 comment required	422 COMMENT_REQUIRED	**PASS**	{"type": "https://errors.alphahelion.io/pic/COMMENT_REQUIRED", "title": "COMMENT_REQUIRED", "status": 422, "detail": "A comment is required to reject or return", "code": "COMMENT_REQUIRED", "instance": "/v1/tasks/01a1b6b...
11	F2-08 RETURN with comment	cc.owner	`POST /v1/tasks/01a1b6b9-6c0a-7033-afc3-7f87dd406885/complete`	200 RETURNED	200	**PASS**	{"instance_id": "01a1b6b9-6a4c-70d2-a8d3-386a8e4f2c16", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2313 \u00b7 Cost Object Owner", "object_type": "REQUISITION", "object_id": "01a1b6b9-6a46-771d-820c-63771ca...
12	F2-08 status RETURNED	requester	`GET /v1/requisitions/REQ-2313`	RETURNED	200: ok RETURNED	**PASS**	{"number": "REQ-2313", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": "Line 2 tooling stock", "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant"...

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
13	F2 edit returned line	requester	`PATCH /v1/requisitions/REQ-2313/lines/10`	200	200	**PASS**	{"status": "RETURNED", "reapproval": false}
14	F2-07 resubmit	requester	`POST /v1/requisitions/REQ-2313/submit`	IN_APPROVAL	200: ok IN_APPROVAL	**PASS**	{"number": "REQ-2313", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": "Line 2 tooling stock", "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant"...
15	F2-08 PR1: REQ_APPROVAL task (COST_OBJECT_OWNER) APPROVE	cc.owner	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-6a4c-70d2-a8d3-386a8e4f2c16", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2313 \u00b7 Cost Object Own...
16	F2-09 approved	requester	`GET /v1/requisitions/REQ-2313`	APPROVED, requisition.approved	200: ok APPROVED; ok event approved	**PASS**	{"number": "REQ-2313", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": "Line 2 tooling stock", "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant"...
17	A10 requisition sent to ERP	system	`integration_message`	PURCHASE_REQUISITION message (per A10: none)	[]	**PARTIAL**	no PURCHASE_REQUISITION message queued (A10)
18	F2-02 free-text line ₹6 L, no quotes	requester	`POST /v1/requisitions`	201 DRAFT	201	**PASS**	{"number": "REQ-2314", "origin": "MANUAL", "origin_ref": null, "purpose_code": "MAINTENANCE", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status...
19	F2-07 quote policy refuses (no quotes)	requester	`POST /v1/requisitions/REQ-2314/submit`	422 RULE_VIOLATION quote policy	422 RULE_VIOLATION	**PASS**	{"type": "https://errors.alphahelion.io/pic/RULE_VIOLATION", "title": "RULE_VIOLATION", "status": 422, "detail": "line 10: 3 quote(s) from different suppliers, each with its reference and date, required for a free text l...
20	F2-02 same line with 3 quotes	requester	`POST /v1/requisitions`	201	201	**PASS**	{"number": "REQ-2315", "origin": "MANUAL", "origin_ref": null, "purpose_code": "MAINTENANCE", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status...
21	F2-07 submit over a soft budget (CC 4200 = ₹1 L)	requester	`POST /v1/requisitions/REQ-2315/submit`	soft budget stop asks for a justification (422) or warns	422 BUDGET_JUSTIFICATION_REQUIRED	**PASS**	{"type": "https://errors.alphahelion.io/pic/BUDGET_JUSTIFICATION_REQUIRED", "title": "BUDGET_JUSTIFICATION_REQUIRED", "status": 422, "detail": "COST_CENTER 4200: soft stop, add a justification", "code": "BUDGET_JUSTIFICA...
22	F2-07 submit with justification	requester	`POST /v1/requisitions/REQ-2315/submit`	IN_APPROVAL with chain COO, DEPT_HEAD	200: ok IN_APPROVAL; ok chain	**PASS**	{"number": "REQ-2315", "origin": "MANUAL", "origin_ref": null, "purpose_code": "MAINTENANCE", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status...
23	F2-08 PR2 (first approver): REQ_APPROVAL task (COST_OBJECT_OWNER) APPROVE	cc.owner	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-80bb-75f9-adfe-f9b20c2ae9a3", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2315 \u00b7 Cost Object Own...
24	F2-08 dept head rejects	dept.head	`POST /v1/tasks/01a1b6b9-8397-7460-8db6-e8b36f32328d/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-80bb-75f9-adfe-f9b20c2ae9a3", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2315 \u00b7 Dept Head", "object_type": "REQUISITION", "object_id": "01a1b6b9-80b5-74f0-a348-f8e46190e132", ...
25	F2-08 REJECTED, budget released	requester	`GET /v1/requisitions/REQ-2315`	REJECTED	200: ok REJECTED	**PASS**	{"number": "REQ-2315", "origin": "MANUAL", "origin_ref": null, "purpose_code": "MAINTENANCE", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status...
26	F2-08 budget commitment released on reject	system	`budget`	CC 4200 committed 0	[('4100', 0), ('4200', 0), ('W-100', 0)]	**PASS**	
27	F2 cancel a REJECTED requisition	requester	`POST /v1/requisitions/REQ-2315/cancel`	409 illegal transition	409 ILLEGAL_TRANSITION	**PASS**	{"type": "https://errors.alphahelion.io/pic/ILLEGAL_TRANSITION", "title": "ILLEGAL_TRANSITION", "status": 409, "detail": "Requisition REQ-2315 is REJECTED; 'cancel' is not allowed from this state", "code": "ILLEGAL_TRANS...
28	F2-02 high-value material PR (₹25 L)	requester	`POST /v1/requisitions`	201	201	**PASS**	{"number": "REQ-2316", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...
29	F2-07 submit high value	requester	`POST /v1/requisitions/REQ-2316/submit`	chain COO, DEPT_HEAD, PROCUREMENT_HEAD, FINANCE_CONTROLLER	200: ok 4-step chain	**PASS**	{"number": "REQ-2316", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
30	F2-08 PR3: REQ_APPROVAL task (COST_OBJECT_OWNER) APPROVE	cc.owner	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-861a-71e8-8d0c-66e221b52392", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2316 \u00b7 Cost Object Own...
31	F2-08 PR3: REQ_APPROVAL task (DEPT_HEAD) APPROVE	dept.head	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-861a-71e8-8d0c-66e221b52392", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2316 \u00b7 Dept Head", "ob...
32	F2-08 PR3: REQ_APPROVAL task (PROCUREMENT_HEAD) APPROVE	proc.head	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-861a-71e8-8d0c-66e221b52392", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2316 \u00b7 Procurement Hea...
33	F2-08 PR3: REQ_APPROVAL task (FINANCE_CONTROLLER) APPROVE	fin1	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-861a-71e8-8d0c-66e221b52392", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2316 \u00b7 Finance Control...
34	F2-09 PR3 approved	requester	`GET /v1/requisitions/REQ-2316`	APPROVED	200: ok APPROVED	**PASS**	{"number": "REQ-2316", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...
35	A12 SOW form line, long text (3782 chars) as the frontend sends it	requester	`POST /v1/requisitions`	422 length check (or stored), never 500	500 {"type": "https://errors.alphahelion.io/pic/INTERNAL_ERROR", "title": "INTERNAL_ERROR", "status": 500, "detail": "An unexpected error occurred. Quote the tracel...	**BROKEN**	unhandled DB error 500 on the text the SOW form sends (Appendix A12 CONFIRMED)
36	F2-02 template SOW line (short text)	requester	`POST /v1/requisitions`	201; template defaults applied; SOW attributes stored	201	**PASS**	{"number": "REQ-2317", "origin": "MANUAL", "origin_ref": null, "purpose_code": "SERVICES", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status": ...
37	A12 SOW attributes stored on the line	system	`requisition line`	attributes kept	line keys with attr: [] value=null template=None category=PROFESSIONAL-SERVICES	**PARTIAL**	structured SOW attributes and template code are not kept on the line (A12)
38	F2-07 submit SOW requisition	requester	`POST /v1/requisitions/REQ-2317/submit`	IN_APPROVAL	200	**PASS**	{"number": "REQ-2317", "origin": "MANUAL", "origin_ref": null, "purpose_code": "SERVICES", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status": ...
39	F2-02 free-text reusable line -> MDM request	requester	`POST /v1/requisitions`	201; MRQ raised in parallel for the reusable line	201	**PASS**	{"number": "REQ-2318", "origin": "MANUAL", "origin_ref": null, "purpose_code": "MAINTENANCE", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status...
40	M03 link: reusable line raised MRQ	system	`material_request`	MRQ for line 10	[('MRQ-7736', 'IN_STEWARDSHIP')]	**PASS**	
41	A13 promote free-text line 20 (API)	requester	`POST /v1/requisitions/REQ-2318/lines/20/promote-material`	201 MRQ	201	**PASS**	{"number": "MRQ-7737", "status": "IN_STEWARDSHIP", "description": "Cable gland PG16 nickel plated brass IP68", "type": "NON_STOCK", "material_group": "ELECTRONICS", "category": "ELECTRONICS", "base_uom": "EA", "valuation...
42	A13 promote twice refused	requester	`POST /v1/requisitions/REQ-2318/lines/20/promote-material`	409 ALREADY_PROMOTED	409 ALREADY_PROMOTED	**PASS**	{"type": "https://errors.alphahelion.io/pic/ALREADY_PROMOTED", "title": "ALREADY_PROMOTED", "status": 409, "detail": "Line 20 already has material request MRQ-7737", "code": "ALREADY_PROMOTED", "instance": "/v1/requisiti...
43	F2 cancel DRAFT requisition	requester	`POST /v1/requisitions/REQ-2318/cancel`	200 CANCELLED, requisition.cancelled	200: ok CANCELLED	**PASS**	{"number": "REQ-2318", "origin": "MANUAL", "origin_ref": null, "purpose_code": "MAINTENANCE", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status...
44	F2-02 requisition without purpose code	requester	`POST /v1/requisitions`	201 draft (or 422)	201	**PASS**	{"number": "REQ-2319", "origin": "MANUAL", "origin_ref": null, "purpose_code": null, "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status": "DRAFT...
45	F2-07 submit without purpose (P01)	requester	`POST /v1/requisitions/REQ-2319/submit`	422 purpose code / MANDATORY_FIELDS_MISSING	422 RULE_VIOLATION	**PASS**	{"type": "https://errors.alphahelion.io/pic/RULE_VIOLATION", "title": "RULE_VIOLATION", "status": 422, "detail": "A purpose code is mandatory before submission", "code": "RULE_VIOLATION", "instance": "/v1/requisitions/RE...
46	A11 hosted catalogue item maintenance API	admin	`POST /v1/catalogue-items`	none exists (A11)	404	**NOT BUILT**	no command (Appendix A11 CONFIRMED)

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
47	REQ-04 buyer workbench	buyer	`GET /v1/buyer-workbench`	approved lines incl. PR1, PR3	200: ok PR1 listed	**PASS**	[{"requisition": "REQ-2313", "line": 10, "origin": "MANUAL", "description": "Weld neck flange DN100", "itemType": "MATERIAL", "purchaseType": "STOCK", "material": "M-400001", "category": "FORGINGS", "priceEstMinor": 1250...
48	REQ-04 workbench needs req:buyer	requester	`GET /v1/buyer-workbench`	403	403 FORBIDDEN	**PASS**	{"type": "https://errors.alphahelion.io/pic/FORBIDDEN", "title": "FORBIDDEN", "status": 403, "detail": "Missing scope(s): req:buyer", "code": "FORBIDDEN", "instance": "/v1/buyer-workbench", "traceId": "edfcd5d71cefe7c2e9...
49	F2 change-after-approval: requisition	requester	`POST /v1/requisitions`	201	201	**PASS**	{"number": "REQ-2321", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...
50	F2 change-after-approval: submit	requester	`POST /v1/requisitions/REQ-2321/submit`	200	200	**PASS**	{"number": "REQ-2321", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...
51	F2 change-after-approval: approval: REQ_APPROVAL task (COST_OBJECT_OWNER) APPROVE	cc.owner	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6ba-2bd4-71db-9d63-dab94af5c21a", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2321 \u00b7 Cost Object Own...
52	F2 change-after-approval: approved	requester	`GET /v1/requisitions/REQ-2321`	APPROVED	APPROVED	**PASS**	
53	F2-08 change value after approval -> re-approval	requester	`PATCH /v1/requisitions/REQ-2321/lines/10`	back to IN_APPROVAL with a new round	200	**PASS**	{"status": "IN_APPROVAL", "reapproval": true}
54	F2-08 status after value change	system	`requisition`	IN_APPROVAL	IN_APPROVAL	**PASS**	
55	F2-08 change need date only (no re-approval)	requester	`PATCH /v1/requisitions/REQ-2316/lines/10`	200, stays APPROVED (or 409 when in sourcing)	200	**PASS**	{"status": "APPROVED", "reapproval": false}
56	F2-02 guided search finds the contract line	requester	`GET /v1/guided-buying/search`	recommended CONTRACT channel with today's price	200: ok contract hit	**PASS**	{"query": "weld neck flange", "searchOrder": ["CONTRACT", "CATALOGUE", "MATERIAL", "FREE_TEXT"], "recommendedChannel": "CONTRACT", "results": {"contract": [{"channel": "CONTRACT", "contract": "SA-5520", "line": 10, "cont...
57	guided search channel	requester	`GET /v1/guided-buying/search`	recommended = CONTRACT	recommended=CONTRACT contractHit={"channel": "CONTRACT", "contract": "SA-5520", "line": 10, "contractLineId": "01a1b6b9-d251-7340-a3fa-f86e5c9b923f", "material": "M-400001", "materialGroup": "F...	**INFO**	
58	F2-02 CONTRACT line requisition (₹12,300)	requester	`POST /v1/requisitions`	201, price from contract	201	**PASS**	{"number": "REQ-2327", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...
59	F2-08 auto-approval (empty chain: contract line, <= ₹25k)	requester	`POST /v1/requisitions/REQ-2327/submit`	APPROVED immediately, approval_chain []	200: ok APPROVED; ok empty chain	**PASS**	{"number": "REQ-2327", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...

F3 sourcing (M05)

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
1	F3-01 bundle candidates	category.manager	`GET /v1/sourcing/bundle-candidates`	the approved PR line is a candidate	200: ok PR listed	**PASS**	{"category": null, "categoryPlan": null, "origin": null, "items": [{"requisition": "REQ-2311", "line": 10, "origin": "MRP", "category": "FORGINGS", "description": "Forged steel weld neck flange DN100 PN16 ASTM A105", "ma...
2	F3-02 invite a supplier not qualified for the category	category.manager	`POST /v1/sourcing/events`	422 P06/SRC-INVITE	422 RULE_VIOLATION	**PASS**	{"type": "https://errors.alphahelion.io/pic/RULE_VIOLATION", "title": "RULE_VIOLATION", "status": 422, "detail": "SUP-1004 is NOT_REGISTERED for FORGINGS; only Qualified suppliers (or Conditional with waiver) may be invi...

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
3	F3-02 weights must total 100 (create)	category.manager	`POST /v1/sourcing/events`	draft may be saved; weights 110 refused at publish (SRC-WEIGHTS)	201	**PASS**	{"number": "RFQ-2231", "type": "RFQ", "title": "Weld neck flanges DN100 - annual rate", "category": "FORGINGS", "status": "DRAFT", "owner_id": "01a1b6b8-8b2b-76fc-be89-82c38cf5f617", "currency": "INR", "round": 1, "weigh...
4	F3-02 publish with weights 110 refused	category.manager	`POST /v1/sourcing/events/RFQ-2231/publish`	422 SRC-WEIGHTS	422 RULE_VIOLATION	**PASS**	{"type": "https://errors.alphahelion.io/pic/RULE_VIOLATION", "title": "RULE_VIOLATION", "status": 422, "detail": "Scoring weights must sum to 100 (currently 110)", "code": "RULE_VIOLATION", "instance": "/v1/sourcing/even...
5	F3-13 cancel draft	category.manager	`POST /v1/sourcing/events/RFQ-2231/cancel`	200 CANCELLED, lines back to OPEN	200	**PASS**	{"number": "RFQ-2231", "type": "RFQ", "title": "Weld neck flanges DN100 - annual rate", "category": "FORGINGS", "status": "CANCELLED", "owner_id": "01a1b6b8-8b2b-76fc-be89-82c38cf5f617", "currency": "INR", "round": 1, "w...
6	F3-02 create event from PR line	category.manager	`POST /v1/sourcing/events`	201 DRAFT RFQ-..., PR line IN_SOURCING	201: ok DRAFT; ok PR line IN_SOURCING	**PASS**	{"number": "RFQ-2232", "type": "RFQ", "title": "Weld neck flanges DN100 - annual rate", "category": "FORGINGS", "status": "DRAFT", "owner_id": "01a1b6b8-8b2b-76fc-be89-82c38cf5f617", "currency": "INR", "round": 1, "weigh...
7	F3 portal cannot see a draft	SUPPLIER_USER A	`GET /v1/sourcing/events/RFQ-2232`	404	404 NOT_FOUND	**PASS**	{"type": "https://errors.alphahelion.io/pic/NOT_FOUND", "title": "NOT_FOUND", "status": 404, "detail": "Sourcing event RFQ-2232 not found", "code": "NOT_FOUND", "instance": "/v1/sourcing/events/RFQ-2232", "traceId": "79a...
8	F3-02 publish with a stale If-Match	category.manager	`POST /v1/sourcing/events/RFQ-2232/publish`	412	412	**PASS**	{"type": "https://errors.alphahelion.io/pic/STALE_WRITE", "title": "STALE_WRITE", "status": 412, "detail": "The document...
9	F3-02 publish (weights lock)	category.manager	`POST /v1/sourcing/events/RFQ-2232/publish`	200 OPEN, SOURCING_CLOSE timer, sourcing.event.published	200: ok OPEN; ok event published; ok close timer	**PASS**	{"number": "RFQ-2232", "type": "RFQ", "title": "Weld neck flanges DN100 - annual rate", "category": "FORGINGS", "status": "OPEN", "owner_id": "01a1b6b8-8b2b-76fc-be89-82c38cf5f617", "currency": "INR", "round": 1, "weight...
10	F3-02 publish again refused	category.manager	`POST .../publish`	409	409	**PASS**	
11	F3-02 invitees notified	system	`notification`	invitation notification per supplier	[]	**PARTIAL**	
12	F3-04 clarification question	SUPPLIER_USER B	`POST /v1/sourcing/events/RFQ-2232/clarifications`	201	201	**PASS**	{"seq": 1, "kind": "QUESTION", "question": "Is the PN16 flange to be supplied with mill test certificate EN 10204 3.1?", "answer": null, "status": "OPEN", "askedAt": "2026-11-11T10:00:00+05:30", "answeredAt": null, "mine...
13	F3-04 answer to all invitees	category.manager	`POST /v1/sourcing/events/RFQ-2232/clarifications/1/answer`	200 sourcing.clarification.answered	200	**PASS**	{"seq": 1, "kind": "QUESTION", "question": "Is the PN16 flange to be supplied with mill test certificate EN 10204 3.1?", "answer": "Yes, EN 10204 3.1 certificates are required with every lot.", "status": "ANSWERED", "ask...
14	F3-05 sealed bid A	SUPPLIER_USER A	`POST /v1/sourcing/events/RFQ-2232/bids`	201 SEALED, only hash readable	201: ok SEALED	**PASS**	{"event": "RFQ-2232", "round": 1, "status": "SEALED", "hash": "cce1c6536800a78511cb15e9817e257f4934fd70abaded644741d8bda90fde95", "edits": 0, "submittedAt": "2026-11-11T10:00:00+05:30"}
15	F3-05 sealed bid B	SUPPLIER_USER B	`POST /v1/sourcing/events/RFQ-2232/bids`	201 SEALED, only hash readable	201: ok SEALED	**PASS**	{"event": "RFQ-2232", "round": 1, "status": "SEALED", "hash": "1fc9b73637b80e86c7661236df3e015facf52bb6cbbae41c8e3e5e4c81c778d8", "edits": 0, "submittedAt": "2026-11-11T10:00:00+05:30"}
16	F3-05 sealed bid C	SUPPLIER_USER C	`POST /v1/sourcing/events/RFQ-2232/bids`	201 SEALED, only hash readable	201: ok SEALED	**PASS**	{"event": "RFQ-2232", "round": 1, "status": "SEALED", "hash": "e1952044e83a418125a57c731ee922c9358671076d00c762a42d0d4bbc633672", "edits": 0, "submittedAt": "2026-11-11T10:00:00+05:30"}
17	F3-05 bid stored sealed	system	`bid table`	encrypted payload + SHA-256	[('SEALED', 64), ('SEALED', 64), ('SEALED', 64)] cols=['sealed_payload', 'payload_hash']	**PASS**	
18	F3-05 uninvited supplier bid refused + logged	SUPPLIER_USER D (not invited)	`POST /v1/sourcing/events/RFQ-2232/bids`	403 NOT_INVITED + audit UNINVITED_BID_REFUSED	403 NOT_INVITED: ok audit	**PASS**	{"type": "https://errors.alphahelion.io/pic/NOT_INVITED", "title": "NOT_INVITED", "status": 403, "detail": "SUP-1004 is not invited to RFQ-2232", "code": "NOT_INVITED", "instance": "/v1/sourcing/events/RFQ-2232/bids", "t...
19	F3-06 early read of bids refused	category.manager	`GET /v1/sourcing/events/RFQ-2232/bids`	423 BIDS_SEALED + audit BID_ACCESS_DENIED	423 BIDS_SEALED: ok audit BID_ACCESS_DENIED	**PASS**	{"type": "https://errors.alphahelion.io/pic/BIDS_SEALED", "title": "BIDS_SEALED", "status": 423, "detail": "Bids for RFQ-2232 stay sealed until 2026-11-13T11:30:00+00:00", "code": "BIDS_SEALED", "instance": "/v1/sourcing...
20	F3-06 early comparison refused	category.manager	`GET /v1/sourcing/events/RFQ-2232/comparison`	423	423 BIDS_SEALED	**PASS**	{"type": "https://errors.alphahelion.io/pic/BIDS_SEALED", "title": "BIDS_SEALED", "status": 423, "detail": "Bids for RFQ-2232 stay sealed until 2026-11-13T11:30:00+00:00", "code": "BIDS_SEALED", "instance": "/v1/sourcing...

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
21	F3-06 close before close time	category.manager	`POST /v1/sourcing/events/RFQ-2232/close`	409 CLOSE_TIME_NOT_REACHED	409 CLOSE_TIME_NOT_REACHED	**PASS**	{"type": "https://errors.alphahelion.io/pic/CLOSE_TIME_NOT_REACHED", "title": "CLOSE_TIME_NOT_REACHED", "status": 409, "detail": "RFQ-2232 closes at 2026-11-13T11:30:00+00:00", "code": "CLOSE_TIME_NOT_REACHED", "instance...
22	F3-05 late bid refused + logged	SUPPLIER_USER C	`POST /v1/sourcing/events/RFQ-2232/bids`	refused LATE_BID_REFUSED	409 EVENT_CLOSED: ok audit LATE_BID_REFUSED	**PASS**	{"type": "https://errors.alphahelion.io/pic/EVENT_CLOSED", "title": "EVENT_CLOSED", "status": 409, "detail": "RFQ-2232 closed at 2026-11-13T11:30:00+00:00", "code": "EVENT_CLOSED", "instance": "/v1/sourcing/events/RFQ-22...
23	F3-06 close timer fired	worker	`timer SOURCING_CLOSE`	CLOSED (two-person opening pending)	status=CLOSED	**PASS**	
24	F3-06 read before two-person opening	category.manager	`GET /v1/sourcing/events/RFQ-2232/bids`	423 still sealed	423 BIDS_SEALED	**PASS**	{"type": "https://errors.alphahelion.io/pic/BIDS_SEALED", "title": "BIDS_SEALED", "status": 423, "detail": "Bids for RFQ-2232 stay sealed until 2026-11-13T11:30:00+00:00 and are then opened by two people", "code": "BIDS_...
25	F3-06 first opener	category.manager	`POST /v1/sourcing/events/RFQ-2232/open`	200 first opening recorded	200	**PASS**	{"event": "RFQ-2232", "unsealed": 0, "openers": 1, "remaining": 1, "awaitingOpening": true}
26	F3-06 same person again refused	category.manager	`POST /v1/sourcing/events/RFQ-2232/open`	409 ALREADY_OPENED_BY_YOU	409 ALREADY_OPENED_BY_YOU	**PASS**	{"type": "https://errors.alphahelion.io/pic/ALREADY_OPENED_BY_YOU", "title": "ALREADY_OPENED_BY_YOU", "status": 409, "detail": "You have already opened these bids: a second, different person must open them too", "code": ...
27	F3-06 second opener unseals	category.manager2	`POST /v1/sourcing/events/RFQ-2232/open`	200 UNDER_EVALUATION, BIDS_UNSEALED, sourcing.bids.unsealed	200: ok event unsealed	**PASS**	{"event": "RFQ-2232", "unsealed": 3, "competitive": true, "matrix": [{"supplier": "SUP-1003", "supplierId": "01a1b6b8-a9ff-755f-8603-8d185fcbde85", "round": 1, "bidId": "01a1b6b9-b09a-757e-a2d2-6f0076cb2d52", "submittedA...
28	F3-06 bids readable	category.manager	`GET /v1/sourcing/events/RFQ-2232/bids`	200 3 bids	200	**PASS**	[{"supplier": "SUP-1003", "supplierId": "01a1b6b8-a9ff-755f-8603-8d185fcbde85", "round": 1, "bidId": "01a1b6b9-b09a-757e-a2d2-6f0076cb2d52", "submittedAt": "2026-11-11T04:30:00+00:00", "validUntil": "2027-03-09", "lateDe...
29	F3-07 scoring before declaration refused	category.manager	`POST /v1/sourcing/events/RFQ-2232/scores`	refused: declaration first	422 DECLARATION_REQUIRED	**PASS**	{"type": "https://errors.alphahelion.io/pic/DECLARATION_REQUIRED", "title": "DECLARATION_REQUIRED", "status": 422, "detail": "Declare any conflict of interest in RFQ-2232 before you score its bids", "code": "DECLARATION_...
30	F3-07 conflict declaration (owner)	category.manager	`POST /v1/sourcing/events/RFQ-2232/declarations`	201	201	**PASS**	{"user": "category.manager@fa7.example", "hasConflict": false, "declaredAt": "2026-11-13T18:00:00+05:30"}
31	F3-07 scores (owner only)	category.manager	`POST /v1/sourcing/events/RFQ-2232/scores`	200	200	**PASS**	{"event": "RFQ-2232", "matrix": [{"supplier": "SUP-1003", "supplierId": "01a1b6b8-a9ff-755f-8603-8d185fcbde85", "round": 1, "bidId": "01a1b6b9-b09a-757e-a2d2-6f0076cb2d52", "submittedAt": "2026-11-11T04:30:00+00:00", "va...
32	F3-07 comparison (TCO, ranking, technical gate)	category.manager	`GET /v1/sourcing/events/RFQ-2232/comparison`	C fails technical gate (40 < 50)	200: ok gate shown	**PASS**	{"event": "RFQ-2232", "status": "UNDER_EVALUATION", "round": 1, "scoringMethod": "LOWEST_RATIO", "currency": "INR", "referenceDate": "2026-11-10", "competitive": true, "minBidders": 2, "validBidders": 3, "baselineMinor":...
33	F3-06 scenarios	category.manager	`POST /v1/sourcing/events/RFQ-2232/scenarios`	LOWEST_COST, BEST_VALUE, CAPACITY_CONSTRAINED; C excluded	200	**PASS**	[{"id": "01a1b6b9-c407-768e-a9f0-271b2651de90", "kind": "LOWEST_COST", "feasible": true, "allocations": [{"supplier": "SUP-1002", "supplierId": "01a1b6b8-9edb-7621-8e39-bb6b890e9a3d", "share": "1.0000", "monthlyQty": "20...
34	F3-06 scenario allocations	system	`scenarios`	BEST_VALUE -> A, LOWEST_COST -> B (C gated out)	[('LOWEST_COST', [('SUP-1002', '1.0000')], True), ('BEST_VALUE', [('SUP-1001', '1.0000')], True), ('CAPACITY_CONSTRAINED', [('SUP-1002', '1.0000')], True)]	**PASS**	
35	SoD-05 sole evaluator proposes award	category.manager	`POST /v1/sourcing/events/RFQ-2232/award`	409 SOD_CONFLICT	409 SOD_CONFLICT	**PASS**	{"type": "https://errors.alphahelion.io/pic/SOD_CONFLICT", "title": "SOD_CONFLICT", "status": 409, "detail": "SOD-05: The sole evaluator of an event may not awa...
36	F3-07 second evaluator declaration	quality.inspector	`POST /v1/sourcing/events/RFQ-2232/declarations`	201	201	**PASS**	{"user": "quality.inspector@fa7.example", "hasConflict": false, "declaredAt": "2026-11-13T18:00:00+05:30"}
37	F3-07 second evaluator scores	quality.inspector	`POST /v1/sourcing/events/RFQ-2232/scores`	200	200	**PASS**	{"event": "RFQ-2232", "matrix": [{"supplier": "SUP-1003", "supplierId": "01a1b6b8-a9ff-755f-8603-8d185fcbde85", "round": 1, "bidId": "01a1b6b9-b09a-757e-a2d2-6f0076cb2d52", "submittedAt": "2026-11-11T04:30:00+00:00", "va...

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
38	F3-10 award non-lowest without rationale refused	category.manager	`POST /v1/sourcing/events/RFQ-2232/award`	422 rationale (NOT_LOWEST)	422 RULE_VIOLATION	**PASS**	{"type": "https://errors.alphahelion.io/pic/RULE_VIOLATION", "title": "RULE_VIOLATION", "status": 422, "detail": "Award to a non-lowest bid needs a written rationale", "code": "RULE_VIOLATION", "instance": "/v1/sourcing/...
39	F3-10 propose award with rationale	category.manager	`POST /v1/sourcing/events/RFQ-2232/award`	200 AWARD_PENDING, AWARD_APPROVAL chain (3 Cr: CM -> CPO -> FC)	200: ok AWARD_PENDING	**PASS**	{"number": "RFQ-2232", "type": "RFQ", "title": "Weld neck flanges DN100 - annual rate", "category": "FORGINGS", "status": "AWARD_PENDING", "owner_id": "01a1b6b8-8b2b-76fc-be89-82c38cf5f617", "currency": "INR", "round": 1...
40	F3-11 award approval tasks	system	`task`	AWARD_APPROVAL CATEGORY_MANAGER first, proposer excluded	[('AWARD_APPROVAL', 'CATEGORY_MANAGER', 'OPEN')]	**PASS**	
41	F3-11 proposer has no approval task	category.manager	`GET /v1/tasks`	none	0	**PASS**	
42	F3-11 award approval: AWARD_APPROVAL task (CATEGORY_MANAGER) APPROVE	category.manager2	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-a7e7-734b-b6fa-cb9ece0968f9", "kind": "AWARD_APPROVAL", "name": "Award Approval \u00b7 RFQ-2232 \u00b7 Category Ma...
43	F3-11 award approval: AWARD_APPROVAL task (CPO) APPROVE	cpo	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-a7e7-734b-b6fa-cb9ece0968f9", "kind": "AWARD_APPROVAL", "name": "Award Approval \u00b7 RFQ-2232 \u00b7 Cpo", "obje...
44	F3-11 award approval: AWARD_APPROVAL task (FINANCE_CONTROLLER) APPROVE	fin1	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-a7e7-734b-b6fa-cb9ece0968f9", "kind": "AWARD_APPROVAL", "name": "Award Approval \u00b7 RFQ-2232 \u00b7 Finance Con...
45	F3-12 AWARDED	category.manager	`GET /v1/sourcing/events/RFQ-2232`	AWARDED, PR line SOURCED	200: ok AWARDED	**PASS**	{"number": "RFQ-2232", "type": "RFQ", "title": "Weld neck flanges DN100 - annual rate", "category": "FORGINGS", "status": "AWARDED", "owner_id": "01a1b6b8-8b2b-76fc-be89-82c38cf5f617", "currency": "INR", "round": 1, "wei...
46	F3-12 events	system	`outbox`	sourcing.event.awarded, sourcing.award.regret x2	['sourcing.award.regret', 'sourcing.bid.submitted', 'sourcing.bids.unsealed', 'sourcing.clarification.answered', 'sourcing.event.awarded', 'sourcing.event.published', 'task.assigned']	**PASS**	
47	F3-12 regret letters	system	`notification SOURCING_NOT_AWARDED`	B and C portal users	[('SOURCING_NOT_AWARDED', 'RFQ-2232', 'accounts@c-fa7.example'), ('SOURCING_NOT_AWARDED', 'RFQ-2232', 'accounts@b-fa7.example')]	**PASS**	
48	F3-12 hand-off: contract draft from award	consumer contract-from-award	`contract table`	SCHEDULING_AGREEMENT DRAFT for A	[('SA-5520', 'SCHEDULING_AGREEMENT', 'DRAFT', 'SUP-1001')]	**PASS**	
49	F3-12 hand-off view	category.manager	`GET /v1/sourcing/events/RFQ-2232/handoff`	contracts, prices pending, poReleaseEligible false	200	**PASS**	{"event": "RFQ-2232", "status": "AWARDED", "decisions": [{"status": "APPROVED", "valueMinor": 3000000000, "exceptions": [{"code": "NOT_LOWEST", "detail": "the award includes a bid that is not the lowest eligible price"},...
50	B6 RFP numbering	category.manager	`POST /v1/sourcing/events`	RFP number series	201	**PASS**	{"number": "RFQ-2233", "type": "RFP", "title": "RFP numbering probe", "category": "FORGINGS", "status": "DRAFT", "owner_id": "01a1b6b8-8b2b-76fc-be89-82c38cf5f617", "currency": "INR", "round": 1, "weights_locked": false,...
51	B6 every event type numbered RFQ-	system	`sourcing event`	RFP-...	RFQ-2233 type=RFP	**PARTIAL**	cosmetic: RFP numbered RFQ- (B6)
52	F3-12 savings recorded as NEGOTIATED	consumer savings-negotiated	`savings_initiative`	initiative NEGOTIATED/CONTRACTED	[('SV-0412', 'CONTRACTED')]	**PASS**	

F4 contract (M07)

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
1	F4-02 award draft: clauses, obligations, price validity	contract.manager	`GET /v1/contracts/SA-5520`	DRAFT, clauses from CLM-PLAYBOOK, validity from 1st of next month, price validity unpublished	200: ok DRAFT; ok clauses; ok price validity unpublished	**PASS**	{"number": "SA-5520", "type": "SCHEDULING_AGREEMENT", "title": "Forged steel weld neck flange DN100 PN16 ASTM A105 \u2013 Aravind Forge Works Pvt Ltd", "supplier_id": "01a1b6b8-9558-7755-b4bf-3e74eb841130", "company_code...
2	F4-02 award contract terms	system	`contract`	SA, price 1250, 12 months	type=SCHEDULING_AGREEMENT from=2026-12-01 to=2027-11-30 lines=[('24000', None)] owner=01a1b6b8-8b2b-76fc-be89-82c38cf5f617	**INFO**	
3	F4-03 AI review -> IN_NEGOTIATION	contract.manager	`POST /v1/contracts/SA-5520/review`	200 risk, IN_NEGOTIATION	200	**PASS**	{"risk": 0, "riskBand": "LOW", "findings": [], "aiRef": "agent-run:01a1b6b9-d888-7136-b8e8-a37d08f05b3d"}
4	F4-04 supplier redline (portal)	SUPPLIER_USER A	`POST /v1/contracts/SA-5520/redlines`	200 NON_STANDARD pending	200 PAYMENT	**PASS**	{"contract_id": "01a1b6b9-d24a-763f-8dfe-91f5fb55280f", "code": "PAYMENT", "title": "Payment terms and MSME compliance", "text": "Payment N45 from receipt of a correct invoice.", "position": "NON_STANDARD", "risk_score":...
5	F4-05 submit with pending redline refused	contract.manager	`POST /v1/contracts/SA-5520/submit`	409 REDLINE_REVIEW_PENDING	409 REDLINE_REVIEW_PENDING	**PASS**	{"type": "https://errors.alphahelion.io/pic/REDLINE_REVIEW_PENDING", "title": "REDLINE_REVIEW_PENDING", "status": 409, "detail": "Review the supplier redline(s) on PAYMENT before submitting SA-5520 for approval", "code":...
6	F4-04 review redline: ACCEPT as FALLBACK	contract.manager	`POST /v1/contracts/SA-5520/redlines/PAYMENT/review`	200 FALLBACK risk 30	200 PAYMENT	**PASS**	{"contract_id": "01a1b6b9-d24a-763f-8dfe-91f5fb55280f", "code": "PAYMENT", "title": "Payment terms and MSME compliance", "text": "Payment N45 from receipt of a correct invoice.", "position": "FALLBACK", "risk_score": 30,...
7	F4-05 submit	contract.manager	`POST /v1/contracts/SA-5520/submit`	IN_APPROVAL, chain LEGAL, FINANCE_CONTROLLER, CPO (value >= 1 Cr)	200: ok chain	**PASS**	{"status": "IN_APPROVAL", "chain": ["LEGAL", "FINANCE_CONTROLLER", "CPO"]}
8	F4-06 contract approval: CONTRACT_APPROVAL task (LEGAL) APPROVE	legal	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-d30a-73bb-951a-b0969698b5b9", "kind": "CONTRACT_APPROVAL", "name": "Contract Approval \u00b7 SA-5520 \u00b7 Legal"...
9	F4-06 contract approval: CONTRACT_APPROVAL task (FINANCE_CONTROLLER) APPROVE	fin1	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-d30a-73bb-951a-b0969698b5b9", "kind": "CONTRACT_APPROVAL", "name": "Contract Approval \u00b7 SA-5520 \u00b7 Financ...
10	F4-06 contract approval: CONTRACT_APPROVAL task (CPO) APPROVE	cpo	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-d30a-73bb-951a-b0969698b5b9", "kind": "CONTRACT_APPROVAL", "name": "Contract Approval \u00b7 SA-5520 \u00b7 Cpo", ...
11	F4-06 OUT_FOR_SIGNATURE	contract.manager	`GET /v1/contracts/SA-5520`	OUT_FOR_SIGNATURE	200: ok OUT_FOR_SIGNATURE	**PASS**	{"number": "SA-5520", "type": "SCHEDULING_AGREEMENT", "title": "Forged steel weld neck flange DN100 PN16 ASTM A105 \u2013 Aravind Forge Works Pvt Ltd", "supplier_id": "01a1b6b8-9558-7755-b4bf-3e74eb841130", "company_code...
12	F4-08 activate before signatures refused	contract.manager	`POST /v1/contracts/SA-5520/activate`	422 CLM-ACTIVATE	422 RULE_VIOLATION	**PASS**	{"type": "https://errors.alphahelion.io/pic/RULE_VIOLATION", "title": "RULE_VIOLATION", "status": 422, "detail": "All signatures are required before activation", "code": "RULE_VIOLATION", "instance": "/v1/contracts/SA-55...
13	F4-07 buyer signature	contract.manager	`POST /v1/contracts/SA-5520/signatures`	200 signedBy BUYER	200	**PASS**	{"contract": "SA-5520", "signedBy": ["BUYER"]}
14	F4-07 supplier signature (portal)	SUPPLIER_USER A	`POST /v1/contracts/SA-5520/signatures`	200 signedBy BUYER, SUPPLIER	200	**PASS**	{"contract": "SA-5520", "signedBy": ["BUYER", "SUPPLIER"]}
15	F4-08 activate	contract.manager	`POST /v1/contracts/SA-5520/activate`	ACTIVE, price validity published, timers, contract.activated	200: ok ACTIVE; ok published; ok timers	**PASS**	{"number": "SA-5520", "type": "SCHEDULING_AGREEMENT", "title": "Forged steel weld neck flange DN100 PN16 ASTM A105 \u2013 Aravind Forge Works Pvt Ltd", "supplier_id": "01a1b6b8-9558-7755-b4bf-3e74eb841130", "company_code...
16	F4-08 activation events/handoff	system	`outbox/quota/source list`	contract.activated, quota line, source list entry	outbox=['contract.activated', 'task.assigned'] quota=[(1,)] source_list=[(1,)]	**PASS**	
17	F3-12 hand-off after activation	category.manager	`GET /v1/sourcing/events/RFQ-2232/handoff`	prices published, quota, source list, poReleaseEligible true	200: ok poReleaseEligible	**PASS**	{"event": "RFQ-2232", "status": "AWARDED", "decisions": [{"status": "APPROVED", "valueMinor": 3000000000, "exceptions": [{"code": "NOT_LOWEST", "detail": "the award includes a bid that is not the lowest eligible price"},...
18	B5 contract posted to ERP	system	`integration_message`	(B5: contracts are not posted)	[]	**PARTIAL**	contract not posted to ERP (B5)

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
19	B6 award contract numbering	system	`contract`	SA- for a scheduling agreement	SA-5520	**INFO**	
20	F4-10 add obligation	contract.manager	`POST /v1/contracts/SA-5520/obligations`	201	201	**PASS**	{"contract_id": "01a1b6b9-d24a-763f-8dfe-91f5fb55280f", "party": "SUPPLIER", "description": "Submit quarterly capacity confirmation", "due_date": "2026-12-19", "status": "OPEN", "kind": "OBLIGATION", "completed_at": null...
21	F4-10 complete obligation	contract.manager	`POST /v1/obligations/01a1b6b9-e99a-766f-9bbf-6355e8e42bde/complete`	200 on time	200	**PASS**	{"contract_id": "01a1b6b9-d24a-763f-8dfe-91f5fb55280f", "party": "SUPPLIER", "description": "Submit quarterly capacity confirmation", "due_date": "2026-12-19", "status": "DONE", "kind": "OBLIGATION", "completed_at": "202...
22	B4 MAT-B contract with A: draft	contract.manager	`POST /v1/contracts`	201 DRAFT	201	**PASS**	{"number": "SA-5521", "type": "QUANTITY_CONTRACT", "title": "B4 MAT-B contract with A contract", "supplier_id": "01a1b6b8-9558-7755-b4bf-3e74eb841130", "company_code": "1000", "status": "DRAFT", "valid_from": "2026-11-17...
23	B4 MAT-B contract with A: submit	contract.manager	`POST /v1/contracts/SA-5521/submit`	200 IN_APPROVAL	200	**PASS**	{"status": "IN_APPROVAL", "chain": ["CONTRACT_MANAGER"]}
24	B4 MAT-B contract with A: approval: CONTRACT_APPROVAL task (CONTRACT_MANAGER) APPROVE	contract.manager2	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6ba-3d67-70a7-ae6e-852ef9d0fba1", "kind": "CONTRACT_APPROVAL", "name": "Contract Approval \u00b7 SA-5521 \u00b7 Contra...
25	B4 MAT-B contract with A: activate	contract.manager	`POST /v1/contracts/SA-5521/activate`	200 ACTIVE	200: ok ACTIVE	**PASS**	{"number": "SA-5521", "type": "QUANTITY_CONTRACT", "title": "B4 MAT-B contract with A contract", "supplier_id": "01a1b6b8-9558-7755-b4bf-3e74eb841130", "company_code": "1000", "status": "ACTIVE", "valid_from": "2026-11-1...
26	B4 MAT-B contract with B: draft	contract.manager	`POST /v1/contracts`	201 DRAFT	201	**PASS**	{"number": "SA-5522", "type": "QUANTITY_CONTRACT", "title": "B4 MAT-B contract with B contract", "supplier_id": "01a1b6b8-9edb-7621-8e39-bb6b890e9a3d", "company_code": "1000", "status": "DRAFT", "valid_from": "2026-11-17...
27	B4 MAT-B contract with B: submit	contract.manager	`POST /v1/contracts/SA-5522/submit`	200 IN_APPROVAL	200	**PASS**	{"status": "IN_APPROVAL", "chain": ["CONTRACT_MANAGER"]}
28	B4 MAT-B contract with B: approval: CONTRACT_APPROVAL task (CONTRACT_MANAGER) APPROVE	contract.manager2	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6ba-4aa4-7703-bbdb-2a519b2edafc", "kind": "CONTRACT_APPROVAL", "name": "Contract Approval \u00b7 SA-5522 \u00b7 Contra...
29	B4 MAT-B contract with B: activate	contract.manager	`POST /v1/contracts/SA-5522/activate`	200 ACTIVE	200: ok ACTIVE	**PASS**	{"number": "SA-5522", "type": "QUANTITY_CONTRACT", "title": "B4 MAT-B contract with B contract", "supplier_id": "01a1b6b8-9edb-7621-8e39-bb6b890e9a3d", "company_code": "1000", "status": "ACTIVE", "valid_from": "2026-11-1...
30	F4-09 consumption alert at 80 %	consumer	`outbox/notification`	contract.consumption.threshold + CONTRACT_CONSUMPTION notification	events=['contract.consumption.threshold'] notif= [('CONTRACT_CONSUMPTION', 'SA-5520', 'category.manager@fa7.example'), ('CONTRACT_CONSUMPTION', 'SA-5520', 'contract.manager2@fa7.example'), ('CONTRACT_CONSUMPTION', 'SA-5520', 'contract.manager@fa7.example')]	**PASS**	
31	F4-09b index reset outside the band	contract.manager	`POST /v1/contracts/SA-5520/index-resets`	INDEX_RESET_ADJUSTED, new price-validity period	200	**PASS**	{"contract": "SA-5520", "changePct": "10.00", "adjustPct": "7.00", "lines": [{"line": 10, "oldPriceMinor": 125000, "newPriceMinor": 133750}]}
32	F4-09b price periods after index reset	system	`price_validity`	new period from 2027-01-01, older untouched	[('2026-12-01', '2027-11-30', 125000, 'AWARD'), ('2027-01-01', '2027-11-30', 133750, 'INDEX_RESET_ADJUSTED')]	**INFO**	
33	F4 short NDA (expiry probe): draft	contract.manager	`POST /v1/contracts`	201 DRAFT	201	**PASS**	{"number": "SA-5523", "type": "NDA", "title": "F4 short NDA (expiry probe) contract", "supplier_id": "01a1b6b8-a9ff-755f-8603-8d185fcbde85", "company_code": "1000", "status": "DRAFT", "valid_from": "2026-11-17", "valid_t...

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
34	F4 short NDA (expiry probe): submit	contract.manager	`POST /v1/contracts/SA-5523/submit`	200 IN_APPROVAL	200	**PASS**	{"status": "IN_APPROVAL", "chain": ["CONTRACT_MANAGER"]}
35	F4 short NDA (expiry probe): approval: CONTRACT_APPROVAL task (CONTRACT_MANAGER) APPROVE	contract.manager2	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6ba-cd1f-7225-a7d2-1047c4876342", "kind": "CONTRACT_APPROVAL", "name": "Contract Approval \u00b7 SA-5523 \u00b7 Contra...
36	F4 short NDA (expiry probe): activate	contract.manager	`POST /v1/contracts/SA-5523/activate`	200 ACTIVE	200: ok ACTIVE	**PASS**	{"number": "SA-5523", "type": "NDA", "title": "F4 short NDA (expiry probe) contract", "supplier_id": "01a1b6b8-a9ff-755f-8603-8d185fcbde85", "company_code": "1000", "status": "ACTIVE", "valid_from": "2026-11-17", "valid_...
37	F4-01 manual contract draft	contract.manager	`POST /v1/contracts`	201 DRAFT	201	**PASS**	{"number": "SA-5524", "type": "QUANTITY_CONTRACT", "title": "Flanges second source - quantity contract", "supplier_id": "01a1b6b8-9edb-7621-8e39-bb6b890e9a3d", "company_code": "1000", "status": "DRAFT", "valid_from": "20...
38	B6 manual quantity contract numbering	system	`contract`	own series per type	SA-5524 (type QUANTITY_CONTRACT)	**PARTIAL**	cosmetic: quantity contract numbered SA- (B6)
39	F4-05 submit (below 1 Cr) by contract manager	contract.manager	`POST /v1/contracts/SA-5524/submit`	chain [CONTRACT_MANAGER], owner excluded	200: ok chain CM	**PASS**	{"status": "IN_APPROVAL", "chain": ["CONTRACT_MANAGER"]}
40	F4-06 owner has no approval task	contract.manager	`GET /v1/tasks`	none	0	**PASS**	
41	F4-06 approval: CONTRACT_APPROVAL task (CONTRACT_MANAGER) APPROVE	contract.manager2	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6ba-d7d4-7633-9e89-ac2f81765de6", "kind": "CONTRACT_APPROVAL", "name": "Contract Approval \u00b7 SA-5524 \u00b7 Contra...
42	F4-07 e-signature envelope (MOCK provider)	contract.manager	`POST /v1/contracts/SA-5524/signature-envelopes`	201 envelope MOCK-..., signers BUYER, SUPPLIER	201	**PASS**	{"provider": "MOCK", "envelopeId": "MOCK-0C0AA7B6E6FB4D75", "reference": "PIC:FA7:SA-5524", "status": "SENT", "documentHash": "2d1c444e13ec33d4d6e1d64906ffd3d0a4159b65256dc2de0ec4548dd3792268", "signers": [{"party": "BUY...
43	F4-07 provider webhooks (both signers)	MOCK provider	`POST /v1/esign/webhook`	200 each, signedBy both	[200, 200, 200]	**PASS**	
44	F4-08 activate (demo signatures)	contract.manager	`POST /v1/contracts/SA-5524/activate`	200 ACTIVE	200: ok ACTIVE; ok demoSignature	**PASS**	{"number": "SA-5524", "type": "QUANTITY_CONTRACT", "title": "Flanges second source - quantity contract", "supplier_id": "01a1b6b8-9edb-7621-8e39-bb6b890e9a3d", "company_code": "1000", "status": "ACTIVE", "valid_from": "2...
45	F4-11 amendment preview (VARIATION)	contract.manager	`POST /v1/contracts/SA-5520/amendments/preview`	value delta + chain	200	**PASS**	{"contract": "SA-5520", "kind": "VARIATION", "effectiveFrom": "2026-11-17", "fromVersion": 1, "changes": {"lines": [{"line": 10, "unitPriceMinor": 123000}]}, "before": {"version": 1, "lines": [{"line": 10, "description":...
46	F4-11 amendment create+submit	contract.manager	`POST /v1/contracts/SA-5520/amendments`	201 IN_APPROVAL	201	**PASS**	{"number": "CA-101", "contract_id": "01a1b6b9-d24a-763f-8dfe-91f5fb55280f", "kind": "VARIATION", "changes": {"lines": [{"line": 10, "unitPriceMinor": 123000}], "valueDeltaMinor": -9360000, "approvalChain": ["CONTRACT_MAN...
47	F4-11 amendment approval: CONTRACT_AMENDMENT task (CONTRACT_MANAGER) APPROVE	contract.manager2	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-d30a-73bb-951a-b0969698b5b9", "kind": "CONTRACT_AMENDMENT", "name": "Contract Amendment \u00b7 SA-5520 \u00b7 Cont...
48	F4-11 amendment applied	contract.manager	`GET /v1/contracts/SA-5520/amendments/CA-101`	EFFECTIVE, contract version +1, new price period	200: ok EFFECTIVE	**PASS**	{"number": "CA-101", "contract_id": "01a1b6b9-d24a-763f-8dfe-91f5fb55280f", "kind": "VARIATION", "changes": {"band": "below", "lines": [{"line": 10, "unitPriceMinor": 123000}], "ruleSet": "CONTRACT-AMENDMENT@2026.1", "ap...
49	F4-13 rebate agreement	contract.manager	`POST /v1/contracts/SA-5520/rebates`	201 ACTIVE, accrual 0 (no invoices)	201	**PASS**	{"number": "RBA-101", "contract_id": "01a1b6b9-d24a-763f-8dfe-91f5fb55280f", "supplier_id": "01a1b6b8-9558-7755-b4bf-3e74eb841130", "title": "Volume rebate flanges", "basis": "VALUE", "period": "QUARTERLY", "tier_mode": ...

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
50	F4-12 terminate for cause	contract.manager	`POST /v1/contracts/SA-5524/terminate`	201 TERMINATION amendment, chain LEGAL, CPO	201	**PASS**	{"number": "CA-102", "contract_id": "01a1b6ba-d777-72f2-ae98-b07a488cede2", "kind": "TERMINATION", "changes": {"forCause": true, "valueDeltaMinor": -238000000, "approvalChain": ["LEGAL", "CPO"], "band": "termination", "r...
51	F4-12 termination approval: CONTRACT_AMENDMENT task (LEGAL) APPROVE	legal	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6ba-d7d4-7633-9e89-ac2f81765de6", "kind": "CONTRACT_AMENDMENT", "name": "Contract Amendment \u00b7 SA-5524 \u00b7 Lega...
52	F4-12 termination approval: CONTRACT_AMENDMENT task (CPO) APPROVE	cpo	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6ba-d7d4-7633-9e89-ac2f81765de6", "kind": "CONTRACT_AMENDMENT", "name": "Contract Amendment \u00b7 SA-5524 \u00b7 Cpo"...
53	F4-12 TERMINATED	contract.manager	`GET /v1/contracts/SA-5524`	TERMINATED, prices end, contract.terminated	200: ok TERMINATED; ok event	**PASS**	{"number": "SA-5524", "type": "QUANTITY_CONTRACT", "title": "Flanges second source - quantity contract", "supplier_id": "01a1b6b8-9edb-7621-8e39-bb6b890e9a3d", "company_code": "1000", "status": "TERMINATED", "valid_from"...
54	F4-12 renewal notice (timer at valid_to-120)	worker	`timer CONTRACT_RENEWAL`	EXPIRING + CONTRACT_RENEWAL task	status=EXPIRING tasks=['CONTRACT_RENEWAL', 'CONTRACT_RENEWAL']	**PASS**	
55	B5 CONTRACT_EXPIRING notification	system	`notification`	(B5: never produced)	[]	**PARTIAL**	no CONTRACT_EXPIRING notification although the contract went EXPIRING (B5)
56	F4-13 rebate accrue (period ended)	contract.manager	`POST /v1/rebates/RBA-101/accrue`	200	200	**PASS**	{"number": "RBA-101", "contract_id": "01a1b6b9-d24a-763f-8dfe-91f5fb55280f", "supplier_id": "01a1b6b8-9558-7755-b4bf-3e74eb841130", "title": "Volume rebate flanges", "basis": "VALUE", "period": "QUARTERLY", "tier_mode": ...
57	F4-13 rebate accruals	system	`rebate_accrual`	accrual rows (0 without invoices)	[('01a1b6bb-27c9-75e5-a3bd-229374f1739f', 'OPEN', 0)]	**INFO**	
58	F4-13 rebate settle	contract.manager	`POST /v1/rebate-accruals/{id}/settle`	credit memo to ERP once (needs accrual > 0)	422 {"type": "https://errors.alphahelion.io/pic/RULE_VIOLATION", "title": "RULE_VIOLATION", "status": 422, "detail": "The period FY2027-28-Q2 ends on 2027...	**PARTIAL**	
59	F4-12 renewal decision -> EXTENSION draft	contract.manager2	`POST /v1/tasks/01a1b6bb-2104-7190-b9e6-11487b9d68ec/complete`	200 EXTENSION amendment drafted	200	**PASS**	{"instance_id": "01a1b6b9-d30a-73bb-951a-b0969698b5b9", "kind": "CONTRACT_RENEWAL", "name": "Contract Renewal \u00b7 SA-5520 \u00b7 Contract Manager", "object_type": "CONTRACT", "object_id": "01a1b6b9-d24a-763f-8dfe-91f5...
60	F4-12 extension amendment	system	`amendments`	EXTENSION (DRAFT or IN_APPROVAL)	[('CA-103', 'DRAFT')]	**PASS**	
61	F4-12 extension approval: CONTRACT_AMENDMENT task (LEGAL) APPROVE	legal	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-d30a-73bb-951a-b0969698b5b9", "kind": "CONTRACT_AMENDMENT", "name": "Contract Amendment \u00b7 SA-5520 \u00b7 Lega...
62	F4-12 extension approval: CONTRACT_AMENDMENT task (FINANCE_CONTROLLER) APPROVE	fin1	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-d30a-73bb-951a-b0969698b5b9", "kind": "CONTRACT_AMENDMENT", "name": "Contract Amendment \u00b7 SA-5520 \u00b7 Fina...
63	F4-12 extended contract	system	`contract`	ACTIVE, later valid_to, never RENEWED	status=ACTIVE valid_to=2028-11-30	**PASS**	
64	B5 RENEWED state	system	`contract`	(B5: no caller)	status=ACTIVE	**CONFIRMED**	
65	F4-12 expiry timer -> EXPIRED	worker	`timer CONTRACT_EXPIRY`	EXPIRED after valid_to	status=EXPIRED valid_to=2027-01-08	**PASS**	
66	F4-13 rebate accrue for an ended period (Q3 FY26-27)	contract.manager	`POST /v1/rebates/RBA-101/accrue?on=2026-12-15`	200 accrual	200 {"number": "RBA-101", "contract_id": "01a1b6b9-d24a-763f-8dfe-91f5fb55280f", "supplier_id": "01a1b6b8-9558-7755-b4bf-3e74eb841130", "title": "Volume rebate flanges", "basis": "VALUE", "period": "QUART...	**PASS**	
67	F4-13 accrual rows	system	`rebate_accrual`	one per period	[('01a1b6bb-27c9-75e5-a3bd-229374f1739f', 'OPEN'), ('01a1b6bb-3712-7523-8d64-516246b4a3de', 'OPEN')]	**INFO**	

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
68	F4-13 settle accrual 01a1b6bb-27c9-75e5-a3bd-229374f1739f	contract.manager	`POST /v1/rebate-accruals/{id}/settle`	credit memo to ERP (or refused when 0 / period open)	422 {"type": "https://errors.alphahelion.io/pic/RULE_VIOLATION", "title": "RULE_VIOLATION", "status": 422, "detail": "The period FY2027-28-Q2 ends on 2027-09-30: settle after it has ended", "code": "RULE_...	**INFO**	
69	F4-13 settle accrual 01a1b6bb-3712-7523-8d64-516246b4a3de	contract.manager	`POST /v1/rebate-accruals/{id}/settle`	credit memo to ERP (or refused when 0 / period open)	422 {"type": "https://errors.alphahelion.io/pic/RULE_VIOLATION", "title": "RULE_VIOLATION", "status": 422, "detail": "No rebate is due for FY2026-27-Q3: the tier was not reached", "code": "RULE_VIOLATION"...	**INFO**	
70	F4-13 rebate credit memo to ERP	system	`integration_message`	credit memo when > 0	[]	**INFO**	

F5 purchase order (M08)

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
1	F5 demand (MAT-A 300 EA): requisition	requester	`POST /v1/requisitions`	201	201	**PASS**	{"number": "REQ-2320", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...
2	F5 demand (MAT-A 300 EA): submit	requester	`POST /v1/requisitions/REQ-2320/submit`	200	200	**PASS**	{"number": "REQ-2320", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...
3	F5 demand (MAT-A 300 EA): approval: REQ_APPROVAL task (COST_OBJECT_OWNER) APPROVE	cc.owner	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-ea4e-739c-8327-3a3c862d08a8", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2320 \u00b7 Cost Object Own...
4	F5 demand (MAT-A 300 EA): approval: REQ_APPROVAL task (DEPT_HEAD) APPROVE	dept.head	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6b9-ea4e-739c-8327-3a3c862d08a8", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2320 \u00b7 Dept Head", "ob...
5	F5 demand (MAT-A 300 EA): approved	requester	`GET /v1/requisitions/REQ-2320`	APPROVED	APPROVED	**PASS**	
6	B1 PO create outside the buyer's plants	buyer.p200 (plants P200)	`POST /v1/purchase-orders`	403/404 if ABAC PO_CREATE applied	201 {"id": "PO-77410", "version": 1, "status": "IN_APPROVAL", "inherited": {"price": 1190.0, "paymentTerms": "N60", "incoter...	**PARTIAL**	missing control: no ABAC PO_CREATE plant check (B1)
7	B2 PO source CONTRACT	buyer	`POST /v1/purchase-orders`	only REQUISITION accepted (B2)	422 {"type": "https://errors.alphahelion.io/pic/VALIDATION_FAILED", "title": "VALIDATION_FAILED", "statu...	**NOT BUILT**	no command (Appendix B2 CONFIRMED)
8	B2 PO source MANUAL	buyer	`POST /v1/purchase-orders`	only REQUISITION accepted (B2)	422 {"type": "https://errors.alphahelion.io/pic/VALIDATION_FAILED", "title": "VALIDATION_FAILED", "statu...	**NOT BUILT**	no command (Appendix B2 CONFIRMED)
9	F5-01 create PO from requisition (source determination -> contract)	buyer	`POST /v1/purchase-orders`	201, supplier A, contract price 1250 inherited (P07), terms/tolerances from contract	201: ok price 1250	**PASS**	{"id": "PO-77411", "version": 1, "status": "TRANSMITTED", "inherited": {"price": 1250.0, "paymentTerms": "N60", "incoterms": null, "tolerance": "+2/-5"}, "validation": {"riskScore": 0, "touchless": true, "factors": []}, ...
10	F5-01 PO detail	buyer	`GET /v1/purchase-orders/PO-77411`	price_source CONTRACT, contract linked, lineage	status=TRANSMITTED supplier=SUP-1001 src=['CONTRACT'] touchless=True risk=0 org=None cur=INR lineage=['REQ-2320', 'REQ-2316', 'RFQ-2232', 'SA-5520']	**PASS**	
11	F5-06 released + transmitted	system	`PO status`	TRANSMITTED (touchless or after approval)	TRANSMITTED released_by=agent:TOUCHLESS_PO	**PASS**	
12	F5-06 ERP PURCHASE_ORDER + events	system	`integration/outbox`	PURCHASE_ORDER posted; released, transmitted events	outbox=['purchase_order.released', 'purchase_order.transmitted'] audit=['CREATED', 'DOCUMENT_TRANSMITTED', 'ERP_POSTED', 'RELEASE', 'TRANSMIT', 'VALIDATED'] erp=['PURCHASE_ORDER:POSTED'] tasks=[]	**PASS**	

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
13	F5-06 transmissions	buyer	`GET /v1/purchase-orders/PO-77411/transmissions`	PORTAL transmission with SHA-256	200	**PASS**	[{"id": "01a1b6b9-fd0e-71a8-8ff0-8786b5182266", "version": 1, "channel": "PORTAL", "format": "PORTAL", "recipient": "SUP-1001", "status": "SENT", "sha256": "21cae81a0ce3680c3c91f318a0b7539142a09d88249e3575d500edf515820b8...
14	F4-09 consumption from the PO	contract.manager	`GET /v1/contracts/SA-5520/consumption`	released qty 300 on the contract line	200: ok 300 released	**PASS**	{"contract": "SA-5520", "status": "ACTIVE", "thresholdsHit": [], "lines": [{"line": 10, "description": "Forged steel weld neck flange DN100 PN16 ASTM A105", "targetQty": "24000.000", "releasedQty": "300.000", "releasedVa...
15	F5-07 other supplier cannot see the PO	SUPPLIER_USER B	`GET /v1/purchase-orders/PO-77411`	404	404 NOT_FOUND	**PASS**	{"type": "https://errors.alphahelion.io/pic/NOT_FOUND", "title": "NOT_FOUND", "status": 404, "detail": "Purchase order PO-77411 not found", "code": "NOT_FOUND", "instance": "/v1/purchase-orders/PO-77411", "traceId": "8dd...
16	F5-07 supplier proposes a later date (portal)	SUPPLIER_USER A	`POST /v1/purchase-orders/PO-77411/confirmations`	200 CHANGE_REQUESTED, PO_CHANGE_REVIEW task for the buyer	200: ok CHANGE_REQUESTED	**PASS**	{"status": "TRANSMITTED", "confirmation": "CHANGE_REQUESTED", "outsideTolerance": ["line 10: later date 2026-12-17"]}
17	F5-08 PO_CHANGE_REVIEW in buyer inbox	buyer	`GET /v1/tasks`	task	[('PO_CHANGE_REVIEW', 'BUYER')]	**PASS**	
18	F5-08 buyer accepts the proposal	buyer	`POST /v1/tasks/01a1b6ba-027a-7731-8893-5b2d90ac5bc1/complete`	200 change order v2, re-sent	200	**PASS**	{"instance_id": "01a1b6b9-fc17-74e4-8e25-d6dad3a14586", "kind": "PO_CHANGE_REVIEW", "name": "Po Change Review \u00b7 PO-77411 \u00b7 Buyer", "object_type": "PURCHASE_ORDER", "object_id": "01a1b6b9-fbbc-731b-8ec7-bca2e862...
19	F5-08 change order version	system	`PO`	version 2, TRANSMITTED, need date +7	v2 TRANSMITTED need=['2026-12-17']	**PASS**	
20	F5-07 supplier confirms v2	SUPPLIER_USER A	`POST /v1/purchase-orders/PO-77411/confirmations`	200 CONFIRMED, purchase_order.confirmed	200: ok CONFIRMED	**PASS**	{"status": "CONFIRMED", "confirmation": "CONFIRMED", "hoursToConfirm": "0.00"}
21	F5-08 buyer change order (qty 300 -> 320)	buyer	`POST /v1/purchase-orders/PO-77411/versions`	200 version 3, back to TRANSMITTED, purchase_order.changed	200	**PASS**	{"po": "PO-77411", "version": 3, "status": "TRANSMITTED"}
22	F5-08 version after buyer change	system	`PO`	v3 TRANSMITTED	v3 TRANSMITTED	**PASS**	
23	F5-09 delivery schedule (split)	buyer	`PUT /v1/purchase-orders/PO-77411/lines/10/schedule`	200 2 schedule lines	200	**PASS**	{"po": "PO-77411", "version": 3, "status": "TRANSMITTED", "mode": "IN_PLACE", "schedule": [{"line": 10, "seq": 1, "description": "M-400001", "uom": "EA", "qty": "160", "deliveryDate": "2026-12-17", "confirmedDate": null,...
24	F5-01 PO with supplierId (manual price)	buyer	`POST /v1/purchase-orders`	201 IN_APPROVAL (not touchless: no contract price)	201	**PASS**	{"id": "PO-77412", "version": 1, "status": "IN_APPROVAL", "inherited": {"price": 1190.0, "paymentTerms": "N60", "incoterms": null, "tolerance": "+0/-0"}, "validation": {"riskScore": 0, "touchless": false, "factors": []},...
25	F5-04 not touchless	system	`PO`	IN_APPROVAL	IN_APPROVAL touchless=False risk=0 factors=[]	**PASS**	
26	F5-05 creator excluded from own PO approval	buyer	`GET /v1/tasks`	no task for the creator	0	**PASS**	
27	F5-05 approver rejects	buyer2	`POST /v1/tasks/01a1b6ba-0b4f-728b-9fb4-8c2a3aa4d93e/complete`	200 -> DRAFT	200	**PASS**	{"instance_id": "01a1b6ba-0aa5-70eb-a05b-f6679447ca1b", "kind": "PO_APPROVAL", "name": "Po Approval \u00b7 PO-77412 \u00b7 Buyer", "object_type": "PURCHASE_ORDER", "object_id": "01a1b6ba-0a2c-7089-ba94-f4b60304da13", "ob...
28	F5-05 rejected PO back to DRAFT	system	`PO`	DRAFT	DRAFT	**PASS**	
29	F5-05 resubmit	buyer	`POST /v1/purchase-orders/PO-77412/resubmit`	200 IN_APPROVAL	200	**PASS**	{"po": "PO-77412", "version": 1, "status": "IN_APPROVAL", "totalMinor": 11900000}
30	F5-05 PO2 approval: PO_APPROVAL task (BUYER) APPROVE	buyer2	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6ba-0aa5-70eb-a05b-f6679447ca1b", "kind": "PO_APPROVAL", "name": "Po Approval \u00b7 PO-77412 \u00b7 Buyer", "object_t...
31	F5-06 PO2 transmitted	system	`PO`	TRANSMITTED	TRANSMITTED	**PASS**	

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
32	F5-11 cancel PO2 (nothing shipped)	buyer	`POST /v1/purchase-orders/PO-77412/cancel`	200 CANCELLED, PR qty reopened, purchase_order.cancelled	200	**PASS**	{"po": "PO-77412", "status": "CANCELLED", "from": "TRANSMITTED"}
33	F5-11 cancellation effects	system	`outbox/erp/pr`	cancelled event, ERP cancel, PR line open again	outbox=['purchase_order.cancelled', 'purchase_order.released', 'purchase_order.transmitted', 'task.assigned'] audit=['APPROVAL_APPROVE', 'APPROVAL_REJECT', 'CANCEL', 'CANCELLATION_TRANSMITTED', 'CREATED', 'DOCUMENT_TRANSMITTED', 'ERP_CANCELLED', 'ERP_POSTED', 'REJECT', 'RELEASE', 'RESUBMITTED', 'ROU	**INFO**	
34	SoD-01 setup: buyer-supplier-manager registers supplier E	buyer.sm	`POST /v1/suppliers`	201	201 SUP-1005	**PASS**	{"code": "SUP-1005", "legal_name": "Erode Forge Traders Pvt Ltd", "status": "REGISTERED", "gstin_masked": "\u2022\u2022\u2022\u2022\u2022\u2022\u2022\u2022\u2022\u2022\u2022\u2022\u2022K1Z2", "state_code": "33", "country": "IN", "se...
35	SoD-01 PO for a supplier the buyer created	buyer.sm	`POST /v1/purchase-orders`	409 SOD_CONFLICT	409 SOD_CONFLICT	**PASS**	{"type": "https://errors.alphahelion.io/pic/SOD_CONFLICT", "title": "SOD_CONFLICT", "status": 409, "detail": "SOD-01: Creator of a supplier master may not creat...
36	B3 contract qty at release	-	`-`	re-check at release	not driven (needs a contract amendment racing a PO in approval)	**NOT TESTABLE**	
37	B2 PO org/currency	system	`PO`	org from requisition / supplier	purchasing_org=None currency=INR	**INFO**	
38	B4 demand 1: requisition	requester	`POST /v1/requisitions`	201	201	**PASS**	{"number": "REQ-2322", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...
39	B4 demand 1: submit	requester	`POST /v1/requisitions/REQ-2322/submit`	200	200	**PASS**	{"number": "REQ-2322", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...
40	B4 demand 1: approval: REQ_APPROVAL task (COST_OBJECT_OWNER) APPROVE	cc.owner	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6ba-576d-776a-936c-45f9e4a4be45", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2322 \u00b7 Cost Object Own...
41	B4 demand 1: approved	requester	`GET /v1/requisitions/REQ-2322`	APPROVED	APPROVED	**PASS**	
42	B4 demand 2: requisition	requester	`POST /v1/requisitions`	201	201	**PASS**	{"number": "REQ-2323", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...
43	B4 demand 2: submit	requester	`POST /v1/requisitions/REQ-2323/submit`	200	200	**PASS**	{"number": "REQ-2323", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...
44	B4 demand 2: approval: REQ_APPROVAL task (COST_OBJECT_OWNER) APPROVE	cc.owner	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6ba-7625-764d-a718-c7e3db9d07a3", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2323 \u00b7 Cost Object Own...
45	B4 demand 2: approved	requester	`GET /v1/requisitions/REQ-2323`	APPROVED	APPROVED	**PASS**	
46	B4 demand 3: requisition	requester	`POST /v1/requisitions`	201	201	**PASS**	{"number": "REQ-2324", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
47	B4 demand 3: submit	requester	`POST /v1/requisitions/REQ-2324/submit`	200	200	**PASS**	{"number": "REQ-2324", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...
48	B4 demand 3: approval: REQ_APPROVAL task (COST_OBJECT_OWNER) APPROVE	cc.owner	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6ba-9172-74cd-a942-b7a986e23071", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2324 \u00b7 Cost Object Own...
49	B4 demand 3: approved	requester	`GET /v1/requisitions/REQ-2324`	APPROVED	APPROVED	**PASS**	
50	B4 source determination with 2 live contracts, no quota	buyer	`POST /v1/purchase-orders x3`	deterministic choice (e.g. lowest price / priority) or an explicit choice for the buyer	contracts SA-5521(A @2100) SA-5522(B @1990); picks=[('PO-77413', 'SUP-1001', [210000]), ('PO-77414', 'SUP-1001', [210000]), ('PO-77415', 'SUP-1001', [210000])]	**PARTIAL**	first contract row wins, here the dearer one (B4)
51	F4-09 big call-off demand (19,000 EA): requisition	requester	`POST /v1/requisitions`	201	201	**PASS**	{"number": "REQ-2325", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...
52	F4-09 big call-off demand (19,000 EA): submit	requester	`POST /v1/requisitions/REQ-2325/submit`	200	200	**PASS**	{"number": "REQ-2325", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...
53	F4-09 big call-off demand (19,000 EA): approval: REQ_APPROVAL task (COST_OBJECT_OWNER) APPROVE	cc.owner	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6ba-a8a6-74ef-9def-e59568297253", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2325 \u00b7 Cost Object Own...
54	F4-09 big call-off demand (19,000 EA): approval: REQ_APPROVAL task (DEPT_HEAD) APPROVE	dept.head	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6ba-a8a6-74ef-9def-e59568297253", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2325 \u00b7 Dept Head", "ob...
55	F4-09 big call-off demand (19,000 EA): approval: REQ_APPROVAL task (PROCUREMENT_HEAD) APPROVE	proc.head	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6ba-a8a6-74ef-9def-e59568297253", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2325 \u00b7 Procurement Hea...
56	F4-09 big call-off demand (19,000 EA): approval: REQ_APPROVAL task (FINANCE_CONTROLLER) APPROVE	fin1	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6ba-a8a6-74ef-9def-e59568297253", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2325 \u00b7 Finance Control...
57	F4-09 big call-off demand (19,000 EA): approved	requester	`GET /v1/requisitions/REQ-2325`	APPROVED	APPROVED	**PASS**	
58	F5-01 big call-off PO (not touchless: > 25 L)	buyer	`POST /v1/purchase-orders`	201 IN_APPROVAL	201	**PASS**	{"id": "PO-77416", "version": 1, "status": "IN_APPROVAL", "inherited": {"price": 1250.0, "paymentTerms": "N60", "incoterms": null, "tolerance": "+2/-5"}, "validation": {"riskScore": 15, "touchless": false, "factors": [{"...
59	F5-05 big PO approval: PO_APPROVAL task (BUYER) APPROVE	buyer2	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6ba-be9a-76d9-99e9-5b71bd0bd1c2", "kind": "PO_APPROVAL", "name": "Po Approval \u00b7 PO-77416 \u00b7 Buyer", "object_t...
60	F5-05 big PO approval: PO_APPROVAL task (PROCUREMENT_HEAD) APPROVE	proc.head	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6ba-be9a-76d9-99e9-5b71bd0bd1c2", "kind": "PO_APPROVAL", "name": "Po Approval \u00b7 PO-77416 \u00b7 Procurement Head"...

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
61	F5-06 big PO released	system	`PO`	TRANSMITTED	TRANSMITTED	**PASS**	
62	B4 demand: requisition	requester	`POST /v1/requisitions`	201	201	**PASS**	{"number": "REQ-2326", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...
63	B4 demand: submit	requester	`POST /v1/requisitions/REQ-2326/submit`	200	200	**PASS**	{"number": "REQ-2326", "origin": "MANUAL", "origin_ref": null, "purpose_code": "PRODUCTION", "purpose_text": null, "requester_id": "01a1b6b8-8b2b-76fc-be89-82b880bf4f8e", "company_code": "1000", "plant": "P100", "status"...
64	B4 demand: approval: REQ_APPROVAL task (COST_OBJECT_OWNER) APPROVE	cc.owner	`POST /v1/tasks/{id}/complete`	200	200	**PASS**	{"instance_id": "01a1b6ba-ef1a-70d0-b8b7-a195a91ea5cf", "kind": "REQ_APPROVAL", "name": "Req Approval \u00b7 REQ-2326 \u00b7 Cost Object Own...
65	B4 demand: approved	requester	`GET /v1/requisitions/REQ-2326`	APPROVED	APPROVED	**PASS**	
66	B4 source determination with 2 live contracts	buyer	`POST /v1/purchase-orders (no contractRef)`	deterministic rule (quota or explicit priority)	active contracts=[('SA-5524', 'SUP-1002'), ('SA-5522', 'SUP-1002'), ('SA-5521', 'SUP-1001'), ('SA-5520', 'SUP-1001')]; 201 picked=[('PO-77417', 'SUP-1001', '01a1b6b9-d24a-763f-8dfe-91f5fb55280f')] {"id": "PO-77417", "version": 1, "status": "TRANSMITTED", "inherited": {"price": 1250.0, "paymentTerms"	**INFO**	
67	F5-04 touchless PO from contract-channel PR	buyer	`POST /v1/purchase-orders`	201 touchless, TRANSMITTED	201: ok touchless	**PASS**	{"id": "PO-77418", "version": 1, "status": "TRANSMITTED", "inherited": {"price": 1337.5, "paymentTerms": "N60", "incoterms": null, "tolerance": "+2/-5"}, "validation": {"riskScore": 0, "touchless": true, "factors": []}, ...
68	B9 erp-reconciliation job covers POs?	admin	`GET /v1/admin/jobs`	PO objects in the reconciliation	recon jobs=['erp-reconciliation'] detail=[{"key": "erp-reconciliation", "name": "Nightly ERP reconciliation", "description": "Reconciles the posted invoices, payments and credit memos of every company code with the ERP documents (count, amount, status) and notifies finance when there are differences	**INFO**	
69	B2 purchasing org / currency on every PO	system	`purchase_order`	taken from the requisition / supplier	cols=['purch_org', 'currency'] rows=[('PO-77410', '1000', 'INR'), ('PO-77412', '1000', 'INR'), ('PO-77413', '1000', 'INR'), ('PO-77411', '1000', 'INR'), ('PO-77414', '1000', 'INR'), ('PO-77416', '1000', 'INR')]	**INFO**	

F8 supplier

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
1	[A] F8-01 invitation	supplier.manager	`POST /v1/supplier-invitations`	201, token + link, event supplier.invitation.sent	201: ok token	**PASS**	{"email": "accounts@a-fa7.example", "legal_na "categories": ["FORGINGS"], "invited_by": "01a 82c528e1c00e", "expires_at": "2026-11-23T09:
2	[A] F8-01 open link (public)	anonymous	`GET /v1/public/invitations/FA7.BJ5a6MB6tKLjKGe7R95G67cWWkdoI3RRf7KuBXmTeUo`	200 invitation view	200	**PASS**	{"legalName": "Aravind Forge Works Pvt Ltd", "categories": ["FORGINGS"], "tenant": "Flow Au "FA7", "expiresAt": "2026-11-23T03:30:00+00:(
3	[A] F8-01 self-registration	supplier (public)	`POST /v1/public/invitations/FA7.BJ5a6MB6tKLjKGe7R95G67cWWkdoI3RRf7KuBXmTeUo/register`	201 supplier REGISTERED, qualification started, portal login	201: ok supplier code; ok qualifications started; ok screening CLEAR	**PASS**	{"supplier": "SUP-1001", "legalName": "Aravinc "REGISTERED", "screening": "CLEAR", "qualifica "screeningReviewRequired": false, "login": {"te
4	[A] F8-01 events	system	`outbox`	supplier.registered (+invitation.sent)	topics=['supplier.registered', 'supplier.status.changed']	**PASS**	outbox=['supplier.registered', 'supplier.status.(['CATEGORY_STATUS_CHANGED', 'REGISTERED
5	[A] F8-01 used link refused	anonymous	`POST /v1/public/invitations/FA7.BJ5a6MB6tKLjKGe7R95G67cWWkdoI3RRf7KuBXmTeUo/register`	410 (used)	410 INVITATION_USED	**PASS**	{"type": "https://errors.alphahelion.io/pic/INVI "INVITATION_USED", "status": 410, "detail": "TI used. Sign in to the supplier portal instead.", "
6	[A] F8-02 profile bank+certificate (portal)	SUPPLIER_USER A	`POST /v1/suppliers/SUP-1001/profile`	200 bank change PENDING_CALLBACK, screening run	200: ok bankChange	**PASS**	{"certificates": [{"id": "01a1b6b8-9838-76cd-9f "number": "QMS-A-001", "mandatoryFor": []}], "sanctionsHits": [], "screenedAt": "2026-11-09T

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
7	[A] F8-04 bank call-back	supplier.manager	`POST /v1/bank-changes/BCR-101/callback`	200 PENDING_APPROVAL, 2 BANK_CHANGE tasks for FINANCE_CONTROLLER	200: ok PENDING_APPROVAL	**PASS**	{"number": "BCR-101", "supplier_id": "01a1b6b "bank_account_id": "01a1b6b8-985f-7300-aaf("PENDING_APPROVAL", "requested_by": "01a1 ce198c4606ce"...
8	[A] F8-04 approval task opened	system	`task`	BANK_CHANGE task(s) for FINANCE_CONTROLLER	[('BANK_CHANGE', 'FINANCE_CONTROLLER', 'OPEN')]	**PASS**	
9	[A] F8-04 approve bank (fin1)	fin1	`POST /v1/tasks/01a1b6b8-9965-70bd-b3f3-a540e08f956b/complete`	200	200	**PASS**	{"instance_id": "01a1b6b8-959f-7220-b58f-27 "name": "Bank Change \u00b7 BCR-101 \u00b "BANK_CHANGE", "object_id": "01a1b6b8-986
10	[A] F8-04 approve bank (fin2)	fin2	`POST /v1/tasks/01a1b6b8-9b49-7454-8148-3c433c114d82/complete`	200	200	**PASS**	{"instance_id": "01a1b6b8-959f-7220-b58f-27 "name": "Bank Change \u00b7 BCR-101 \u00b "BANK_CHANGE", "object_id": "01a1b6b8-986
11	[A] F8-04 two different approvers	system	`task`	2 BANK_CHANGE tasks completed by 2 different FCs (sequential)	[('COMPLETED', 'APPROVE', 'fin1@fa7.example'), ('COMPLETED', 'APPROVE', 'fin2@fa7.example')]	**PASS**	
12	[A] F8-04 bank active after dual approval	supplier.manager	`GET /v1/suppliers/SUP-1001/profile`	active account, payable_from = +72h	200: ok active account; ok payable_from set	**PASS**	{"supplier": "SUP-1001", "status": "REGISTERED "certificates": [{"supplier_id": "01a1b6b8-9558- "ISO9001", "number": "QMS-A-001", "issued_c
13	[B] F8-01 invitation	supplier.manager	`POST /v1/supplier-invitations`	201, token + link, event supplier.invitation.sent	201: ok token	**PASS**	{"email": "accounts@b-fa7.example", "legal_na Ltd", "categories": ["FORGINGS"], "invited_by": 82c528e1c00e", "expires_at": "2026-11-23T09:
14	[B] F8-01 open link (public)	anonymous	`GET /v1/public/invitations/FA7.i_kkHJmPi1KQ2usrmB_tqfvRYOn0ospFw3cuOYAClEM`	200 invitation view	200	**PASS**	{"legalName": "Bhavani Precision Forgings Pvt fa7.example", "categories": ["FORGINGS"], "ter "tenantCode": "FA7", "expiresAt": "2026-11-23
15	[B] F8-01 self-registration	supplier (public)	`POST /v1/public/invitations/FA7.i_kkHJmPi1KQ2usrmB_tqfvRYOn0ospFw3cuOYAClEM/register`	201 supplier REGISTERED, qualification started, portal login	201: ok supplier code; ok qualifications started; ok screening CLEAR	**PASS**	{"supplier": "SUP-1002", "legalName": "Bhavan "REGISTERED", "screening": "CLEAR", "qualifica "screeningReviewRequired": false, "login": {"te
16	[B] F8-01 events	system	`outbox`	supplier.registered (+invitation.sent)	topics=['supplier.registered', 'supplier.status.changed']	**PASS**	outbox=['supplier.registered', 'supplier.status.(['CATEGORY_STATUS_CHANGED', 'REGISTERED
17	[B] F8-01 used link refused	anonymous	`POST /v1/public/invitations/FA7.i_kkHJmPi1KQ2usrmB_tqfvRYOn0ospFw3cuOYAClEM/register`	410 (used)	410 INVITATION_USED	**PASS**	{"type": "https://errors.alphahelion.io/pic/INVI "INVITATION_USED", "status": 410, "detail": "TI used. Sign in to the supplier portal instead.", "
18	[B] F8-02 profile bank+certificate (portal)	SUPPLIER_USER B	`POST /v1/suppliers/SUP-1002/profile`	200 bank change PENDING_CALLBACK, screening run	200: ok bankChange	**PASS**	{"certificates": [{"id": "01a1b6b8-a261-73c3-93 "number": "QMS-B-001", "mandatoryFor": []}], "sanctionsHits": [], "screenedAt": "2026-11-09T
19	[B] F8-04 bank call-back	supplier.manager	`POST /v1/bank-changes/BCR-102/callback`	200 PENDING_APPROVAL, 2 BANK_CHANGE tasks for FINANCE_CONTROLLER	200: ok PENDING_APPROVAL	**PASS**	{"number": "BCR-102", "supplier_id": "01a1b6b "bank_account_id": "01a1b6b8-a272-7030-89a "PENDING_APPROVAL", "requested_by": "01a1
20	[B] F8-04 approval task opened	system	`task`	BANK_CHANGE task(s) for FINANCE_CONTROLLER	[('BANK_CHANGE', 'FINANCE_CONTROLLER', 'OPEN')]	**PASS**	
21	[B] F8-04 approve bank (fin1)	fin1	`POST /v1/tasks/01a1b6b8-a31f-741c-a282-e19625df12dc/complete`	200	200	**PASS**	{"instance_id": "01a1b6b8-9efc-7412-8f90-944 "name": "Bank Change \u00b7 BCR-102 \u00b "BANK_CHANGE", "object_id": "01a1b6b8-a27
22	[B] F8-04 approve bank (fin2)	fin2	`POST /v1/tasks/01a1b6b8-a4b9-7482-953d-1b06db17f4f5/complete`	200	200	**PASS**	{"instance_id": "01a1b6b8-9efc-7412-8f90-944 "name": "Bank Change \u00b7 BCR-102 \u00b "BANK_CHANGE", "object_id": "01a1b6b8-a27
23	[B] F8-04 two different approvers	system	`task`	2 BANK_CHANGE tasks completed by 2 different FCs (sequential)	[('COMPLETED', 'APPROVE', 'fin1@fa7.example'), ('COMPLETED', 'APPROVE', 'fin2@fa7.example')]	**PASS**	

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
24	[B] F8-04 bank active after dual approval	supplier.manager	`GET /v1/suppliers/SUP-1002/profile`	active account, payable_from = +72h	200: ok active account; ok payable_from set	**PASS**	{"supplier": "SUP-1002", "status": "REGISTERED "certificates": [{"supplier_id": "01a1b6b8-9edb "ISO9001", "number": "QMS-B-001", "issued_c
25	[C] F8-01 invitation	supplier.manager	`POST /v1/supplier-invitations`	201, token + link, event supplier.invitation.sent	201: ok token	**PASS**	{"email": "accounts@c-fa7.example", "legal_na "categories": ["FORGINGS"], "invited_by": "01a 82c528e1c00e", "expires_at": "2026-11-23T09:
26	[C] F8-01 open link (public)	anonymous	`GET /v1/public/invitations/FA7.QobhdVOpckNjJmEPgvHEsCtjK5P1NG4MV5LvaeLQlZA`	200 invitation view	200	**PASS**	{"legalName": "Chola Hot Forgings Pvt Ltd", "e "categories": ["FORGINGS"], "tenant": "Flow Au "FA7", "expiresAt": "2026-11-23T03:30:00+00:(
27	[C] F8-01 self-registration	supplier (public)	`POST /v1/public/invitations/FA7.QobhdVOpckNjJmEPgvHEsCtjK5P1NG4MV5LvaeLQlZA/register`	201 supplier REGISTERED, qualification started, portal login	201: ok supplier code; ok qualifications started; ok screening CLEAR	**PASS**	{"supplier": "SUP-1003", "legalName": "Chola H "REGISTERED", "screening": "CLEAR", "qualifica "screeningReviewRequired": false, "login": {"te
28	[C] F8-01 events	system	`outbox`	supplier.registered (+invitation.sent)	topics=['supplier.registered', 'supplier.status.changed']	**PASS**	outbox=['supplier.registered', 'supplier.status.(['CATEGORY_STATUS_CHANGED', 'REGISTERED
29	[C] F8-01 used link refused	anonymous	`POST /v1/public/invitations/FA7.QobhdVOpckNjJmEPgvHEsCtjK5P1NG4MV5LvaeLQlZA/register`	410 (used)	410 INVITATION_USED	**PASS**	{"type": "https://errors.alphahelion.io/pic/INVI "INVITATION_USED", "status": 410, "detail": "TH used. Sign in to the supplier portal instead.", "
30	[C] F8-02 profile bank+certificate (portal)	SUPPLIER_USER C	`POST /v1/suppliers/SUP-1003/profile`	200 bank change PENDING_CALLBACK, screening run	200: ok bankChange	**PASS**	{"certificates": [{"id": "01a1b6b8-b1c3-759b-9c "number": "QMS-C-001", "mandatoryFor": []}], "sanctionsHits": [], "screenedAt": "2026-11-09T
31	[C] F8-04 bank call-back	supplier.manager	`POST /v1/bank-changes/BCR-103/callback`	200 PENDING_APPROVAL, 2 BANK_CHANGE tasks for FINANCE_CONTROLLER	200: ok PENDING_APPROVAL	**PASS**	{"number": "BCR-103", "supplier_id": "01a1b6b "bank_account_id": "01a1b6b8-b1d5-71e4-88 "PENDING_APPROVAL", "requested_by": "01a1
32	[C] F8-04 approval task opened	system	`task`	BANK_CHANGE task(s) for FINANCE_CONTROLLER	[('BANK_CHANGE', 'FINANCE_CONTROLLER', 'OPEN')]	**PASS**	
33	[C] F8-04 approve bank (fin1)	fin1	`POST /v1/tasks/01a1b6b8-b325-7142-83f8-466f2be0cfa2/complete`	200	200	**PASS**	{"instance_id": "01a1b6b8-aa20-779c-a324-09 "BANK_CHANGE", "name": "Bank Change \u0(Controller", "object_type": "BANK_CHANGE", " 92e3-a57125acf...
34	[C] F8-04 approve bank (fin2)	fin2	`POST /v1/tasks/01a1b6b8-b4ae-739d-9bab-acbea3a2bf0a/complete`	200	200	**PASS**	{"instance_id": "01a1b6b8-aa20-779c-a324-09 "BANK_CHANGE", "name": "Bank Change \u0(Controller", "object_type": "BANK_CHANGE", " 92e3-a57125acf...
35	[C] F8-04 two different approvers	system	`task`	2 BANK_CHANGE tasks completed by 2 different FCs (sequential)	[('COMPLETED', 'APPROVE', 'fin1@fa7.example'), ('COMPLETED', 'APPROVE', 'fin2@fa7.example')]	**PASS**	
36	[C] F8-04 bank active after dual approval	supplier.manager	`GET /v1/suppliers/SUP-1003/profile`	active account, payable_from = +72h	200: ok active account; ok payable_from set	**PASS**	{"supplier": "SUP-1003", "status": "REGISTERED "certificates": [{"supplier_id": "01a1b6b8-a9ff-7 "ISO9001", "number": "QMS-C-001", "issued_c
37	[A] F8-05 supplier 360 (qualification open)	supplier.manager	`GET /v1/suppliers/SUP-1001`	qualification row for the invited category	200 SUP-1001: ok qualification exists	**PASS**	{"code": "SUP-1001", "legal_name": "Aravind F "REGISTERED", "gstin_masked": "\u2022\u2022\u2022\u2022\u2022\u2022\u2 "state_code": "29", "country": "IN", "se...
38	[A] F8-05 verify certificate ISO9001	quality.inspector	`POST /v1/certificates/01a1b6b8-9838-76cd-9fa1-6e8d5e27de17/verify`	200 verified	200	**PASS**	{"supplier_id": "01a1b6b8-9558-7755-b4bf-3e "number": "QMS-A-001", "issued_on": "2025-0 "mandatory_for": [], "verified": true, "status": "\
39	[A] F8-05 assessment	supplier.manager	`POST /v1/qualifications/01a1b6b8-96b9-76f9-822f-e506df7676bd/assessment`	200	200	**PASS**	{"supplier_id": "01a1b6b8-9558-7755-b4bf-3e "status": "OPEN", "questionnaire": {"capacity": "audit_score": "80", "sample_approval": null, "c

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
40	[A] F8-06 decision QUALIFIED	supplier.manager2	`POST /v1/qualifications/01a1b6b8-96b9-76f9-822f-e506df7676bd/decision`	200; supplier ACTIVE; ERP SUPPLIER_MASTER {code}|CREATE queued	200	**PASS**	{"supplier": "SUP-1001", "category": "FORGINC "ruleSetVersion": "SUPPLIER-QUAL@2026.1", "
41	[A] F8-06 supplier ACTIVE + category QUALIFIED	supplier.manager	`GET /v1/suppliers/SUP-1001`	status ACTIVE, category QUALIFIED	200 SUP-1001: ok ACTIVE; ok QUALIFIED	**PASS**	{"code": "SUP-1001", "legal_name": "Aravind F "gstin_masked": "\u2022\u2022\u2022\u2022\u2022\u2022\u2 "state_code": "29", "country": "IN", "segmen...
42	[A] F8-06 ERP vendor create queued	system	`integration_message`	SUPPLIER_MASTER {code}|CREATE	[('SUPPLIER_MASTER', 'SUP-1001', 'POSTED', 'SUP-1001|CREATE', '300001')]	**PASS**	
43	[B] F8-05 supplier 360 (qualification open)	supplier.manager	`GET /v1/suppliers/SUP-1002`	qualification row for the invited category	200 SUP-1002: ok qualification exists	**PASS**	{"code": "SUP-1002", "legal_name": "Bhavani P "REGISTERED", "gstin_masked": "\u2022\u2022\u2022\u2022\u2022\u2022\u2 "state_code": "33", "country": "I...
44	[B] F8-05 verify certificate ISO9001	quality.inspector	`POST /v1/certificates/01a1b6b8-a261-73c3-934a-bbfeec0af0d1/verify`	200 verified	200	**PASS**	{"supplier_id": "01a1b6b8-9edb-7621-8e39-bb "number": "QMS-B-001", "issued_on": "2025-0 "mandatory_for": [], "verified": true, "status": "\
45	[B] F8-05 assessment	supplier.manager	`POST /v1/qualifications/01a1b6b8-9fac-7614-81b8-594aba3ffbf2/assessment`	200	200	**PASS**	{"supplier_id": "01a1b6b8-9edb-7621-8e39-bb "FORGINGS", "status": "OPEN", "questionnaire "questionnaire_score": "82", "audit_score": "80
46	[B] F8-06 decision QUALIFIED	supplier.manager2	`POST /v1/qualifications/01a1b6b8-9fac-7614-81b8-594aba3ffbf2/decision`	200; supplier ACTIVE; ERP SUPPLIER_MASTER {code}|CREATE queued	200	**PASS**	{"supplier": "SUP-1002", "category": "FORGINC "ruleSetVersion": "SUPPLIER-QUAL@2026.1", "
47	[B] F8-06 supplier ACTIVE + category QUALIFIED	supplier.manager	`GET /v1/suppliers/SUP-1002`	status ACTIVE, category QUALIFIED	200 SUP-1002: ok ACTIVE; ok QUALIFIED	**PASS**	{"code": "SUP-1002", "legal_name": "Bhavani P "ACTIVE", "gstin_masked": "\u2022\u2022\u2022\u2022\u2022\u2022\u2 "state_code": "33", "country": "IN", ...
48	[B] F8-06 ERP vendor create queued	system	`integration_message`	SUPPLIER_MASTER {code}|CREATE	[('SUPPLIER_MASTER', 'SUP-1002', 'POSTED', 'SUP-1002|CREATE', '300002')]	**PASS**	
49	[C] F8-05 supplier 360 (qualification open)	supplier.manager	`GET /v1/suppliers/SUP-1003`	qualification row for the invited category	200 SUP-1003: ok qualification exists	**PASS**	{"code": "SUP-1003", "legal_name": "Chola Ho "REGISTERED", "gstin_masked": "\u2022\u2022\u2022\u2022\u2022\u2022\u2 "state_code": "33", "country": "IN", "seg...
50	[C] F8-05 verify certificate ISO9001	quality.inspector	`POST /v1/certificates/01a1b6b8-b1c3-759b-9cea-cc0ab1069ad5/verify`	200 verified	200	**PASS**	{"supplier_id": "01a1b6b8-a9ff-755f-8603-8d1 "number": "QMS-C-001", "issued_on": "2025-0 "mandatory_for": [], "verified": true, "status": "\
51	[C] F8-05 assessment	supplier.manager	`POST /v1/qualifications/01a1b6b8-ab2e-7268-8647-d53e74b16d89/assessment`	200	200	**PASS**	{"supplier_id": "01a1b6b8-a9ff-755f-8603-8d1 "status": "OPEN", "questionnaire": {"capacity": "audit_score": "80", "sample_approval": null, "c
52	[C] F8-06 decision QUALIFIED	supplier.manager2	`POST /v1/qualifications/01a1b6b8-ab2e-7268-8647-d53e74b16d89/decision`	200; supplier ACTIVE; ERP SUPPLIER_MASTER {code}|CREATE queued	200	**PASS**	{"supplier": "SUP-1003", "category": "FORGINC "ruleSetVersion": "SUPPLIER-QUAL@2026.1", "
53	[C] F8-06 supplier ACTIVE + category QUALIFIED	supplier.manager	`GET /v1/suppliers/SUP-1003`	status ACTIVE, category QUALIFIED	200 SUP-1003: ok ACTIVE; ok QUALIFIED	**PASS**	{"code": "SUP-1003", "legal_name": "Chola Ho "gstin_masked": "\u2022\u2022\u2022\u2022\u2022\u2022\u2 "state_code": "33", "country": "IN", "segment...
54	[C] F8-06 ERP vendor create queued	system	`integration_message`	SUPPLIER_MASTER {code}|CREATE	[('SUPPLIER_MASTER', 'SUP-1003', 'POSTED', 'SUP-1003|CREATE', '300003')]	**PASS**	

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
55	[A] F8-06 ERP vendor posted (after tick)	worker	`integration_message`	POSTED with erp_doc_no	[('SUPPLIER_MASTER', 'SUP-1001', 'POSTED', 'SUP-1001|CREATE', '300001')]	**PASS**	
56	[B] F8-06 ERP vendor posted (after tick)	worker	`integration_message`	POSTED with erp_doc_no	[('SUPPLIER_MASTER', 'SUP-1002', 'POSTED', 'SUP-1002|CREATE', '300002')]	**PASS**	
57	[C] F8-06 ERP vendor posted (after tick)	worker	`integration_message`	POSTED with erp_doc_no	[('SUPPLIER_MASTER', 'SUP-1003', 'POSTED', 'SUP-1003|CREATE', '300003')]	**PASS**	
58	[D] F8-01 direct registration	supplier.manager	`POST /v1/suppliers`	201 REGISTERED, SOD-01 recorded	201 SUP-1004: ok REGISTERED	**PASS**	{"code": "SUP-1004", "legal_name": "Deccan M "REGISTERED", "gstin_masked": "\u2022\u2022\u2022

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
70	[D] F8-12 bank change on a BLOCKED supplier refused	supplier.manager	`POST /v1/suppliers/SUP-1004/bank-changes`	409 SUPPLIER_BLOCKED (guide: supplier not BLOCKED)	201: status 201 not in (409,)	**BROKEN**	{"number": "BCR-105", "supplier_id": "01a1b6b "bank_account_id": "01a1b6b8-df2b-7238-94c "PENDING_CALLBACK", "requested_by": "01a1 82c528e1c00e"...
71	[D] F8-12 unblock	supplier.manager	`POST /v1/suppliers/SUP-1004/unblock`	200 previous status restored	200: ok not BLOCKED	**PASS**	{"supplier": "SUP-1004", "status": "ACTIVE", "pa
72	[D] F8-12 unblock again refused	supplier.manager	`POST /v1/suppliers/SUP-1004/unblock`	409 not blocked	409 ILLEGAL_TRANSITION	**PASS**	{"type": "https://errors.alphahelion.io/pic/ILLE "ILLEGAL_TRANSITION", "status": 409, "detail": 'unblock' is not allowed from this state", "code
73	[A] B8 portal bank change after onboarding	SUPPLIER_USER A	`POST /v1/suppliers/SUP-1001/bank-changes`	refused (portal cannot request; B8)	403 FORBIDDEN	**PASS**	{"type": "https://errors.alphahelion.io/pic/FOR "status": 403, "detail": "Missing scope(s): suppl "instance": "/v1/suppliers/SUP-1001/bank-cha
74	[A] B8 portal profile bank after onboarding	SUPPLIER_USER A	`POST /v1/suppliers/SUP-1001/profile`	refused BANK_CHANGE_PROCESS_REQUIRED	409 BANK_CHANGE_PROCESS_REQUIRED	**PASS**	{"type": "https://errors.alphahelion.io/pic/BAN "title": "BANK_CHANGE_PROCESS_REQUIRED" SUP-1001 has ended: bank details change onl
75	[A] F8-04 bank change after onboarding (manager)	supplier.manager	`POST /v1/suppliers/SUP-1001/bank-changes`	201 BCR PENDING_CALLBACK, SOD-02 recorded	201	**PASS**	{"number": "BCR-106", "supplier_id": "01a1b6b "bank_account_id": "01a1b6b8-e08f-768c-a66 "PENDING_CALLBACK", "requested_by": "01a1 82c528e1c00e"...
76	[A] F8-04 second pending bank change refused	supplier.manager2	`POST /v1/suppliers/SUP-1001/bank-changes`	409 BANK_CHANGE_PENDING	201: status 201 not in (409,)	**BROKEN**	{"number": "BCR-107", "supplier_id": "01a1b6b "bank_account_id": "01a1b6b8-e180-76b0-be: "PENDING_CALLBACK", "requested_by": "01a1 82c64544a680"...
77	[A] F8-04 call-back (other manager)	supplier.manager2	`POST /v1/bank-changes/BCR-106/callback`	200	200	**PASS**	{"number": "BCR-106", "supplier_id": "01a1b6b "bank_account_id": "01a1b6b8-e08f-768c-a66 "PENDING_APPROVAL", "requested_by": "01a1 82c528e1c00e"...
78	[A] F8-04 second call-back refused	supplier.manager2	`POST /v1/bank-changes/BCR-106/callback`	409	409 ILLEGAL_TRANSITION	**PASS**	{"type": "https://errors.alphahelion.io/pic/ILLE "ILLEGAL_TRANSITION", "status": 409, "detail": PENDING_APPROVAL; 'callback' is not allowed
79	[A] F8-04 approve bank change (fin1)	fin1	`POST /v1/tasks/01a1b6b8-e260-701d-b608-fdfc9fd9885b/complete`	200	200	**PASS**	{"instance_id": "01a1b6b8-e0a5-7327-a650-a4 "name": "Bank Change \u00b7 BCR-106 \u00b "BANK_CHANGE", "object_id": "01a1b6b8-e09
80	[A] F8-04 approve bank change (fin2)	fin2	`POST /v1/tasks/01a1b6b8-e418-7778-84b3-7374e1953e08/complete`	200	200	**PASS**	{"instance_id": "01a1b6b8-e0a5-7327-a650-a4 "name": "Bank Change \u00b7 BCR-106 \u00b "BANK_CHANGE", "object_id": "01a1b6b8-e09
81	[A] F8-04 bank change to ERP	system	`integration_message`	SUPPLIER_MASTER {code}|BANK|{BCR}	[('SUPPLIER_MASTER', 'SUP-1001', 'POSTED', 'SUP-1001|BANK|BCR-106', '300005')]	**PASS**	
82	[A] MDM-04 TAX master change (portal)	SUPPLIER_USER A	`POST /v1/suppliers/SUP-1001/master-changes`	201 SMC IN_APPROVAL, 2 tasks	201	**PASS**	{"number": "SMC-101", "supplier_id": "01a1b6 "field_group": "TAX", "before_masked": {"gstin "\u2022\u2022\u2022\u2022\u2022\u2022\u2022\u2 "pan": "\u2022\u2022...
83	[A] MDM-04 approval tasks	system	`task`	SUPPLIER_MASTER_CHANGE for FINANCE_CONTROLLER and SUPPLIER_MANAGER	[('SUPPLIER_MASTER_CHANGE', 'FINANCE_CONTROLLER', 'OPEN'), ('SUPPLIER_MASTER_CHANGE', 'SUPPLIER_MANAGER', 'OPEN')]	**PASS**	
84	B7 ABAC on master-change tasks	supplier.mgr.mro / fin.other	`GET /v1/tasks`	out-of-scope approvers should not see the task	MRO-only SM sees 1, company-9999 FC sees 1	**PARTIAL**	missing control: out-of-scope approver/reque

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
85	[A] MDM-04 approve (fin1)	fin1	`POST /v1/tasks/01a1b6b8-e8ba-73f1-b09d-79238dc0a32e/complete`	200	200	**PASS**	{"instance_id": "01a1b6b8-e87d-72c2-bf83-f8 "SUPPLIER_MASTER_CHANGE", "name": "Supp \u00b7 Finance Controller", "object_type": "SU "object_id": "...
86	[A] MDM-04 approve (supplier.manager)	supplier.manager	`POST /v1/tasks/01a1b6b8-e8ca-73c6-9cea-3287f1d9b2f4/complete`	200	200	**PASS**	{"instance_id": "01a1b6b8-e87d-72c2-bf83-f8 "SUPPLIER_MASTER_CHANGE", "name": "Supp \u00b7 Supplier Manager", "object_type": "SU "01...
87	[A] MDM-04 change APPLIED	supplier.manager	`GET /v1/suppliers/SUP-1001/master-changes`	APPLIED; supplier.master.changed; SUPPLIER_MASTER update to ERP	200: ok APPLIED; ok event supplier.master.changed; ok ERP update	**PASS**	[{"number": "SMC-101", "supplier_id": "01a1b6 "field_group": "TAX", "before_masked": {"pan": "\u2022\u2022\u2022\u2022\u2022\u202223 "\u2022\u2022\u2022\u2022\u2022\u2022\u2
88	B7 ABAC: out-of-scope manager requests master change	supplier.mgr.mro	`POST /v1/suppliers/SUP-1001/master-changes`	should be refused (403/404) for a manager scoped to MRO / plant P999 / company 9999	201	**PARTIAL**	missing control: out-of-scope approver/reque 102", "supplier_id": "01a1b6b8-9558-7755-b4 "CONTACT", "before_masked": {"email": "a\u2 "phone": "\u2022\u2022\u2022\u2022\u2022
89	[A] F8-08 scorecard measures refresh	supplier.manager	`POST /v1/suppliers/SUP-1001/scorecard-measures/refresh`	200	200	**PASS**	[]
90	[A] F8-08 supplier performance read	supplier.manager	`GET /v1/supplier-performance`	200	200	**PASS**	[]
91	B8 master change field group COMMERCIAL	supplier.manager	`POST /v1/suppliers/SUP-1001/master-changes`	a command to change legal name / terms	422 {"type": "https://errors.alphahelion.io/pic/VALIDATION_FAILED", "title": "VALIDATION_FAILED", "status": 422, "detail": "fieldGroup: String s...	**NOT BUILT**	no field group for legal name / payment terms CONFIRMED)
92	B8 master change field group LEGAL_NAME	supplier.manager	`POST /v1/suppliers/SUP-1001/master-changes`	a command to change legal name / terms	422 {"type": "https://errors.alphahelion.io/pic/VALIDATION_FAILED", "title": "VALIDATION_FAILED", "status": 422, "detail": "fieldGroup: String s...	**NOT BUILT**	no field group for legal name / payment terms CONFIRMED)
93	B9 scheduled job catalogue	admin	`GET /v1/admin/jobs`	jobs for risk scan, sanctions re-screen, rebate accrual, offer expiry	200 jobs=['audit-archive', 'erp-reconciliation', 'idempotency-purge', 'inventory-reservations', 'mrp-schedule', 'payment-due', 'retention-purge', 'scorecard-monthly', 'spend-cube-refresh', 'spend-forecast-snapshot', 'spend-quality', 'spend-sla'] matches={'risk': [], 'sanction': [], 'rebate': [], 'of	**CONFIRMED**	

M01 spend

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
1	setup: ERP spend history load	migration	`kit.spend_history`	lines loaded	33 lines	**PASS**	
2	SPD cube read	category.manager	`GET /v1/spend/overview`	200 totals incl. history	200	**PASS**	{"periods": ["2025-11", "2026-10"], "totalMinor": 4620000000, "addressableMinor": 4620000000, "addressablePct": "100.0", "underManagementPct": "45.5", "contractedPct": "45.5", "maverickPct": "0.0", "classificationAccurac...
3	admin re-evaluate (classification + maverick)	admin	`POST /v1/admin/spend/reevaluate`	200	200	**PASS**	{"auto": 0, "review": 0, "hits": 0}
4	duplicates scan	mdm.purchasing	`POST /v1/spend/duplicates/scan`	cluster Shakti Forge (name similarity)	200	**PASS**	{"new": 1, "known": 0}

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
5	duplicates list	mdm.purchasing	`GET /v1/spend/duplicates`	clusters	200	**PASS**	{"clusters": [{"key": "c9fd077742b81f24a086", "suppliers": [{"code": "V-9001", "name": "Shakti Forge Industries Pvt Ltd", "spendMinor": 3850000000, "parent": null}, {"code": "V-9002", "name": "Shakti Forge Industries Pri...
6	duplicate clusters	system	`GET /v1/spend/duplicates`	Shakti pair	[('c9fd077742b81f24a086', 'OPEN', [])]	**PASS**	
7	duplicates propose merge	mdm.purchasing	`POST /v1/spend/duplicates/c9fd077742b81f24a086/propose-merge`	201 PROPOSED, SPEND_SUPPLIER_MERGE task for MDM_STEWARD	201	**PASS**	{"cluster_key": "c9fd077742b81f24a086", "supplier_codes": ["V-9001", "V-9002"], "reasons": ["similar name (1.00)"], "score": "1", "spend_group_minor": 4400000000, "status": "PROPOSED", "survivor_code": "V-9001", "proposa...
8	duplicate merge task	system	`task`	SPEND_SUPPLIER_MERGE	[('SPEND_SUPPLIER_MERGE', 'MDM_STEWARD', 'OPEN')]	**PASS**	
9	duplicate merge: proposer excluded	mdm.purchasing	`GET /v1/tasks`	no task for the proposer	0	**PASS**	
10	duplicate merge approved by second steward	mdm.quality	`POST /v1/tasks/01a1b6ba-2208-75d2-beee-2b3f4b7d2f18/complete`	200 MERGE_APPROVED	200	**PASS**	{"instance_id": null, "kind": "SPEND_SUPPLIER_MERGE", "name": "Spend Supplier Merge \u00b7 DUP-C9FD0777 \u00b7 Mdm Steward", "object_type": "SPEND_DUPLICATE_CLUSTER", "object_id": "01a1b6ba-20c0-750f-a141-b34516829d96", ...
11	duplicate merge effect	system	`supplier`	V-9002 linked to V-9001 as corporate parent	[('V-9002', True), ('V-9001', False)]	**PASS**	
12	maverick worklist	category.manager	`GET /v1/spend/maverick`	hits from rules	200	**PASS**	{"rules": [{"code": "NON_CATALOGUE", "name": "Non-catalogue purchase where a catalogue exists", "description": "Free-text purchase in a material group that has an active catalogue.", "severity": "Medium", "owner_role": "...
13	maverick hits on history	system	`maverick_hit`	hits exist	0 hits []	**INFO**	ERP-history spend lines are not rule-evaluated (by design); disposition tested on AXN seeded hits below.
14	steward classification queue	mdm.purchasing	`GET /v1/spend/classification-reviews`	queue	200	**PASS**	{"categories": [{"code": "ALU-DIECAST", "name": "Aluminium die castings"}, {"code": "CAPEX-MACHINERY", "name": "Machinery and equipment"}, {"code": "ELECTRONICS", "name": "Electronics"}, {"code": "FACILITIES", "name": "F...
15	classification reviews present	system	`classification_review`	reviews (history lines have confidence 1.0)	0	**INFO**	
16	SPD-07 generate category plans	category.manager	`POST /v1/category-plans`	DRAFT plans CP-101... for FORGINGS, FASTENERS	200	**PASS**	[{"number": "CP-101", "category": "FASTENERS", "fiscal_year": "2026-27", "spend_minor": 220000000, "profit_impact": "2.24", "supply_risk": "6.5", "kraljic": "BOTTLENECK", "evidence": {"direct": true, "qualifiedSuppliers"...
17	category plans generated	system	`category_plan`	plans	[('CP-101', 'FASTENERS', 'DRAFT'), ('CP-102', 'FORGINGS', 'DRAFT')]	**PASS**	
18	SPD-07 edit plan	category.manager	`PATCH /v1/category-plans/CP-101`	200	200	**PASS**	{"number": "CP-101", "category": "FASTENERS", "fiscal_year": "2026-27", "spend_minor": 220000000, "profit_impact": "2.24", "supply_risk": "6.5", "kraljic": "BOTTLENECK", "evidence": {"hhi": 10000, "direct": true, "window...
19	SPD-07 submit plan	category.manager	`POST /v1/category-plans/CP-101/submit`	IN_APPROVAL, CATEGORY_PLAN_APPROVAL task for procurement head	200: ok IN_APPROVAL	**PASS**	{"number": "CP-101", "category": "FASTENERS", "fiscal_year": "2026-27", "spend_minor": 220000000, "profit_impact": "2.24", "supply_risk": "6.5", "kraljic": "BOTTLENECK", "evidence": {"hhi": 10000, "direct": true, "window...
20	SPD-07 owner approves own plan (SoD)	category.manager	`POST /v1/category-plans/CP-101/approve`	refused (owner excluded)	403 FORBIDDEN	**PASS**	{"type": "https://errors.alphahelion.io/pic/FORBIDDEN", "title": "FORBIDDEN", "status": 403, "detail": "Category plans are approved by the procurement...
21	SPD-07 approval task	proc.head	`GET /v1/tasks`	CATEGORY_PLAN_APPROVAL	['CATEGORY_PLAN_APPROVAL']	**PASS**	
22	SPD-07 approve plan (procurement head)	proc.head	`POST /v1/tasks/01a1b6ba-270c-733a-b595-5d68d18c7319/complete`	200 APPROVED	200	**PASS**	{"instance_id": null, "kind": "CATEGORY_PLAN_APPROVAL", "name": "Category Plan Approval \u00b7 CP-101 \u00b7 Procurement Head", "object_type": "CATEGORY_PLAN", "object_id": "01a1b6ba-25f0-710f-b460-e02603fc3b66", "object...
23	SPD-07 plan APPROVED	category.manager	`GET /v1/category-plans/CP-101`	APPROVED	200: ok APPROVED	**PASS**	{"number": "CP-101", "category": "FASTENERS", "fiscal_year": "2026-27", "spend_minor": 220000000, "profit_impact": "2.24", "supply_risk": "6.5", "kraljic": "BOTTLENECK", "evidence": {"hhi": 10000, "direct": true, "window...

#	Step	Actor	API call	Expected	Actual	Verdict	Evidence
24	SPD-07 revise approved plan	category.manager	`POST /v1/category-plans/CP-101/revise`	201 new DRAFT version	201	**PASS**	{"number": "CP-103", "category": "FASTENERS", "fiscal_year": "2026-27", "spend_minor": 220000000, "profit_impact": "2.24", "supply_risk": "6.5", "kraljic": "BOTTLENECK", "evidence": {"hhi": 10000, "direct": true, "window...
25	SPD-07 reject plan	proc.head	`POST /v1/category-plans/CP-102/reject`	200 REJECTED	200: ok REJECTED	**PASS**	{"number": "CP-102", "category": "FORGINGS", "fiscal_year": "2026-27", "spend_minor": 4400000000, "profit_impact": "10", "supply_risk": "3.95", "kraljic": "LEVERAGE", "evidence": {"hhi": 7812, "direct": true, "window": [...
26	SPD-07 pipeline links	category.manager	`GET /v1/category-plans/CP-101/pipeline-links`	sourcing events of the category	200	**PASS**	{"plan": "CP-101", "category": "FASTENERS", "sourcingEvents": [], "savingsInitiatives": []}
27	AXN maverick worklist	category.manager@AXN	`GET /v1/spend/maverick`	OPEN hits	200 open CM hits=45	**PASS**	
28	AXN maverick: comment < 5 chars refused	category.manager@AXN	`POST /v1/spend/maverick-hits/{id}/disposition`	(422,)	422 422	**PASS**	{"type": "https://errors.alphahelion.io/pic/COMMENT_REQUIRED", "title": "COMMENT_REQUIRED", "status": 422, "detail": "Every maverick disposition needs...
29	AXN maverick: FALSE_POSITIVE without reason code refused	category.manager@AXN	`POST /v1/spend/maverick-hits/{id}/disposition`	(422,)	422 422	**PASS**	{"type": "https://errors.alphahelion.io/pic/REASON_CODE_REQUIRED", "title": "REASON_CODE_REQUIRED", "status": 422, "detail": "FALSE_POSITIVE needs a r...
30	AXN maverick: FALSE_POSITIVE with reason -> DISPOSED	category.manager@AXN	`POST /v1/spend/maverick-hits/{id}/disposition`	(200,)	200 DISPOSED	**PASS**	{"number": "MAV-1398", "rule_code": "OFF_CONTRACT", "spend_line_id": "01a1b692-923b-76cf-a61a-f4c25e4dabcd", "invoice_ref": "ERP-5105000310", "po_ref"...
31	AXN maverick: dispose a closed hit -> 409	category.manager@AXN	`POST /v1/spend/maverick-hits/{id}/disposition`	(409,)	409 409	**PASS**	{"type": "https://errors.alphahelion.io/pic/ILLEGAL_TRANSITION", "title": "ILLEGAL_TRANSITION", "status": 409, "detail": "MAV-1398 is already DISPOSED...
32	AXN maverick: RECOVERY -> RECOVERY_OPEN	category.manager@AXN	`POST /v1/spend/maverick-hits/{id}/disposition`	(200,)	200 RECOVERY_OPEN	**PASS**	{"number": "MAV-1092", "rule_code": "OFF_CONTRACT", "spend_line_id": "01a1b692-9237-71bd-a298-dfc67ee37c50", "invoice_ref": "ERP-5105000072", "po_ref"...
33	AXN maverick: recovery above hit amount refused	category.manager@AXN	`POST .../resolve`	422	422	**PASS**	{"type": "https://errors.alphahelion.io/pic/RECOVERY_EXCEEDS_HIT", "title": "RECOVERY_EXCEEDS_HIT", "status": 422, "deta...
34	AXN maverick: resolve recovery	category.manager@AXN	`POST .../resolve`	200 RESOLVED	200 RESOLVED	**PASS**	{"number": "MAV-1092", "rule_code": "OFF_CONTRACT", "spend_line_id": "01a1b692-9237-71bd-a298-dfc67ee37c50", "invoice_re...
35	AXN steward queue	mdm.purchasing@AXN	`GET /v1/spend/classification-reviews`	OPEN reviews	200 3	**PASS**	
36	AXN steward decision (accept proposal)	mdm.purchasing@AXN	`POST /v1/spend/classification-reviews/{id}/decision`	200 ACCEPTED	200 ACCEPTED	**PASS**	{"spend_line_id": "01a1b693-0f6b-7436-b148-deee376ab1d6", "invoice_ref": "HTS/1188", "description": "Preventive maintenance visit \u2013 die casting c...
37	AXN steward reclassify + teach keyword	mdm.purchasing@AXN	`POST .../decision`	200 RECLASSIFIED (or PENDING_SECOND / 409 KEYWORD_CONFLICT)	200 RECLASSIFIED	**PASS**	{"spend_line_id": "01a1b692-9288-74fa-9e13-79ea131c43c3", "invoice_ref": "ERP-5105002495", "description": "Training programme for quality team", "prop...

B. Receive to pay, AI and platform (M09 to M12, Platform)

F6 Receiving

Step	Actor	API call	Expected	Actual	Verdict	Evidence
1a ASN without e-way bill above Rs 50,000	SUPPLIER_USER	`POST /v1/asns`	422 EWAY_BILL_REQUIRED	422 EWAY_BILL_REQUIRED Consignment value 59,000.00 exceeds 50,000.00: an e-way bill is required (eway_bill_no, vehicle_no, eway_bill_valid_until (or distance_km) missing)	**PASS**	
1b ASN shipping above ordered qty (tolerance 0%)	SUPPLIER_USER	`POST /v1/asns`	422 RCV-TOLERANCE	422 RULE_VIOLATION Line 10: shipping 120 exceeds ordered quantity with tolerance	**PASS**	RCV-TOLERANCE
1c ASN e-way bill in wrong format	SUPPLIER_USER	`POST /v1/asns`	422 EWAY-BILL format	422 RULE_VIOLATION E-way bill number 1234 does not match the required format (12 digits)	**PASS**	

Step	Actor	API call	Expected	Actual	Verdict	Evidence
1d ASN with e-way bill, vehicle, 2 batches	SUPPLIER_USER	`POST /v1/asns`	201 IN_TRANSIT; event asn.received	201 {"number": "ASN-61207", "status": "IN_TRANSIT", "eway_bill_valid_until": "2026-11-11T23:59:59+05:30", "eway_bill_no": "123456789012"}	**PASS**	events=['asn.received(unpublished)'] poSteps=['F5-09:COMPLETED', 'F5-10:COMPLETED']
1e ASN idempotent replay	SUPPLIER_USER	`POST /v1/asns (same Idempotency-Key)`	same ASN replayed	201 {"number": "ASN-61207"}	**PASS**	replay returned the same ASN; route-level replay sets body flag, no Idempotent-Replayed header
1f ASN by another supplier on this PO	SUPPLIER_USER (other)	`POST /v1/asns`	403/404 SUPPLIER_SCOPE	403 FORBIDDEN You can only act on your own company's documents	**PASS**	
2a GR by the PO creator (buyer)	BUYER	`POST /v1/goods-receipts`	403 (no gr:post; SOD-03 by role design)	403 FORBIDDEN Missing scope(s): gr:post	**PASS**	
2b GR over-delivery 101/100 (tolerance 0%)	RECEIVER	`POST /v1/goods-receipts`	422 RCV-TOLERANCE	422 RULE_VIOLATION Line 10: over-delivery beyond +0% is blocked at receipt (101.000 > 100.000)	**PASS**	RCV-TOLERANCE
2c GR of batch/expiry-managed goods without batch	RECEIVER	`POST /v1/goods-receipts`	422 RCV-LOT	422 RULE_VIOLATION Line 10: FLB-QM is lot-controlled; capture batch / lot number, expiry date at receipt	**PASS**	RCV-LOT
2d GR with shelf life below minShelfLifeDays (30)	RECEIVER	`POST /v1/goods-receipts`	422 RCV-LOT shelf life	422 RULE_VIOLATION Line 10: lot BX has 10 days of shelf life left; FLB-QM needs at least 30 days at receipt	**PASS**	
2e GR naming an ASN of another PO	RECEIVER	`POST /v1/goods-receipts`	422 ASN_PO_MISMATCH	422 ASN_PO_MISMATCH ASN ASN-61207 was shipped against PO-77416, not PO-77419; post the receipt against that order or without the ASN	**PASS**	
3 GR with 2 batches + expiry into quality hold	RECEIVER	`POST /v1/goods-receipts`	201 QUALITY_HOLD; lots; QUALITY stock 100; ASN RECEIVED; event goods_receipt.posted; no ERP msg yet	201 {"number": "GR-90231", "status": "QUALITY_HOLD"}	**PASS**	lots=[('QL-2211', '40', 'OPEN'), ('QL-2210', '60', 'OPEN')] stock={'0001/QUALITY/B1': 60.0, '0001/QUALITY/B2': 40.0} asn=[{'status': 'RECEIVED'}] events=['goods_receipt.posted(unpublished)'] erp=[] moves=['101:QUALITY:60.0', '101:QUALITY:40.0']
6 GR 20 + UD 18/2 (credit note expected)	RECEIVER, QUALITY_INSPECTOR	`POST /v1/goods-receipts; POST .../usage-decision`	PARTIALLY_REJECTED; RD CREATED	200 {"status": "PARTIALLY_REJECTED", "returnDelivery": "RD-4410"}	**PASS**	
7a GR of serial-managed goods with too few serials	RECEIVER	`POST /v1/goods-receipts`	422 RCV-LOT	422 RULE_VIOLATION Line 10: FLB-SER is lot-controlled; capture 3 serial number(s) (got 2) at receipt	**PASS**	
7b GR 2 serials	RECEIVER	`POST /v1/goods-receipts`	201 ACCEPTED (not QM); ERP GOODS_RECEIPT	201 {"number": "GR-90233", "status": "ACCEPTED"}	**PASS**	erp=[{'message_type': 'GOODS_RECEIPT', 'idempotency_key': 'GR|GR-90233', 'status': 'POSTED', 'erp_doc_no': '5000071231', 'attempts': 1}]
7c GR repeating a received serial	RECEIVER	`POST /v1/goods-receipts`	422 RCV-LOT already received	422 RULE_VIOLATION Line 10: serial number(s) SN-2 already received	**PASS**	
8a GR 100 FLB-RAW (no QM) against ASN	RECEIVER	`POST /v1/goods-receipts`	201 ACCEPTED; 101; ERP GOODS_RECEIPT posted	201 {"number": "GR-90234", "status": "ACCEPTED"}	**PASS**	moves=['101:UNRESTRICTED:100.0'] erp=[{'message_type': 'GOODS_RECEIPT', 'idempotency_key': 'GR|GR-90234', 'status': 'POSTED', 'erp_doc_no': '5000071232', 'attempts': 1}] steps=['F6-01:COMPLETED', 'F6-02:COMPLETED', 'F6-03:COMPLETED', 'F6-05:COMPLETED', 'F6-08:COMPLETED', 'F6-09:COMPLETED']
8b GR reversal (102)	RECEIVER	`POST /v1/goods-receipts/{ref}/reverse`	200 REVERSED; 102; ASN IN_TRANSIT; PO qty reopened; ERP GR|..|102; event goods_receipt.reversed	200 {"status": "REVERSED"}	**PASS**	moves=['101:UNRESTRICTED:100.0', '102:UNRESTRICTED:-100.0'] asn=[{'status': 'IN_TRANSIT'}] poLine=[{'received_qty': '0.000'}] erp=[{'message_type': 'GOODS_RECEIPT', 'idempotency_key': 'GR|GR-90234', 'status': 'POSTED', 'erp_doc_no': '5000071232', 'attempts': 1}, {'message_type': 'GOODS_RECEIPT', 'idempotency_key': 'GR|GR-90234|102', 'status': 'P
8c reverse twice	RECEIVER	`POST .../reverse`	409 ILLEGAL_TRANSITION	409 ILLEGAL_TRANSITION Goods receipt GR-90234 is REVERSED; 'reverse' is not allowed from this state	**PASS**	
8d re-receive after reversal	RECEIVER	`POST /v1/goods-receipts`	201 ACCEPTED	201 {"number": "GR-90235", "status": "ACCEPTED"}	**PASS**	

Step	Actor	API call	Expected	Actual	Verdict	Evidence
9a SES above the PO line value (11 of 10 days)	SUPPLIER_USER	`POST /v1/service-entries`	422 RCV-SES	422 RULE_VIOLATION Line 10: service value exceeds the PO line (0.00 approved, 0.00 in approval, 110,000.00 on this sheet; line value 100,000.00)	**PASS**	
9b SES submit (4 days)	SUPPLIER_USER	`POST /v1/service-entries`	201 SUBMITTED; SES_APPROVAL task for SERVICE_OWNER	201 {"number": "SES-3301", "status": "SUBMITTED", "total_minor": 4000000}	**PASS**	tasks=[{'kind': 'SES_APPROVAL', 'candidate_role': 'SERVICE_OWNER', 'status': 'OPEN', 'decision': None, 'sequence': 1}]
9c SES approve by the PO creator	BUYER	`POST /v1/service-entries/{ref}/approve`	403	403 FORBIDDEN Missing scope(s): ses:approve	**PASS**	
9d SES approve by SERVICE_OWNER	SERVICE_OWNER	`POST /v1/service-entries/{ref}/approve`	200 APPROVED; PO line approved value; event service_entry.approved	200 {"status": "APPROVED"}	**PASS**	events=['task.assigned', 'service_entry.approved'] poLine=[{'ses_approved_minor': 4000000}] tasks=[{'kind': 'SES_APPROVAL', 'candidate_role': 'SERVICE_OWNER', 'status': 'COMPLETED', 'decision': 'APPROVE', 'sequence': 1}] erp=[]
9e SES reject via task inbox	SERVICE_OWNER	`POST /v1/tasks/{id}/complete REJECT`	SES REJECTED	200 {} ses=[{'status': 'REJECTED'}]	**PASS**	
9f resubmit a REJECTED SES	SUPPLIER_USER	`(no route)`	a submit/resubmit command	routes=['GET,POST /v1/service-entries', 'POST /v1/service-entries/{ref}/approve']	**NOT BUILT**	guide: REJECTED ->submit-> SUBMITTED, but the API has only create (auto-submit) and approve
10a GR consignment 50	RECEIVER	`POST /v1/goods-receipts`	201; CONSIGNMENT stock 50 (supplier-owned)	201 {"number": "GR-90236", "status": "ACCEPTED"}	**PASS**	stock={'0001/CONSIGNMENT/': 50.0} erp=[{'message_type': 'GOODS_RECEIPT', 'idempotency_key': 'GR|GR-90236', 'status': 'POSTED', 'erp_doc_no': '5000071235', 'attempts': 1}]
10b consignment withdrawal 12 to a cost centre (201K)	RECEIVER	`POST /v1/consignment/withdrawals`	201; 201K; liability at PO price; event consignment.withdrawn	201 {"movement": "201K", "qty": "12", "liabilityMinor": 240000}	**PASS**	body={'material': 'FLB-CON', 'plant': 'P100', 'storageLocation': '0001', 'qty': '12', 'purpose': 'CONSUMPTION', 'movement': '201K', 'usageRef': 'PROD-ORD-501', 'costObject': 'COST_CENTER:6100', 'liabilityMinor': 240000, 'lots': [{'grLine': '01a1b6a4-3d78- stock={'0001/CONSIGNMENT/': 38.0} events=['consignment.withdrawn(unpublished)']
10c withdrawal above consignment stock	RECEIVER	`POST /v1/consignment/withdrawals`	422 CONSIGNMENT_STOCK_SHORT	422 CONSIGNMENT_STOCK_SHORT Only 38.000 EA FLB-CON in consignment stock at P100/0001; 100 requested	**PASS**	
10d consignment settlement (ERS self-billing)	AP_CLERK	`POST /v1/consignment/settlements`	201 invoice match ERS for 12 x Rs 200	201 {"number": "ERS-FLB-CONS-202611", "status": "MATCHED", "match_type": "ERS", "net_minor": 240000, "gross_minor": 283200}	**PASS**	
10e ERS approve by its builder	AP_CLERK	`POST /v1/invoices/{ref}/approve`	409 SOD-08	409 SOD_CONFLICT SOD-08: Whoever captured an invoice may not approve it (four-eyes)	**PASS**	
10f ERS approve (other clerk) and post	AP_CLERK	`POST .../approve; POST .../post`	APPROVED then POSTED; ERP SUPPLIER_INVOICE	200 {"status": "APPROVED"} / 200 {"status": "POSTED", "erpDocument": "5100004421"}	**PASS**	key=FLB|1000|FLB-CONS|ERS-FLB-CONS-202611 erp=[{'message_type': 'SUPPLIER_INVOICE', 'idempotency_key': 'FLB|1000|FLB-CONS|ERS-FLB-CONS-202611', 'status': 'POSTED', 'erp_doc_no': '5100004421', 'attempts': 1}]
11a read planned components	RECEIVER	`GET /v1/purchase-orders/{ref}/components`	FLB-CMP 40 (2 per part)	200 {} {'po': 'PO-77423', 'lines': [{'line': 10, 'components': [{'qty': '40', 'uom': 'EA', 'qtyPer': '2', 'material': 'FLB-CMP', 'materialId': '01a1b69b-2e49-77cc-90d9-59e12a3e8cfd', 'description': 'Audit ma	**PASS**	
11b GR of subcontracted part before components were provided	RECEIVER	`POST /v1/goods-receipts`	422 COMPONENTS_MISSING	422 COMPONENTS_MISSING PO-77423/10: receiving 20 consumes 40 FLB-CMP, but only 0 are at FLB-SUB; provide the components first (541)	**PASS**	
11c provide components (541)	RECEIVER	`POST /v1/purchase-orders/{ref}/components`	201; 541 own -> subcontractor; event subcontracting.components_provided	201 {} {'po': 'PO-77423', 'line': 10, 'subcontractor': 'FLB-SUB', 'components': [{'material': 'FLB-CMP', 'qty': '40', 'batch': '', 'from': 'P100/0001'}], 'at	**PASS**	stockCMP={'0001/UNRESTRICTED/': 60.0, 'FLB-SUB/SUBCONTRACTOR/': 40.0} events=['purchase_order.released', 'task.assigned', 'purchase_order.transmitted', 'purchase_order.confirmed', 'subcontracting.components_provided(unpublished)']
11d GR of the finished part consumes components (543)	RECEIVER	`POST /v1/goods-receipts`	201; 101 FG +20; 543 CMP -40 at subcontractor	201 {"number": "GR-90238", "status": "ACCEPTED"}	**PASS**	543 SUBCONTRACTOR -40 FLB-CMP posted with reference GR-90238/10 (ledger query by line reference); FG +20; stock at FLB-SUB 40 -> 0; ERP GOODS_RECEIPT POSTED
12 lot trace of batch B1	QUALITY_INSPECTOR	`GET /v1/lots/{material}/{batch}/trace`	receipts, stock on hand	200 {}	**PASS**	

Step	Actor	API call	Expected	Actual	Verdict	Evidence
13 M09 KPIs after the flow	RECEIVER	`GET /v1/kpis?module=M09`	values computed	200 {}	**PASS**	{'M09.DOCK_TO_STOCK': '0.0', 'M09.INVENTORY_TURNS': '0.1', 'M09.QUALITY_REJECTION': '0.4', 'M09.RECEIPT_ACCURACY': '100.0', 'M09.SES_APPROVAL': '0.0', 'M09.STOCK_ADJUSTMENT_VALUE': '0', 'M09.STOCK_COUNT_ACCURACY': None}
4a UD with accepted+rejected != lot qty	QUALITY_INSPECTOR	`POST /v1/inspection-lots/{ref}/usage-decision`	422 QM-UD	422 RULE_VIOLATION Accepted + rejected must equal the lot quantity 60.000	**PASS**	
4b UD rejection without defect code	QUALITY_INSPECTOR	`POST .../usage-decision`	422 QM-UD	422 RULE_VIOLATION Rejections need a defect code and a reason	**PASS**	
4c UD by the receiver	RECEIVER	`POST .../usage-decision`	403	403 FORBIDDEN Missing scope(s): quality:decide	**PASS**	
4d UD accept lot 1 (60) - receipt still has an open lot	QUALITY_INSPECTOR	`POST .../usage-decision`	200; 321 to unrestricted; GR stays QUALITY_HOLD; no ERP msg yet	200 {"status": "QUALITY_HOLD", "openLots": ["QL-2211"]}	**PASS**	moves=['321:QUALITY:-60.0', '321:UNRESTRICTED:60.0'] erp=[]
4e UD partial reject lot 2 (35 ok / 5 POROSITY) - last lot	QUALITY_INSPECTOR	`POST .../usage-decision`	200; GR PARTIALLY_REJECTED; RD created; 350+122; event quality.usage_decision; ERP GOODS_RECEIPT; CAPA	200 {"status": "PARTIALLY_REJECTED", "returnDelivery": "RD-4411", "openLots": []}	**PASS**	gr=PARTIALLY_REJECTED moves=['321:QUALITY:-35.0', '321:UNRESTRICTED:35.0', '350:QUALITY:-5.0', '350:BLOCKED:5.0'] rdMoves=['122:BLOCKED:-5.0'] stock= {'0001/BLOCKED/B2': 0.0, '0001/BLOCKED/C1': 0.0, '0001/QUALITY/B1': 0.0, '0001/QUALITY/B2': 0.0, '0001/QUALITY/C1': 0.0, '0001/UNRESTRICTED/B1': 60.0, '0001/UNRESTRICTED/B2': 35.0, '0001/UNRESTRICTED/C
5a request replacement on the return	QUALITY_INSPECTOR	`POST /v1/return-deliveries/{ref}/replacement`	200 REPLACEMENT_DUE; PO line reopened; event return_delivery.replacement_requested	200 {"status": "REPLACEMENT_DUE", "replacement_required": true}	**PASS**	events=['return_delivery.replacement_requested(unpublished)'] poLine=[{'qty': '100.000', 'received_qty': '95.000', 'rejected_qty': '0.000', 'accepted_qty': '95.000'}] po=PARTIALLY_RECEIVED
5b replacement ASN for the 5 returned	SUPPLIER_USER	`POST /v1/asns`	201 IN_TRANSIT	201 {"number": "ASN-61209", "status": "IN_TRANSIT"}	**PASS**	
5c GR of the replacement	RECEIVER	`POST /v1/goods-receipts`	201 QUALITY_HOLD	201 {"number": "GR-90239", "status": "QUALITY_HOLD"}	**PASS**	
5d UD accept replacement fully	QUALITY_INSPECTOR	`POST .../usage-decision`	200; GR ACCEPTED; ERP GOODS_RECEIPT	200 {"status": "ACCEPTED"}	**PASS**	gr=ACCEPTED erp=[{'message_type': 'GOODS_RECEIPT', 'idempotency_key': 'GR|GR-90239', 'status': 'POSTED', 'erp_doc_no': '5000071239', 'attempts': 1}]
5e return delivery closes after the replacement arrived (C1)	-	`DB pic.return_delivery`	status closed (e.g. REPLACED/CLOSED)	[{'number': 'RD-4411', 'status': 'REPLACEMENT_DUE', 'replacement_required': True, 'credit_note_expected': False}] po=RECEIVED	**PARTIAL**	return delivery never leaves REPLACEMENT_DUE: no command or consumer closes it
5f reverse a GR whose lot was rejected and returned	RECEIVER	`POST /v1/goods-receipts/{ref}/reverse`	409/422 refused (use return)	409 ILLEGAL_TRANSITION Goods receipt GR-90231 is PARTIALLY_REJECTED; 'reverse' is not allowed from this state	**PASS**	

Inventory

Step	Actor	API call	Expected	Actual	Verdict	Evidence
1 read /v1/stock	INVENTORY_CONTROLLER	`GET /v1/stock?material=FLB-INV&plant=P100`	200 with data	200 items=1	**PASS**	{'items': [{'id': '01a1b6a7-c896-77d9-80dd-4c8cdc1b276d', 'material': 'FLB-INV', 'description': 'Audit material FLB-INV', 'uom': 'EA', 'materialGroup': 'GENERAL', 'plant': 'P100', 'storageLocation': '
1 read /v1/stock/summary	INVENTORY_CONTROLLER	`GET /v1/stock/summary`	200 with data	200 items=None	**PASS**	{'asOf': '2026-11-16', 'plant': None, 'plants': ['P100'], 'byType': {'CONSIGNMENT': {'rows': 1, 'qty': '38', 'valueMinor': 760000}, 'UNRESTRICTED': {'rows': 9, 'qty': '600', 'valueMinor': 11200000}},
1 read /v1/stock/alerts	INVENTORY_CONTROLLER	`GET /v1/stock/alerts`	200 with data	200 items=None	**PASS**	{'items': [], 'total': 0, 'nextCursor': None, 'countsByType': {}}
1 read /v1/stock/materials/FLB-INV	INVENTORY_CONTROLLER	`GET /v1/stock/materials/FLB-INV?plant=P100`	200 with data	200 items=None	**PASS**	{'material': {'number': 'FLB-INV', 'description': 'Audit material FLB-INV', 'uom': 'EA', 'type': 'STOCK', 'materialGroup': 'GENERAL', 'mrpType': 'PD', 'status': 'ACTIVE', 'batchManaged': False, 'expir
1 read /v1/stock/movements	INVENTORY_CONTROLLER	`GET /v1/stock/movements?material=FLB-INV`	200 with data	200 items=1	**PASS**	{'items': [{'id': '01a1b6a7-c8ab-7651-92a4-fe33f5da995c', 'createdAt': '2026-11-13T04:30:00+00:00', 'material': 'FLB-INV', 'description': 'Audit material FLB-INV', 'uom': 'EA', 'plant': 'P100', 'stora

Step	Actor	API call	Expected	Actual	Verdict	Evidence
1 movements CSV export	INVENTORY_CONTROLLER	`GET /v1/stock/movements/export`	200 text/csv; audited	200 text/csv; charset=utf-8	**PASS**	Posted at,Material,Description,UoM,Plant,Storage location,Stock type,Batch,Movement,Movement text,Quantity,Value (INR),Reason code,Posting ref,Referen
2 transfer 10 (Rs 1,000 < auto-post limit)	RECEIVER	`POST /v1/stock-postings submit=true`	201 POSTED at once; 311; ERP STOCK_POSTING STP|...; event stock_posting.posted	201 {"number": "STP-1001", "status": "POSTED"}	**PASS**	moves=['311:UNRESTRICTED:-10.0', '311:UNRESTRICTED:10.0'] erp=[{'message_type': 'STOCK_POSTING', 'idempotency_key': 'STP|STP-1001', 'status': 'POSTED', 'erp_doc_no': '4900000001', 'attempts': 1}] events=['stock_posting.submitted(unpublished)', 'stock_posting.approved(unpublished)', 'stock_posting.posted(unpublished)'] stock={'0001/UNRESTRICTED/':
3a block 5	RECEIVER	`POST /v1/stock-postings BLOCK`	POSTED; 344; BLOCKED 5	201 {"number": "STP-1002", "status": "POSTED"}	**PASS**	moves=['344:UNRESTRICTED:-5.0', '344:BLOCKED:5.0'] erp=[{'message_type': 'STOCK_POSTING', 'idempotency_key': 'STP|STP-1002', 'status': 'POSTED', 'erp_doc_no': '4900000002', 'attempts': 1}]
3b unblock 5	RECEIVER	`POST /v1/stock-postings UNBLOCK`	POSTED; 343; BLOCKED 0	201 {"number": "STP-1003", "status": "POSTED"}	**PASS**	moves=['343:BLOCKED:-5.0', '343:UNRESTRICTED:5.0']
3c release from quality via posting	RECEIVER	`POST /v1/stock-postings RELEASE_QUALITY`	422 USE_USAGE_DECISION	422 USE_USAGE_DECISION Stock in quality inspection is released (or rejected) by the usage decision of its inspection lot (POST /v1/inspection-lots/{lot}/usage-decision); a stock posti	**PASS**	
3d transfer more than on hand	RECEIVER	`POST /v1/stock-postings`	422 INSUFFICIENT_STOCK	422 INSUFFICIENT_STOCK FLB-INV: only 290.000 unrestricted at P100/0001; 5000 needed; the stock is in batch(es) (none): name the batch	**PASS**	
4a scrap 3 created and submitted (SCRAP never auto-posts)	INVENTORY_CONTROLLER	`POST /v1/stock-postings; POST .../submit`	SUBMITTED; approval task for INVENTORY_CONTROLLER excl. maker	201 {"number": "STP-1004", "status": "DRAFT"} / 200 {"status": "SUBMITTED"}	**PASS**	tasks=[{'kind': 'STOCK_POSTING_APPROVAL', 'candidate_role': 'INVENTORY_CONTROLLER', 'status': 'OPEN', 'decision': None, 'sequence': 1}]
4b maker approves own scrap	INVENTORY_CONTROLLER (maker)	`POST /v1/stock-postings/{ref}/approve`	409 SOD_CONFLICT SOD-10	409 SOD_CONFLICT SOD-10: Whoever submitted a stock posting may not approve it (maker-checker)	**PASS**	
4c receiver approves	RECEIVER	`POST .../approve`	403	403 FORBIDDEN Missing scope(s): inventory:approve	**PASS**	
4d approve with stale If-Match	INVENTORY_CONTROLLER	`POST .../approve If-Match stale`	412 STALE_WRITE	412 STALE_WRITE The document is at version 3; your If-Match is 999. Reload it and retry.	**PASS**	
4e approve by second controller	INVENTORY_CONTROLLER	`POST .../approve`	200 APPROVED; task completed	200 {"status": "APPROVED"}	**PASS**	tasks=[{'kind': 'STOCK_POSTING_APPROVAL', 'candidate_role': 'INVENTORY_CONTROLLER', 'status': 'COMPLETED', 'decision': 'APPROVE', 'sequence': 1}]
4f post scrap	RECEIVER	`POST /v1/stock-postings/{ref}/post`	POSTED; 551; ERP STOCK_POSTING; event	200 {"status": "POSTED"}	**PASS**	moves=['551:UNRESTRICTED:-3.0'] erp=[{'message_type': 'STOCK_POSTING', 'idempotency_key': 'STP|STP-1004', 'status': 'POSTED', 'erp_doc_no': '4900000004', 'attempts': 1}] events=['stock_posting.approved(unpublished)', 'stock_posting.posted(unpublished)', 'stock_posting.submitted(unpublished)', 'task.assigned(unpublished)'] audit=['CREATED', 'SUBMIT
5a count created, book frozen	INVENTORY_CONTROLLER	`POST /v1/physical-inventories`	201 PLANNED, frozenAt	201 {"number": "PHI-1001", "status": "PLANNED", "frozenAt": "2026-11-16T11:00:00+05:30"}	**PASS**	lines=[{'lineNo': 10, 'material': 'FLB-INV', 'description': 'Audit material FLB-INV', 'uom': 'EA', 'batch': '', 'stockType': 'UNRESTRICTED', 'unlisted': False, 'countStatus': 'PENDING', 'countedQty': None,
5b counter sees a blind count	RECEIVER	`GET /v1/physical-inventories/{ref}`	blind=true, no book qty	blind=True line={'lineNo': 10, 'material': 'FLB-INV', 'description': 'Audit material FLB-INV', 'uom': 'EA', 'batch': '', 'stockType': 'UNRESTRICTED', 'unlisted': False, 'countStatus': 'PENDING', 'countedQty': None, '	**PASS**	
5c blind count 270 (book 282: outside 2%)	RECEIVER	`PUT .../counts`	COUNTING; recount required	200 {"status": "COUNTING"}{'lineNo': 10, 'material': 'FLB-INV', 'description': 'Audit material FLB-INV', 'uom': 'EA', 'batch': '', 'stockType': 'UNRESTRICTED', 'unlisted': False, 'countStatus': 'RECOUNT_REQUIRED', 'countedQty	**PASS**	
5d recount by the same counter	RECEIVER	`PUT .../counts`	403 RECOUNT_SAME_COUNTER	403 RECOUNT_SAME_COUNTER FLB-INV: the recount is made by someone other than the first counter	**PASS**	

Step	Actor	API call	Expected	Actual	Verdict	Evidence
5e recount by a second counter	RECEIVER	`PUT .../counts`	COUNTED	200 {"status": "COUNTED"}	**PASS**	
5f submit count	RECEIVER	`POST .../submit`	PENDING_APPROVAL; approval task	200 {"status": "PENDING_APPROVAL"}	**PASS**	tasks=[{'kind': 'PHYSICAL_INVENTORY_APPROVAL', 'candidate_role': 'INVENTORY_CONTROLLER', 'status': 'OPEN', 'decision': None, 'sequence': 1}]
5g counter approves	RECEIVER	`POST .../approve`	403 (no inventory:approve; SOD-11)	403 FORBIDDEN Missing scope(s): inventory:approve	**PASS**	
5h approve count difference	INVENTORY_CONTROLLER(/FINANCE)	`POST .../approve`	APPROVED	200 {"status": "APPROVED"}	**PASS**	tasks=[{'kind': 'PHYSICAL_INVENTORY_APPROVAL', 'candidate_role': 'INVENTORY_CONTROLLER', 'status': 'COMPLETED', 'decision': 'APPROVE', 'sequence': 1}]
5i post differences	INVENTORY_CONTROLLER	`POST .../post`	POSTED; 702 -10; ERP STOCK_POSTING PHI|...; event physical_inventory.posted	200 {"status": "POSTED"}	**PASS**	moves=['702:UNRESTRICTED:-15.0'] erp=[{'message_type': 'STOCK_POSTING', 'idempotency_key': 'PHI|PHI-1001', 'status': 'POSTED', 'erp_doc_no': '4900000005', 'attempts': 1}] events=['task.assigned(unpublished)', 'physical_inventory.posted(unpublished)'] stock={'0001/BLOCKED/': 0.0, '0001/UNRESTRICTED/': 272.0}
5j SOD-11: create a counter+approver user	TENANT_ADMIN	`POST /v1/admin/users`	409 SOD_CONFLICT SOD-11	422 VALIDATION_FAILED Role combination not allowed: SOD-11 (counting stock and approving the count differences)	**PASS**	control works; answered 422 VALIDATION_FAILED, the guide (P3) documents 409 SOD_CONFLICT (cosmetic mismatch)
6a reservation 20	RECEIVER	`POST /v1/reservations`	201 OPEN; RESERVATION demand 20	201 {"number": "RSV-1001", "status": "OPEN"}	**PASS**	demand=[{'reference': 'RSV-1001', 'qty': '20.000', 'type': 'RESERVATION'}] events=['stock_reservation.created(unpublished)']
6b MRP simulation sees the reservation	MRP_CONTROLLER	`POST /v1/mrp/runs SIMULATION`	gross includes 20	201 gross=20.000 on_hand=282.000	**PASS**	{'material': 'FLB-INV', 'plant': 'P100', 'gross': '20.000', 'on_hand': '282.000', 'open_po': '0', 'in_transit': '0', 'firm_receipts': '0', 'safety_stock': '0', 'net': '0', 'proposals': [], 'exceptions
6c partial issue 8	RECEIVER	`POST /v1/reservations/{ref}/issue`	PARTIALLY_ISSUED; 201 -8; ERP RSV|..|8	200 {"status": "PARTIALLY_ISSUED", "issuedQty": "8", "openQty": "12"}	**PASS**	moves=['201:UNRESTRICTED:-8.0'] erp=[{'message_type': 'STOCK_POSTING', 'idempotency_key': 'RSV|RSV-1001|8', 'status': 'POSTED', 'erp_doc_no': '4900000006', 'attempts': 1}]
6d cancel the rest	RECEIVER	`POST /v1/reservations/{ref}/cancel`	CANCELLED; demand 0; MRP gross 0	200 {"status": "CANCELLED", "openQty": "12"}	**PASS**	demand=[{'qty': '0.000'}] mrpGross=0.000 events= ['stock_reservation.issued(unpublished)', 'stock_reservation.created(unpublished)', 'stock_reservation.cancelled(unpublished)']
7 M09 inventory KPIs	INVENTORY_CONTROLLER	`GET /v1/kpis?module=M09`	count accuracy and adjustment value computed	{'M09.DOCK_TO_STOCK': '12.0', 'M09.INVENTORY_TURNS': '0.1', 'M09.QUALITY_REJECTION': '0.9', 'M09.RECEIPT_ACCURACY': '100.0', 'M09.SES_APPROVAL': '0.0', 'M09.STOCK_ADJUSTMENT_VALUE': '180000', 'M09.STOCK_COUNT_ACCURACY': '0.0'}	**PASS**	

F7 Invoice-to-pay

Step	Actor	API call	Expected	Actual	Verdict	Evidence
1a capture e-invoice (portal, valid IRN) 100 x 100 on received PO	SUPPLIER_USER	`POST /v1/invoices channel E_INVOICE`	201 MATCHED 3WAY first-pass; irn_valid; event invoice.captured	201 {"number": "SSPL-A-0001", "status": "MATCHED", "match_type": "3WAY", "first_pass_matched": true, "irn_valid": true}	**PASS**	events=['invoice.captured(unpublished)'] audit= ['CAPTURED', 'E_INVOICE_VERIFIED', 'CHECK_DUPLICATES', 'MATCH', 'MATCHED'] steps=['F7-01:COMPLETED', 'F7-02:COMPLETED', 'F7-03:COMPLETED', 'F7-04:COMPLETED']
1b exact duplicate (same number)	SUPPLIER_USER	`POST /v1/invoices`	409 DUPLICATE_INVOICE	409 DUPLICATE_INVOICE SSPL-A-0001 from FLB-STD is already captured as SSPL-A-0001	**PASS**	
1c same IRN on another number	SUPPLIER_USER	`POST /v1/invoices`	409 DUPLICATE_IRN	409 DUPLICATE_IRN IRN 9f978dcf7c0bf6089c75b49445b3b34ac4aff3cd4f2a9a9fbc20885c00c439cb is already on invoice SSPL-A-0001	**PASS**	
1d OCR variant of the same number (DMN-03)	AP_CLERK	`POST /v1/invoices PDF_OCR`	201 EXCEPTION DUPLICATE, payment block	201 {"number": "sspl a 0001", "status": "EXCEPTION", "payment_block": true} exc= ['DUPLICATE']	**PASS**	

Step	Actor	API call	Expected	Actual	Verdict	Evidence
1e duplicate hold: re-match refused, reject	AP_CLERK	`POST .../match; POST .../reject`	409 DUPLICATE_HOLD; REJECTED	409 DUPLICATE_HOLD sspl a 0001 is held as a possible duplicate of SSPL-A-0001: resolve the hold (not a duplicate) or reject the invoice / 200 {"status": "REJECTED"}	**PASS**	
1f EDI INVOIC 380 through the gateway	TENANT_ADMIN (integration client)	`POST /v1/edi/inbound`	ACCEPTED; CONTRL; invoice MATCHED	200 status=ACCEPTED invoice=[{'number': 'SSPL-E-0001', 'status': 'MATCHED'}]	**PASS**	{'interchange': '7770001', 'sender': 'FLBSTDEDI', 'status': 'ACCEPTED', 'messages': [{'type': 'INVOIC', 'messageRef': '1', 'reference': '7770001-1', 'document': 'SSPL-E-0001', 'outcome': 'ACCEPTED', 'documentType': 'INVOICE', 'object': 'INVOICE', 'nu
1g EDI re-delivery	TENANT_ADMIN	`POST /v1/edi/inbound (same interchange)`	replays stored ack, no new invoice	200 replayed=True	**PASS**	
2a invoice 20 where 18 accepted (4-way)	SUPPLIER_USER	`POST /v1/invoices`	DMN-03 duplicate check runs before the match	201 {"number": "SSPL-Q-0001", "status": "EXCEPTION", "match_type": null, "payment_block": true} exc=[('DUPLICATE', {'similarTo': 'SSPL-A-0001'})]	**PASS**	held as DUPLICATE of SSPL-A-0001: same supplier, same gross 11,800.00, same date (rule DMN-03 amount/date window) - correct per rule; the 4-way quantity path continues in 2a'
2b supplier credit note for 2	SUPPLIER_USER	`POST /v1/invoices/{ref}/credit-notes`	201; event credit_note.received	201 {"number": "SSPL-CN-0001", "gross_minor": 118000}	**PASS**	events=['credit_note.received(unpublished)']
2c PO creator resolves	BUYER	`POST .../exceptions/{id}/resolve`	403 (no scope) / SOD-04	403 FORBIDDEN Missing scope(s): invoice:resolve	**PASS**	
2d0 resolution type not allowed for a DUPLICATE exception	AP_CLERK	`POST .../exceptions/{id}/resolve`	422 INVALID_RESOLUTION	422 INVALID_RESOLUTION A DUPLICATE exception is resolved with DISPUTE, NOT_DUPLICATE inv=EXCEPTION block=True	**PASS**	
3a price 103 vs PO 100 (x50)	SUPPLIER_USER	`POST /v1/invoices`	EXCEPTION PRICE (variance 150 > tolerance 50)	201 {"number": "SSPL-P-0001", "status": "EXCEPTION"} exc=[('PRICE', {'perUnitMinor': 300, 'lineVarianceMinor': 15000})]	**PASS**	
3b AP proposes PRICE_ACCEPTED	AP_CLERK	`POST .../resolve PRICE_ACCEPTED`	PRICE_VARIANCE task for a buyer other than PO creator	200 {} tasks(buyer2)=1 tasks(buyer/PO creator)=0	**PASS**	tasks=[{'kind': 'PRICE_VARIANCE', 'candidate_role': 'BUYER', 'status': 'OPEN', 'decision': None, 'sequence': 1}]
3c buyer approves the variance in the task inbox	BUYER	`POST /v1/tasks/{id}/complete`	PO re-priced 103; invoice MATCHED	200 {} inv=MATCHED poPrice=[{'unit_price_minor': 10300}]	**PASS**	audit=['PRICE_VARIANCE_ACCEPTED']
4a invoice before goods receipt	AP_CLERK	`POST /v1/invoices PDF_OCR`	EXCEPTION MISSING_GR	201 {"number": "SSPL-M-0001", "status": "EXCEPTION"} exc=['MISSING_GR']	**PASS**	
4b GR posted, resolve GR_POSTED	AP_CLERK	`POST .../resolve GR_POSTED`	MATCHED	200 {} inv=MATCHED	**PASS**	
5a supplier finance data: MICRO, Udyam, TDS 194C-OTH	SUPPLIER_MANAGER	`PUT /v1/suppliers/{code}/finance (If-Match)`	200 isMsme	200 {"msmeClass": "MICRO", "tdsSection": "194C-OTH"}	**PASS**	
5b MSME invoice 400 x 100	SUPPLIER_USER	`POST /v1/invoices`	MATCHED	201 {"number": "MTW-0001", "status": "MATCHED"}	**PASS**	
6a capturer approves own invoice	AP_CLERK (capturer)	`POST /v1/invoices/{ref}/approve`	409 SOD_CONFLICT SOD-08	409 SOD_CONFLICT SOD-08: Whoever captured an invoice may not approve it (four-eyes)	**PASS**	
6b approve with stale If-Match	AP_CLERK	`POST .../approve If-Match stale`	412 STALE_WRITE	412 STALE_WRITE The document is at version 7; your If-Match is 999. Reload it and retry.	**PASS**	
6c buyer approves an invoice	BUYER	`POST .../approve`	403	403 FORBIDDEN Missing scope(s): invoice:approve	**PASS**	
6d approvals by a different user (AP / finance)	AP_CLERK, FINANCE_CONTROLLER	`POST /v1/invoices/{ref}/approve`	200 APPROVED each	{'inv_a': 'ap2:200:APPROVED', 'inv_q': 'ap2:409:409', 'inv_m': 'ap2:200:APPROVED', 'inv_e': 'ap1:200:APPROVED', 'inv_ms': 'ap1:200:APPROVED', 'inv_p': 'fin2:200:APPROVED'}	**PASS**	| SSPL-Q-0001 was still on its duplicate hold (409); approved and posted after the hold was released (step 6d/7c')
7a post without Idempotency-Key	AP_CLERK	`POST /v1/invoices/{ref}/post`	428 IDEMPOTENCY_KEY_REQUIRED	428 IDEMPOTENCY_KEY_REQUIRED Send Idempotency-Key: tenant|company code|supplier|invoice number	**PASS**	

Step	Actor	API call	Expected	Actual	Verdict	Evidence
7b post with a wrong key	AP_CLERK	`POST .../post`	409/422	422 RULE_VIOLATION Posting key must be tenant|company code|supplier|invoice number (FLB|1000|FLB-STD|SSPL-A-0001; percent-encoded when it contains spaces or non-ASCII characters)	**PASS**	
7c post approved invoices to ERP (posting key)	AP_CLERK	`POST /v1/invoices/{ref}/post Idempotency-Key=tenant|cc|supplier|no`	POSTED, ERP doc, due date; replay returns same doc; event invoice.posted	{'inv_a': '200:POSTED:5100004422', 'inv_q': '409:409:None', 'inv_m': '200:POSTED:5100004423', 'inv_e': '200:POSTED:5100004424', 'inv_ms': '200:POSTED:5100004425', 'inv_p': '200:POSTED:5100004426'} replay=True	**PASS**	inv_a due=2026-12-17 erp=[{'message_type': 'SUPPLIER_INVOICE', 'idempotency_key': 'FLB|1000|FLB-STD|SSPL-A-0001', 'status': 'POSTED', 'erp_doc_no': '5100004422', 'attempts': 1}] events=['invoice.captured', 'invoice.posted'] steps=['F7-01:COMPLETED', 'F7-02:COMPLETED', 'F7-03:COMPLETED', 'F7-04:COMPLETED', 'F7-06:COMPLETED', 'F7-07:ACTIVE', 'F7-0
7d MSME/TDS at posting	-	`GET /v1/invoices/{ref}`	TDS 2% of 40,000 = 800; due date capped (acceptance + 45d)	tds=194C-OTH base=4000000 wht=80000 due=2026-12-31 capped=True	**PASS**	
7e GR reversal after invoicing	RECEIVER	`POST /v1/goods-receipts/{ref}/reverse`	409 RECEIPT_INVOICED	409 RECEIPT_INVOICED GR-90235 cannot be reversed: invoice SSPL-A-0001 (posted) references its order lines. Reject the invoice first, or use a return delivery for goods already invoi	**PASS**	
2a' DMN-03 amount/date duplicate hold released NOT_DUPLICATE, re-match	AP_CLERK	`POST .../resolve NOT_DUPLICATE`	re-match: credit note already nets the 2 rejected -> MATCHED 4WAY (or QUANTITY if not netted)	200 {} inv=MATCHED match=4WAY open=[]	**PASS**	SSPL-Q-0001 and SSPL-A-0001 have the same gross 11,800 and date: DMN-03 hold is correct
6d/7c' approve and post SSPL-Q-0001	AP_CLERK	`POST .../approve; POST .../post`	APPROVED; POSTED	200 {"status": "APPROVED"} / 200 {"status": "POSTED", "erpDocument": "5100004427"}	**PASS**	
2e return delivery after its credit note and invoice posting (C1)	-	`DB pic.return_delivery`	closed / CREDITED	[{'number': 'RD-4410', 'status': 'CREATED'}]	**PARTIAL**	credit note received, invoice posted, return still CREATED
8a custom role holding payment:run releases a block (C4)	PAY_DESK (custom)	`POST /v1/invoices/{ref}/payment-block/release`	allowed by scope (Blueprint) / refused by role-name check (C4)	403 FORBIDDEN A payment block is lifted by finance	**BROKEN**	service checks 'FINANCE_CONTROLLER' in roles (payables.release_payment_block)
8b finance lifts the payment block	FINANCE_CONTROLLER	`POST .../payment-block/release`	200 block off; audit PAYMENT_BLOCK_LIFTED	200 {"payment_block": false}	**PASS**	
9a early-payment (terms 2/10 N30)	AP_CLERK	`POST /v1/invoices/{ref}/early-payment/offers TERMS`	201 ACCEPTED, discount 2%, due date = pay-on	201 {"status": "ACCEPTED", "discountMinor": 23600} eligible=True	**PASS**	
9b proposal by AP	AP_CLERK	`GET /v1/payments/proposal`	403	403 FORBIDDEN Missing scope(s): payment:run	**PASS**	
9c proposal by finance	FINANCE_CONTROLLER	`GET /v1/payments/proposal`	items with amount = gross - discount - TDS	[{'invoice': 'SSPL-A-0001', 'supplier': 'FLB-STD', 'supplierName': 'Standard Steel Pvt Ltd', 'dueDate': '2026-11-19', 'overdueDays': 0, 'msme': False, 'msmeCapped': False, 'grossMinor': 1180000, 'creditNotesMinor': 0, 'debitNotesMinor': 0, 'retention	**PASS**	
9d run with an empty selection	FINANCE_CONTROLLER	`POST /v1/payments/runs`	422 EMPTY_SELECTION	422 EMPTY_SELECTION The selection of suppliers or invoices is empty: leave the filter out to pay everything that is due, or name what to pay	**PASS**	
9e payment run 1	FINANCE_CONTROLLER	`POST /v1/payments/runs`	PAY COMPLETED, ERP PAYMENT PAY|..., invoice PAID	{'run': 'PRUN-2611', 'paid': [{'invoice': 'SSPL-A-0001', 'payment': 'PAY-880001', 'amountMinor': 1156400, 'clearing': '1500002210', 'integrationStatus': 'POSTED', 'dueDate': '2026-11-19', 'tdsMinor': 0, 'discountMinor': 23600, 'tcsMinor': 0, 'tdsCertificate': None, 'tdsCertBaseMinor': 0, 'tdsBaseMin	**PASS**	inv=PAID/PAID erp=[{'message_type': 'PAYMENT', 'idempotency_key': 'PAY|PAY-880001', 'status': 'POSTED', 'erp_doc_no': '1500002210', 'attempts': 1}] events=[]
9f bank file pain.001	FINANCE_CONTROLLER	`POST /v1/payments/runs/{run}/file`	201 GENERATED; pain.001 content with the payment	201 {"number": "PF-1001", "status": "GENERATED", "sha256": "ff0f3a5834e603d95db33f26533d562fab5398207ee9a5138249666de7dd7e2a"}	**PASS**	<CstmrCdtTrfI
9g bank statement camt.053 import (dry run, apply, re-import)	FINANCE_CONTROLLER	`POST /v1/bank-statements/import? dry_run=true|false`	dry run READY; apply credits PAY with UTR; re-import idempotent	dry=200 {'statementId': 'FLB-STMT-001', 'format': 'CAMT053', 'account': '50200011223344', 'entries': 1, 'counts': {'UNCHANGED': 1}, 'items': [{'line': 1, 'amo apply=200 again=200 pay=[{'bank_status': 'CREDITED', 'bank_utr': 'HDFCUTR001'}]	**PASS**	{'items': [{'file': 'PF-1001', 'line': 1, 'batch': False, 'entry': 'HDFCUTR001', 'state': 'READY', 'action': 'CREDITED', 'detail': 'will be recorded', 'reason': None, 'bankRef': 'HDFCUTR001', 'payment

Step	Actor	API call	Expected	Actual	Verdict	Evidence
10a supplier debit note (freight charge)	SUPPLIER_USER	`POST /v1/invoices/{ref}/debit-notes`	201 PENDING_APPROVAL; buyer-chain task	201 {"number": "DN-FLB-STD-SSPL-DN-0001", "status": "PENDING_APPROVAL", "grossMinor": 59000}	**PASS**	tasks=[{'kind': 'DEBIT_NOTE_APPROVAL', 'candidate_role': 'BUYER', 'status': 'OPEN', 'decision': None, 'sequence': 1}]
10b buyer chain, AP approve, post	BUYER, AP_CLERK	`POST /v1/tasks/..; POST /v1/debit-notes/{ref}/approve; .../post`	APPROVED, POSTED, ERP SUPPLIER_DEBIT_MEMO	200 {"status": "APPROVED"} / 200 {"status": "POSTED", "erpDocument": "5106000001"}	**PASS**	erp=[{'message_type': 'SUPPLIER_DEBIT_MEMO', 'idempotency_key': 'FLB|1000|FLB-STD|SSPL-A-0001|DN|DN-FLB-STD-SSPL-DN-0001', 'status': 'POSTED', 'erp_doc_no': '5106000001', 'attempts': 1}] events= ['debit_note.received(unpublished)', 'task.assigned(unpublished)', 'debit_note.posted(unpublished)']
11a USD invoice	SUPPLIER_USER	`POST /v1/invoices`	MATCHED in USD	201 {"number": "OAP-F-0001", "status": "MATCHED", "currency": "USD", "gross_minor": 14160}	**PASS**	
11b post without an FX rate	AP_CLERK	`POST /v1/invoices/{ref}/post`	409 FX_RATE_MISSING	409 FX_RATE_MISSING No USD->INR exchange rate is valid on 2026-11-27: load it (ADM reference data, fx_rate) before posting	**PASS**	
12a close a period that has not ended	FINANCE_CONTROLLER	`POST /v1/finance/periods/2026-12/close`	409 PERIOD_NOT_ENDED	409 PERIOD_NOT_ENDED Period 2026-12 has not ended yet: it can be closed from 2027-01-01	**PASS**	
12b close November with cut-off and GR/IR snapshot	FINANCE_CONTROLLER	`POST /v1/finance/periods/2026-11/close; GET .../grir-snapshots`	CLOSED; cutoff; GR/IR lines; event finance.period.closed	200 {"status": "CLOSED"} cutoff={'periodEnd': '2026-11-30', 'unpostedInvoices': 1, 'unpostedGrossMinor': 14160, 'unpostedByStatus': {'APPROVED': 1}, 'grirAccrualMinor': 9512000, 'gri snapshots=1	**PASS**	preview={'periodEnd': '2026-11-30', 'unpostedInvoices': 1, 'unpostedGrossMinor': 14160, 'unpostedByStatus': {'APPROVED': 1}, 'grirAccrualMinor': 9512000} snap0={'id': '01a1b6b0-e70f-7744-8d78-12ba61f77329', 'companyCode': '1000', 'period': '2026-11', 'closedAt': '2026-12-02T04:30:00+00:00', 'totalMinor': 9512000, 'lineCount': 6, 'lines': [{'po': 'P
12c post dated into the closed period	AP_CLERK	`POST /v1/invoices/{ref}/post`	409 PERIOD_CLOSED + audit POSTING_REFUSED	409 PERIOD_CLOSED Posting period 2026-11 of company code 1000 is closed: invoice OAP-F-0001 cannot be posted or dated into it. Reopen rule: finance (scope period:close) reopens t	**PASS**	
11c FX rate maintained, post in December	TENANT_ADMIN, AP_CLERK	`POST /v1/admin/fx-rates; POST .../post`	POSTED; fx_rate fixed; group amount INR	201 {} / 200 {"status": "POSTED"} fx=84.1 grossGroup=1190856 posting=2026-12-02	**PASS**	
12d reopen (reason >= 10 chars)	FINANCE_CONTROLLER	`POST /v1/finance/periods/2026-11/reopen`	422 short reason; 200 OPEN	422 VALIDATION_FAILED reason: String should have at least 10 characters / 200 {"status": "OPEN"}	**PASS**	
13a due run by fin2 (approved INV-P)	FINANCE_CONTROLLER	`POST /v1/payments/runs`	INV-P skipped SOD-09; others paid; late debit note paid	paid=['ERS-FLB-CONS-202611', 'SSPL-E-0001', 'SSPL-M-0001', 'SSPL-Q-0001'] skipped= {'SSPL-P-0001': 'SOD-09: Whoever approved an invoice may not release its payment'}	**PASS**	proposal=['ERS-FLB-CONS-202611', 'SSPL-E-0001', 'SSPL-M-0001', 'SSPL-P-0001', 'SSPL-Q-0001'] skippedProp=[]
13b second run by fin pays INV-P	FINANCE_CONTROLLER	`POST /v1/payments/runs`	INV-P paid	paid=['SSPL-P-0001'] skipped=[]	**PASS**	
13c NEFT file for run 2	FINANCE_CONTROLLER	`POST /v1/payments/runs/{run}/file NEFT_CSV`	201 GENERATED	201 {"number": "PF-1002", "status": "GENERATED"}	**PASS**	
13d camt.053 batch entry + returned credit	FINANCE_CONTROLLER	`POST /v1/bank-statements/import`	batch split per payment CREDITED; returned: invoice POSTED/UNPAID, PAYMENT_REVERSAL PAYREV|..., event payment.returned	dry={'statementId': 'FLB-STMT-002', 'format': 'CAMT053', 'account': '50200011223344', 'entries': 4, 'counts': {'READY': 4}, 'items': [{'line': 1, 'amountMinor': 283200, 'bankRef': 'UTRB0', 'valueDate': '2 apply=200 returnedInv=POSTED/UNPAID	**PASS**	erp=[{'message_type': 'PAYMENT_REVERSAL', 'idempotency_key': 'PAYREV|PAY-880005', 'status': 'POSTED', 'erp_doc_no': '1500009000', 'attempts': 1}, {'message_type': 'PAYMENT', 'idempotency_key': 'PAY|PAY-880005', 'status': 'POSTED', 'erp_doc_no': '1500002214', 'attempts': 1}] events=[] pays=
13e ERP reconciliation	AP_CLERK	`POST /v1/reconciliation/runs`	no open differences (the PAYMENT_REVERSAL was posted by the ERP simulator, so no PAYMENT_RETURN_OPEN item)	201 kinds=[]	**PASS**	reconciliation run 201 with 0 items; pic.reconciliation_item empty; PAYREV|PAY-880005 POSTED
14 run with MSME (TDS withheld), USD (FX gain/loss) and the re-payable returned invoice	FINANCE_CONTROLLER	`POST /v1/payments/runs`	all three paid; TDS deducted; FX difference recorded	paid={'SSPL-Q-0001': (1062000, 0, None), 'OAP-F-0001': (14160, 0, None), 'MTW-0001': (4640000, 80000, None), 'SSPL-A-0001': (59000, None, None)} skipped=[]	**PASS**	DB pic.payment: PAY-880008 USD 141.60 fx_rate 84.60 amount_group 11,979.36 fx_gain_loss -70.80 (invoice rate 84.10); PAY-880009 MTW-0001 tds_minor 80,000 (Rs 800); PAY-880007 re-pays the returned SSPL-Q-0001; PAY-880010 the late debit note

Step	Actor	API call	Expected	Actual	Verdict	Evidence
15 GR/IR accrual read	FINANCE_CONTROLLER	`GET /v1/grir`	accepted-not-invoiced lines (e.g. qm2 +1 received 30 Nov)	200 list[5][{'po': 'PO-77416', 'line': 10, 'qty': '100.000', 'accrualMinor': 5000000}, {'po': 'PO-77420', 'line': 10, 'qty': '100.000', 'accrualMinor': 500000}, {'po': 'PO-77418', 'line': 10, 'qty': '2.000', 'accrualMinor': 400000}, {'po': 'PO-77424', 'line': 1	**PASS**	
16 M10 KPIs	FINANCE_CONTROLLER	`GET /v1/kpis?module=M10`	values computed	{'M10.DISCOUNT_CAPTURE': '25.5', 'M10.DPO': '35.7', 'M10.DUPLICATES_PREVENTED': '3', 'M10.EXCEPTION_AGEING': '0', 'M10.EXCEPTION_CYCLE': '6.8', 'M10.FIRST_PASS_MATCH': '75.0', 'M10.MSME_ON_TIME': '100.0', 'M10.TOUCHLESS_INVOICE': '42.9'}	**PASS**	
11-pre foreign-currency PO	BUYER	`POST /v1/purchase-orders (no currency field)`	a PO in USD	every new PO is INR; set to USD with a DB update for the test	**NOT BUILT**	services/order.py create: currency='INR' hard-coded (Appendix A B2, other auditor's module) - the foreign-currency invoice flow cannot start through the API

Spend feed

Step	Actor	API call	Expected	Actual	Verdict	Evidence
S1 posted invoices appear in the spend cube after the outbox drain	worker	`outbox invoice.posted -> spend-cube consumer; DB pic.spend_line`	one line per posted invoice line, period = posting month	missing=[]	**PASS**	{'ERS-FLB-CONS-202611': [('INVOICE', '', '2026-11', 240000, 240000)], 'MTW-0001': [('INVOICE', '', '2026-11', 4000000, 4000000)], 'SSPL-A-0001': [('INVOICE', '', '2026-11', 1000000, 1000000), ('DEBIT_NOTE', 'DN-FLB-STD-SSPL-DN-0001#1', '2026-11', 50000, 50000)], 'SSPL-E-0001': [('INVOICE', '', '2026-11', 300000, 300000)], 'SSPL-M-0001': [('INVOICE'
S2 credit note in spend with negative sign	worker	`credit_note.received -> spend-cube`	negative amount in the note's period	[('CREDIT_NOTE', 'SSPL-CN-0001', '2026-11', -100000, -100000)]	**PASS**	
S3 debit note in spend with positive sign	worker	`debit_note.posted -> spend-cube`	positive amount in the note's period	[('DEBIT_NOTE', 'DN-FLB-STD-SSPL-DN-0001#1', '2026-11', 50000, 50000)]	**PASS**	
S4 USD invoice spend in group currency	worker	`spend-cube`	amount_group_minor in INR at the posting rate	[('INVOICE', '', '2026-11', 12000, 1009200)]	**PASS**	
S5 spend cube read	CATEGORY_MANAGER	`GET /v1/spend/cube`	200 with the November/December spend	200 {}{'dimensions': ['category'], 'filters': {}, 'currency': 'INR', 'totalMinor': 8114200, 'contractedPct': '0.0', 'maverickPct': '34.7', 'totals': {'spendMinor': 8114200, 'docSpendMinor': 7117000, 'lines'	**PASS**	

F9 Savings

Step	Actor	API call	Expected	Actual	Verdict	Evidence
1 create initiative (SHOULD_COST baseline 550)	CATEGORY_MANAGER	`POST /v1/savings/initiatives`	201 IDENTIFIED; audit IDENTIFIED	201 {"number": "SV-0412", "stage": "IDENTIFIED", "baseline_price_minor": 55000}	**PASS**	audit=['IDENTIFIED']
2a another category manager records the negotiation (C4)	CATEGORY_MANAGER (not owner)	`POST /v1/savings/{ref}/negotiate`	403 owner only	403 FORBIDDEN Only the initiative owner records the negotiated price	**PASS**	owner-only rule confirmed at runtime
2b owner records the negotiated price 500	CATEGORY_MANAGER (owner)	`POST /v1/savings/{ref}/negotiate`	200 NEGOTIATED; ledger (550-500) x 1000 = Rs 50,000	200 {"stage": "NEGOTIATED"}	**PASS**	

Step	Actor	API call	Expected	Actual	Verdict	Evidence
3 realisation from the posted invoice (consumer savings-realised)	worker	`invoice.posted -> savings-realised; GET /v1/savings`	REALISED (550-500) x 100 = Rs 5,000; stage REALISING; event savings.realised; agent run	post=200 {"status": "POSTED"} ledger= [{'type': 'NEGOTIATED', 'amount_minor': 5000000, 'counts_toward_budget': False, 'source_ref': 'Rate agreed 2026-12-01'}, {'type': 'REALISED', 'amount_minor': 500000, 'counts_toward_budget': True, 'source_ref': 'INV-SAV-0001/10'}] stage=REALISING	**PASS**	events=['savings.realised'] agentRun=[{'action_taken': 'EXECUTED', 'output': {'realisedMinor': 500000}}]
4a validate without finance role	CATEGORY_MANAGER	`POST /v1/savings/{ref}/validate`	403	403 FORBIDDEN Missing scope(s): savings:validate	**PASS**	
4b owner (also finance) validates own initiative	CATEGORY_MANAGER+FINANCE_CONTROLLER	`POST .../validate`	409 SOD_CONFLICT SOD-07	409 SOD_CONFLICT SOD-07: Savings owner may not validate the same savings	**PASS**	
4c custom role holding savings:validate (C4)	SAV_DESK (custom)	`POST .../validate`	allowed by scope (Blueprint) / refused by role-name check (C4)	403 FORBIDDEN Savings are validated by finance	**BROKEN**	savings.validate checks 'FINANCE_CONTROLLER' in roles
4d validate with stale ETag	FINANCE_CONTROLLER	`POST .../validate If-Match stale`	412	412 STALE_WRITE The document is at version 4; your If-Match is 999. Reload it and retry.	**PASS**	
4e finance validates	FINANCE_CONTROLLER	`POST .../validate`	200 VALIDATED	200 {"stage": "VALIDATED"}	**PASS**	audit=['IDENTIFIED', 'NEGOTIATE', 'REALISE', 'VALIDATE']
5 P&L view counts validated realised savings	FINANCE_CONTROLLER	`GET /v1/savings/pnl?fiscal_year=2026-27`	December validated Rs 5,000; negotiated not counted	200 {'totals': {'realisedMinor': 500000, 'validatedMinor': 500000, 'pendingValidationMinor': 0, 'negotiatedMinor': 5000000, 'avoidanceMinor': 0, 'budgetImpactMinor': 500000, 'planTargetMinor': None, 'attainmentPct': None}, 'months': [{'period': '2026-04', 'realisedMinor': 0, 'validatedMinor': 0, 'pe	**PASS**	
6 read /v1/savings/SV-0412/waterfall	CATEGORY_MANAGER	`GET /v1/savings/SV-0412/waterfall`	200	200 {}	**PASS**	{"initiative":"SV-0412","stage":"VALIDATED","negotiatedMinor":5000000,"contractedMinor":5000000,"realisedMinor":500000,"avoidanceMinor":0,"leakag []],"budgetImpactMinor":500000,"entries":[{"type":"NE
6 read /v1/savings/pipeline	CATEGORY_MANAGER	`GET /v1/savings/pipeline`	200	200 {}	**PASS**	{"filters":{},"stages": [{"stage":"IDENTIFIED","count":0,"forecastMinor":0,"negotiatedMinor":0,"avoidanceMinor":0,"realisedMinor":0,"realisationPct":null}, {"stage":"NEGOTIATED","count":0,"forecastMinor
7 mark an initiative lost	CATEGORY_MANAGER	`POST /v1/savings/{ref}/lose`	LOST	200 {"stage": "LOST"}	**PASS**	

M12 AI & KPIs

Step	Actor	API call	Expected	Actual	Verdict	Evidence
1 Copilot: invoice INV-SAV-0001	AP_CLERK	`POST /v1/copilot/ask`	document lookup with evidence link	200 hit=True answer=INV-SAV-0001 is posted. Gross ₹59,000.00; 0 open exception(s); ERP document 5100004429. Workflow position: F7-08 "Realised savings & scorecard update" conf={'score': 0.95, 'label': 'high', 'note': 'Computed from live transactions'}	**PASS**	evidence=['INV-SAV-0001'] method={'method': 'document lookup + workflow history (SQL)', 'modelVersion': 'copilot-rules-v1', 'dataFreshness': '2026-12-05T
1 Copilot: invoice INV-SAV-0001	AP_CLERK	`POST /v1/copilot/ask`	document lookup with evidence link	200 hit=True answer=INV-SAV-0001 is posted. Gross ₹59,000.00; 0 open exception(s); ERP document 5100004429. Workflow position: F7-08 "Realised savings & scorecard update" conf={'score': 0.95, 'label': 'high', 'note': 'Computed from live transactions'}	**PASS**	evidence=['INV-SAV-0001'] method={'method': 'document lookup + workflow history (SQL)', 'modelVersion': 'copilot-rules-v1', 'dataFreshness': '2026-12-05T
1 Copilot: invoice SSPL-A-0001	AP_CLERK	`POST /v1/copilot/ask`	document lookup with evidence link	200 hit=False answer=I could not match that to a document, supplier or metric. Try: "status of PO-77410", "spend by supplier", "savings", "open invoice exceptions", "my ap conf={'score': 0.2, 'label': 'low', 'note': 'Partial match to your question; check the evidence'}	**BROKEN**	evidence=[] method={'method': 'intent routing (no match)', 'modelVersion': 'copilot-rules-v1', 'dataFreshness': '2026-12-05T10:00:00+05:30'
1 Copilot: invoice MTW-0001	AP_CLERK	`POST /v1/copilot/ask`	document lookup with evidence link	200 hit=False answer=I could not match that to a document, supplier or metric. Try: "status of PO-77410", "spend by supplier", "savings", "open invoice exceptions", "my ap conf={'score': 0.2, 'label': 'low', 'note': 'Partial match to your question; check the evidence'}	**BROKEN**	evidence=[] method={'method': 'intent routing (no match)', 'modelVersion': 'copilot-rules-v1', 'dataFreshness': '2026-12-05T10:00:00+05:30'
1b Copilot: supplier asks about another supplier's invoice	SUPPLIER_USER	`POST /v1/copilot/ask`	not disclosed	200 INV-SAV-0001 was not found or is not visible to you.	**PASS**	
1c Copilot: open invoice exceptions	FINANCE_CONTROLLER	`POST /v1/copilot/ask`	answer + confidence	200 There are no open invoice exceptions.	**PASS**	
1d Copilot answers logged as agent runs	-	`DB pic.agent_run`	>0 COPILOT runs	[{'n': 7}]	**PASS**	
2a agent catalogue	CATEGORY_MANAGER	`GET /v1/agents`	15 agents	200 n=15	**PASS**	
2b on-demand agent run SPEND_CLASSIFICATION	CATEGORY_MANAGER	`POST /v1/agents/{code}/runs`	200 run stored	200 {}{'agent': 'SPEND_CLASSIFICATION', 'result': {'auto': 0, 'review': 0, 'hits': 0}}	**PASS**	
2c on-demand run of an event-driven agent	AP_CLERK	`POST /v1/agents/INVOICE_EXCEPTION/runs`	422 NOT_RUNNABLE	422 NOT_RUNNABLE INVOICE_EXCEPTION runs on its trigger events; on-demand runs exist for DEMAND_SENSING, SOURCING_STRATEGY, SPEND_CLASSIFICATION, SUPPLIER_RISK	**PASS**	
2d INVOICE_EXCEPTION agent drafted on the match exceptions	agent	`DB pic.agent_run`	runs for SSPL-P-0001 / SSPL-M-0001	[{'action_taken': 'DRAFTED', 'object_ref': 'SSPL-P-0001'}, {'action_taken': 'DRAFTED', 'object_ref': 'SSPL-M-0001'}]	**PASS**	
3a kill switch by non-admin	CATEGORY_MANAGER	`PUT /v1/agents/{code}/kill-switch`	403	403 FORBIDDEN Missing scope(s): agents:manage	**PASS**	
3b kill switch on, run refused	TENANT_ADMIN, CATEGORY_MANAGER	`PUT .../kill-switch; POST .../runs`	200; 409 KILL_SWITCH; audit	200 {} / 409 KILL_SWITCH SPEND_CLASSIFICATION is switched off: kill switch is on (Classifier under review)	**PASS**	audit=[{'action': 'KILL_SWITCH_ON'}, {'action': 'RUN_REQUESTED'}]
3c kill switch off, run allowed	TENANT_ADMIN	`PUT .../kill-switch`	200; run 200	200 {} / 200 {}	**PASS**	
3d tenant AI switch off stops agents	TENANT_ADMIN	`PUT /v1/admin/ai/switch (If-Match)`	200; agent run refused; back on	200 {} / run=409 KILL_SWITCH SPEND_CLASSIFICATION is switched off: AI is switched off for this organisation: Tenant AI pause for audit / on=200 {}	**PASS**	
4 KPI explorer /v1/kpis?trend=false	CPO	`GET /v1/kpis?trend=false`	200 with values / registry	200 n=73	**PASS**	{"period":null,"plant":null,"items":[{"code":"M01.CLASSIFICATION_ACCURACY","module":"M01","name":"Classification accuracy %","unit":"PCT","direction":"HIGHER_BETTER","formula":"(auto-classified + stew

Step	Actor	API call	Expected	Actual	Verdict	Evidence
4 KPI explorer /v1/kpis/registry	CPO	`GET /v1/kpis/registry`	200 with values / registry	200 n=78	**PASS**	[{"code":"M01.CLASSIFICATION_ACCURACY","module":"M01","name":"Classification accuracy %","unit":"PCT","direction":"HIGHER_BETTER","formula":"(auto-classified + steward-accepted) / (auto-classified + s
4 KPI explorer /v1/kpis/M10.FIRST_PASS_MATCH	CPO	`GET /v1/kpis/M10.FIRST_PASS_MATCH`	200 with values / registry	200 n=0	**PASS**	{"code":"M10.FIRST_PASS_MATCH","module":"M10","name":"First-pass match %","unit":"PCT","direction":"HIGHER_BETTER","formula":"invoices matched without an exception / invoices matched","source":"invoic
4 KPI explorer /v1/kpis?module=M12&trend=false	CPO	`GET /v1/kpis?module=M12&trend=false`	200 with values / registry	200 n=4	**PASS**	{"period":null,"plant":null,"items":[{"code":"M12.AGENT_ASSISTED","module":"M12","name":"Agent-assisted transactions %","unit":"PCT","direction":"HIGHER_BETTER","formula":"agent-executed decisions / a
5a scheduled jobs for drift / retrain / KPI snapshot (C6, C7)	TENANT_ADMIN	`GET /v1/admin/jobs`	a job for KPI month-end snapshot and drift check	['audit-archive', 'erp-reconciliation', 'idempotency-purge', 'inventory-reservations', 'mrp-schedule', 'payment-due', 'retention-purge', 'scorecard-monthly', 'spend-cube-refresh', 'spend-forecast-snapshot', 'spend-quality', 'spend-sla']	**PARTIAL**	no KPI snapshot or drift job registered
5b manual KPI snapshot	TENANT_ADMIN	`POST /v1/admin/kpis/snapshot`	200 snapshots stored	200 {}{"period":"2026-12","kpis":73,"withoutValue":25}	**PASS**	
5c manual drift check	TENANT_ADMIN	`POST /v1/ai/models/{code}/drift-check`	200 result	200 {}{"code":"demand_forecast","checked":false,"reason":"no trained version is active","alerts":[]}	**PASS**	

Platform

Step	Actor	API call	Expected	Actual	Verdict	Evidence
P4a custom role PAY_DESK (payment:run) + invite + activate	TENANT_ADMIN, invitee	`POST /v1/admin/roles; POST /v1/admin/users; POST /v1/public/user-tokens/{t}/complete`	201 role, 201 user INVITED, 200 activated	role=201 {} activate=200	**PASS**	
P1a provision tenant with tools/tenant_admin.py	platform operator	`tenant_admin.create (CLI)`	tenant, company code, plant, admin INVITED, activation link	{'tenant': 'PLB', 'admin': 'owner@plb.example'}	**PASS**	audit TENANT PROVISIONED=[{'action': 'USER_INVITED'}, {'action': 'PROVISIONED'}]
P1b re-run provisioning	platform operator	`tenant_admin.create`	refused (exit 2)	ToolError	**PASS**	
P1c first admin activates and enrols MFA	TENANT_ADMIN	`GET/POST /v1/public/user-tokens/{t}; POST /v1/auth/token; POST /v1/me/mfa/totp(/confirm)`	link info 200, activate 200, reuse 410, MFA enrolment then token	info=200 act=200 reuse=410 token=403:MFA_ENROLMENT_REQUIRED	**PASS**	
P2 organisation setup (company code, plant, sloc, purch. org, purch. group)	TENANT_ADMIN	`POST /v1/admin/org/{kind}`	201 each; duplicate 409	{'cc': 201, 'plant': 201, 'sloc': 201, 'porg': 201, 'user': 201, 'pgrp': 201, 'dupCC': 409}	**PASS**	companyCodes=['1000', '2000'] plants=['P100', 'P200']
P3 invite and activate a user	TENANT_ADMIN, invitee	`POST /v1/admin/users; POST /v1/public/user-tokens/{t}/complete; POST /v1/auth/token`	INVITED -> active; sign-in; ABAC attributes	activate=200 token=200 roles=['BUYER'] plants=['P200']	**PASS**	
P4b custom roles and SoD scope-pair refusal	TENANT_ADMIN	`POST /v1/admin/roles; PUT /v1/admin/users/{id}/roles`	SOD-03 pair 409, SOD-11 pair 409, platform scope 422, clean role 201, assigning it to a buyer 409	PO+GR=409:SOD_CONFLICT COUNT+APPROVE=409:SOD_CONFLICT admin-scope=422 GR_DESK=201 assign=409:SOD_CONFLICT	**PASS**	
P8a audit chain verify (new tenant)	TENANT_ADMIN	`GET /v1/audit/verify`	valid true	200 {"valid": true}	**PASS**	
P8b tenant isolation (PLB sees none of FLB's invoices)	TENANT_ADMIN (PLB)	`GET /v1/invoices`	0 items	200 n=0	**PASS**	

Step	Actor	API call	Expected	Actual	Verdict	Evidence
P5a publish RCV-TOLERANCE version effective 10 Dec	TENANT_ADMIN	`POST /v1/admin/rules`	201	201 {"key": "RCV-TOLERANCE", "version": "2026.12-FLB", "effective_from": "2026-12-10"}	**PASS**	{"key":"RCV-TOLERANCE","version":"2026.12-FLB","company_code":null,"plant":null,"category":null,"purchase_type":null,"supplier_gro 12-10","effective_to":null,"body":{"ov
P5b same version label again	TENANT_ADMIN	`POST /v1/admin/rules`	409	409 RULE_VERSION_EXISTS RCV-TOLERANCE@2026.12-FLB already exists; give the new version its own label	**PASS**	
P5c effective date applied: PO of 8 Dec keeps 0%, PO of 10 Dec gets 10%	BUYER, RECEIVER	`POST /v1/purchase-orders; POST /v1/goods-receipts 105/100`	old PO refuses 105 (422); new PO accepts 105 (201)	old=422 new=201	**PASS**	[{'number': 'PO-77431', 'tolerance': {'over_pct': '0', 'price_pct': '0', 'under_pct': '0'}, 'rule_s MATRIX@2026.1;RCV-TOLERANCE@2026.1;PO-RISK@2026.1;DMN-02@2026.1'}, {'num {'over_pct': '10', 'price_pct': '0', 'under_pct': '0'}, 'rule_set_vers
P6 workflow definition draft > validate > publish; new GR runs on it	TENANT_ADMIN	`POST /v1/process-definitions/{key}/drafts|validate|publish`	published ACTIVE; new instance uses the new version; earlier instance keeps its version	key=pic.receipt draft=201 v2 valid=True publish=200:ACTIVE newGR=GR-90248 instV=[2] oldV=[1]	**PASS**	
P7a delegation of SERVICE_OWNER approvals	SERVICE_OWNER	`POST /v1/me/delegations`	201; event delegation.created	201 {"id": "01a1b6bb-40ba-707e-8f7a-fad58de2a532", "status": "ACTIVE"}	**PASS**	{"id":"01a1b6bb-40ba-707e-8f7a-fad58de2a532","delegatorId":"01a1b69b-282d-76a2- 50c7a332a71e","delegator":"Sam ServiceOwner","delegatorEmail":"svcowner@flb.exam 3e13-776
P7b delegate sees and approves the task	REQUESTER (delegate)	`GET /v1/tasks; POST /v1/tasks/{id}/complete`	task in delegate inbox; APPROVED; on_behalf_of audited	200 {} ses=[{'status': 'APPROVED'}]	**PASS**	[{'actor': 'u:deleg@flb.example', 'on_behalf_of': None, 'action': 'APPROVE'}, {'actor': 'u:de 'on_behalf_of': 'u:svcowner@flb.example', 'action': 'APPROVAL_APPROVE'}, {'actor': 'u:pc 'on_behalf_of': None, 'action': 'TASK_CREATED'}]
P7c SLA escalation by the worker (reminder, then manager role)	worker	`timers TASK_SLA via POST /v1/admin/ops/tick`	level 1 reminder, level 2 DEPT_HEAD added, task in DEPT_HEAD inbox; TASK_ESCALATED notification	due=[{'due_at': datetime.datetime(2026, 12, 16, 3, 30, tzinfo=datetime.timezone.utc), 'escalation_level': 0}] levels=[(1, 'SERVICE_OWNER', []), (1, 'SERVICE_OWNER', []), (2, 'SERVICE_OWNER', ['escalatedTo'])] depthead inbox=1	**PASS**	notifications=[{'kind': 'TASK_ESCALATED', 'count': 5}]
P8c audit hash chain verification (FLB after all flows)	TENANT_ADMIN	`GET /v1/audit/verify`	valid true	200 {"valid": true, "checked": 753}	**PASS**	{"valid":true,"checked":753,"head":"5e3d30fbdf8831cdbe358c10d2532a286e8ed7dfaf35
P9a idempotent replay and key reuse	RECEIVER	`POST /v1/stock-postings Idempotency-Key`	same doc replayed (Idempotent-Replayed); other body 422 IDEMPOTENCY_KEY_REUSED	first=201 {"number": "STP-1005"} replay=201 {"number": "STP-1005", "replayed": true} hdr=None reuse=422 IDEMPOTENCY_KEY_REUSED This Idempotency-Key was already used with a different request body	**PASS**	
P9b stale If-Match on a master write	SUPPLIER_MANAGER	`PUT /v1/suppliers/{code}/finance If-Match stale`	412 STALE_WRITE	412 STALE_WRITE The document is at version 2; your If-Match is 1. Reload it and retry.	**PASS**	
P10 outbox drained by the worker	TENANT_ADMIN	`POST /v1/admin/ops/tick`	pending events -> 0, no failures	before=3 tick={'events': 3, 'integration': 0, 'timers': 0, 'jobs': 0} after=0 failing=0	**PASS**	
P11 failed ERP message (stock posting STOCK_POSTING) visible in the monitor and replayed	TENANT_ADMIN	`GET /v1/integration/messages?status=RETRY; POST /v1/integration/messages/{id}/replay`	RETRY listed; replay POSTED; replay again 409; audit INTEGRATION_REPLAY	gr=201 msg=[('RETRY', 1)] listed=True replay=200 {"status": "POSTED", "erp_doc_no": "4900000008"} again=409	**PASS**	audit=['INTEGRATION_REPLAY', 'ERP_POSTED']
P12a job run on demand (payment-due): dry run and real run	TENANT_ADMIN	`POST /v1/admin/jobs/{key}/run; GET .../runs`	dry run result; real run SUCCESS; proposal task, never pays	dry=200 {"status": "SUCCESS"} {'horizon': '2026-12-25', 'due': 0, 'totalMinor': 0, 'rule': 'PAYMENT-RUN@2026.1', 'discountWindows': 0, 'action': 'none run=200 {"status": "SUCCESS"} {'horizon': '2026-12-25', 'due': 0, 'totalMinor': 0, 'rule': 'PAYMENT-RUN@2026.1', 'discountWindows': 0, 'action': 'non	**PASS**	task=[]
P12b jobs.run_due picks the due jobs (worker path)	worker	`jobs.run_due(tenant)`	due jobs run once, single-flight	ran=0 last=[('payment-due', 'SUCCESS'), ('payment-due', 'SUCCESS'), ('spend-cube-refresh', 'SUCCESS'), ('spend-cube-refresh', 'SUCCESS'), ('spend-cube-refresh', 'SUCCESS'), ('spend-sla', 'SUCCESS')]	**PASS**	

Step	Actor	API call	Expected	Actual	Verdict	Evidence
P12c inbound ERP sync jobs registered (A14)	TENANT_ADMIN	`GET /v1/admin/jobs`	sync_masters / sync_stock / sync_open_pos / sync_payment_status jobs	['audit-archive', 'erp-reconciliation', 'idempotency-purge', 'inventory-reservations', 'mrp-schedule', 'payment-due', 'retention-purge', 'scorecard-monthly', 'spend-cube-refresh', 'spend-forecast-snapshot', 'spend-quality', 'spend-sla']	**PARTIAL**	no erp-sync job key: syncs run only on demand
P12d inbound ERP sync on demand	TENANT_ADMIN	`POST /v1/admin/erp-sync/{kind}/run`	200 run recorded (simulator)	200 {}{"id":"01a1b6bc-f182-70cc-b8e4-4a0924029c07","number":"SYN-261223-064500-D92D","system":"SAP_S4","kind":"PAYMENT_STATUS","label":"Payment status","mod	**PASS**	
P13 M13 KPIs	TENANT_ADMIN	`GET /v1/kpis?module=M13`	values	{'M13.INTEGRATION_SUCCESS': '100.0', 'M13.MTTR_INTEGRATION': None, 'M13.SOD_OPEN': '5'}	**PASS**	
P14 readiness	-	`GET /readyz`	200 with checks	200 {}{"status":"ready","schema":"0022","checks":{"database":"ok","schema":{"status":"ok","revision":"0022"},"worker":{"status":"absent","instances":0,"hear	**PASS**	